Digital Notes

MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

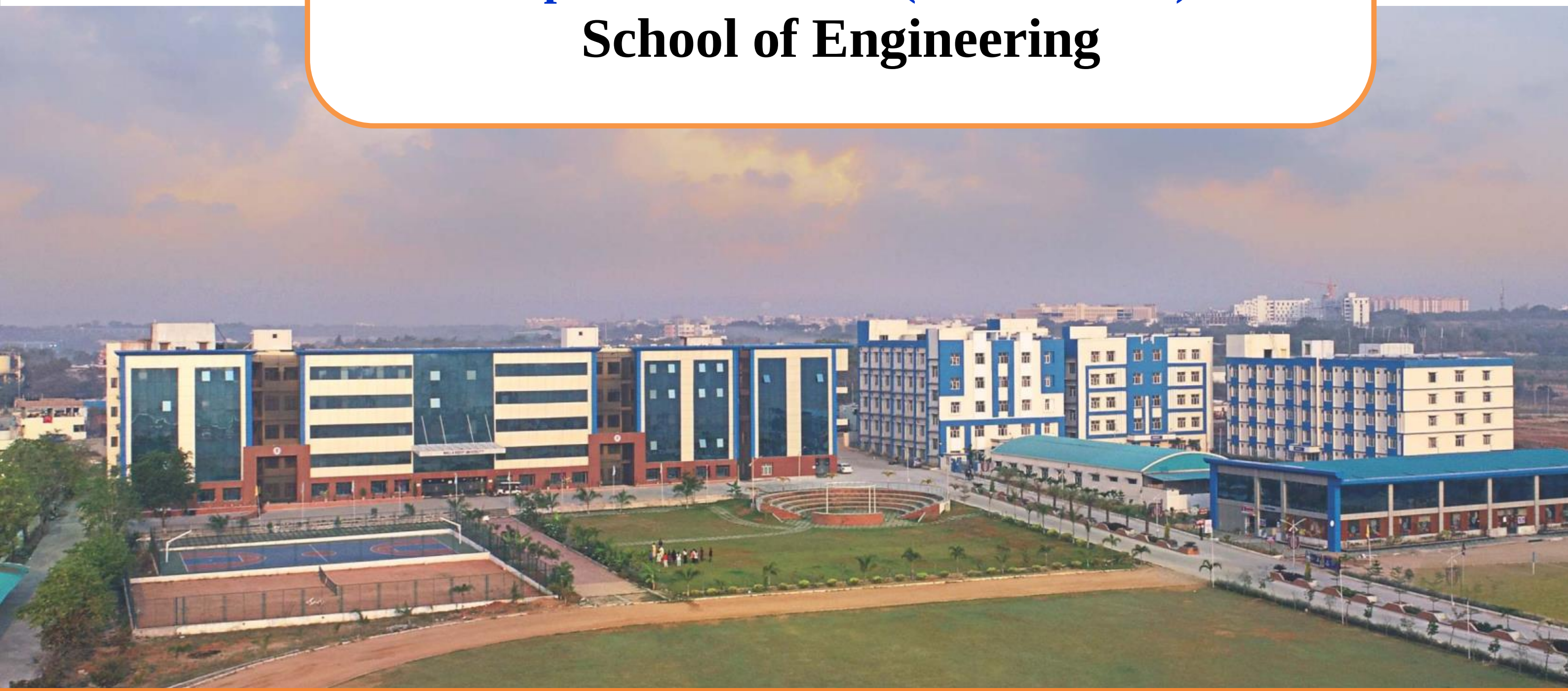
Computer Networks

II B.Tech I Semester

(2024-25)

Department of CSE (Data Science)

School of Engineering



Unit 1

Introduction to Computer Networks

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Introduction

- Each of the past three centuries was dominated by a single new technology.
- The 18th century was the era of the great mechanical systems accompanying the Industrial Revolution.
- The 19th century was the age of the steam engine.
- During the 20th century, the key technology was information gathering, processing, and distribution.
- Among other developments, we saw the installation of worldwide telephone networks, the invention of radio and television, the birth and unprecedented growth of the computer industry, the launching of communication satellites, and, of course, the Internet.
- As a result of rapid technological progress, these areas are rapidly converging in the 21st century and the differences between collecting, transporting, storing, and processing information are quickly disappearing.

- The old model of a single computer serving all of the organization’s computational needs has been replaced by one in which a large number of separate but interconnected computers do the job. These systems are called **computer networks**.
- **Computer network** means a collection of autonomous computers interconnected by a single technology.
- Two computers are said to be interconnected if they are able to exchange information.
- The computers on a network may be linked through cables, telephone links, radio waves, satellites, or infrared light beams.
- Devices in a Network are connected using wired or wireless transmission media such as cable or air.

Basic Set of Rules for Computer Network

- Information must be delivered reliably
- Information must be delivered consistently [unchanging]
- Multiple computers must be able to identity each other

Benefits of computer network:

- Network can increase efficiency
- It can help Standardize Policies, Procedures and Processes
- It Ensures Information consistency and Integrity
- It Ensures information security

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Uses of network

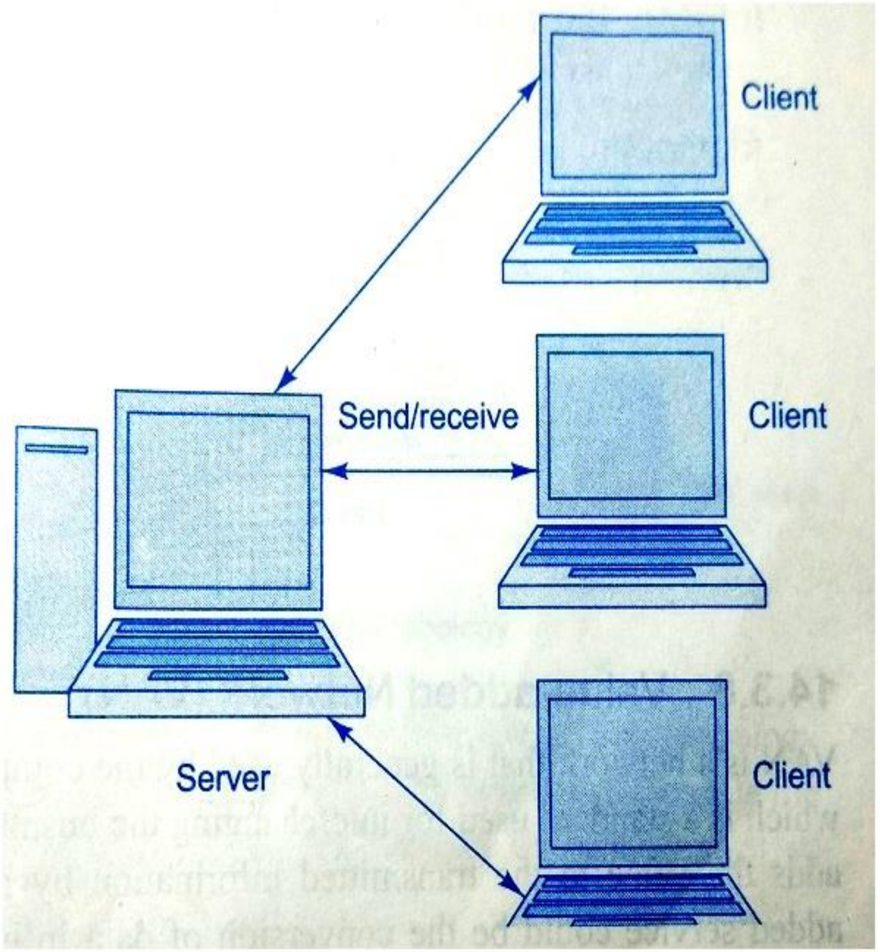
- Simultaneous access to data

Ex: data files, Software, hardware resources

- File Server contain documents which are used by other computers
- Personal communications

Ex: email, conferencing, voice over IP

- Easy data backup



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Applications of Networks

- **Resource Sharing**
  - ✓ Hardware (computing resources, disks, printers)
  - ✓ Software (application software)
- **Information Sharing**
  - ✓ Easy accessibility from anywhere (files, databases)
  - ✓ Search Capability (WWW)
- **Communication**
  - ✓ Email
  - ✓ Message broadcast
- **Remote computing**
- **Distributed processing (GRID Computing)**

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Network Topologies

- Logical layout of wires and equipment is called ***topology***.

Topology: Shape of the network

- The way in which devices are interconnected to form a network is called **network topology**.

- Some of the factors that affect choice of topology for a network are

  - ✓ Cost
  - ✓ Scalability
  - ✓ Flexibility
  - ✓ Reliability
  - ✓ Ease of installation
  - ✓ Ease of maintenance

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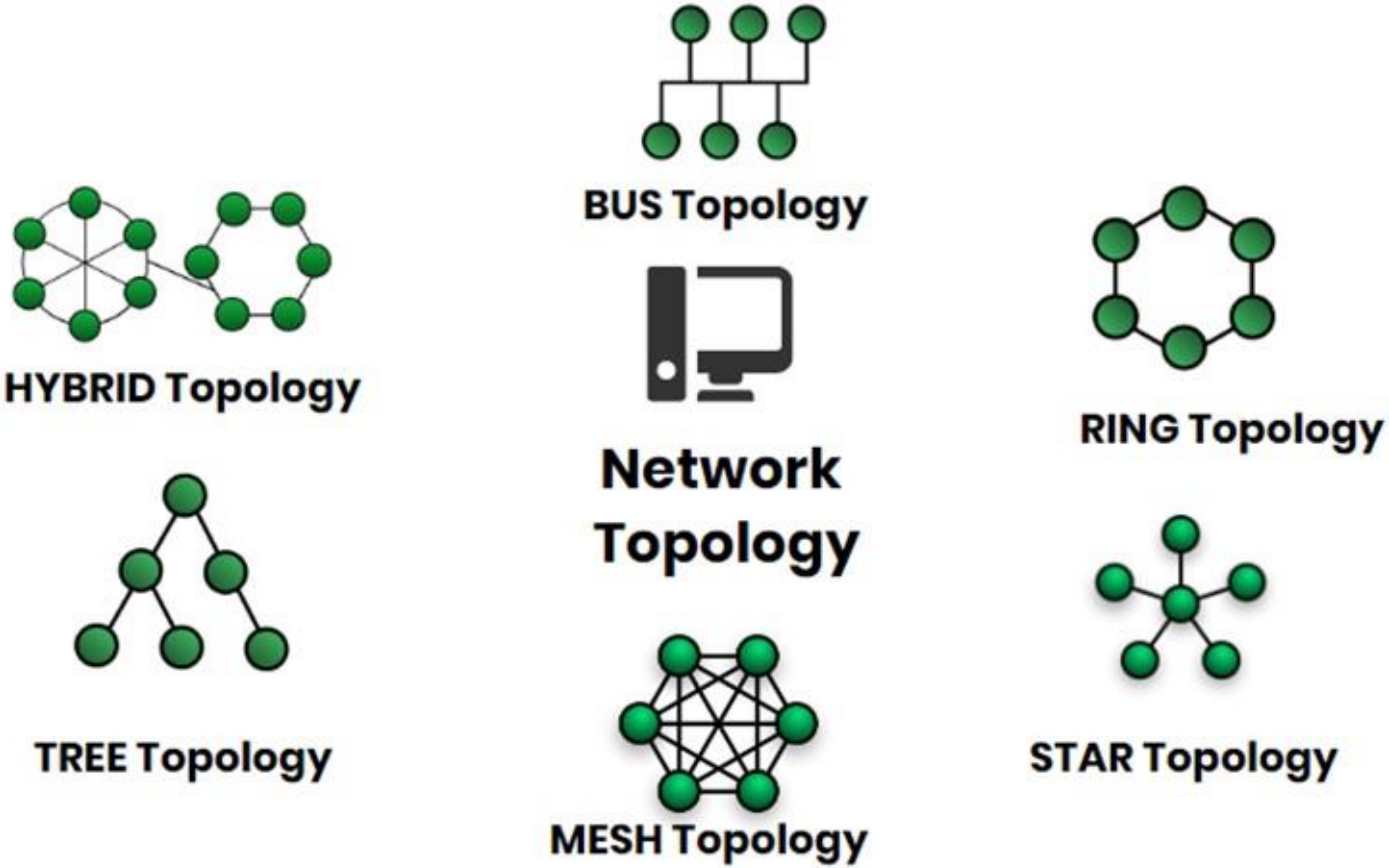
BUS Topology
HYBRID Topology
RING Topology
Network
Topology
TREE Topology
STAR Topology
MESH Topology

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Bus Topology

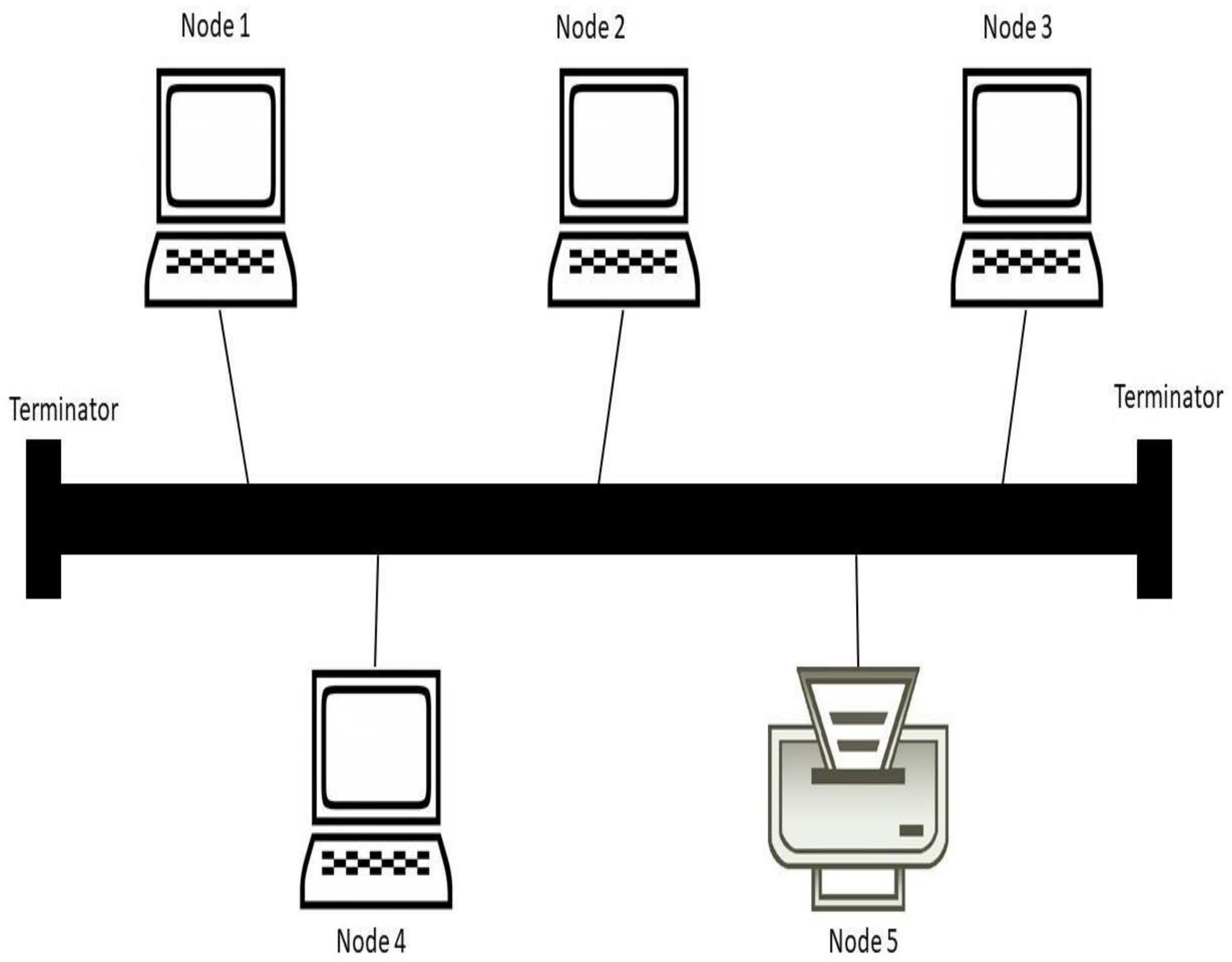
- It is also called linear bus
- One wire connects all nodes

Advantages:

- Easy to setup
- Require less hardware
- Low cost

Disadvantages:

- Slow
- Easy to crash
- If one system effected remaining systems cannot communicate with each other



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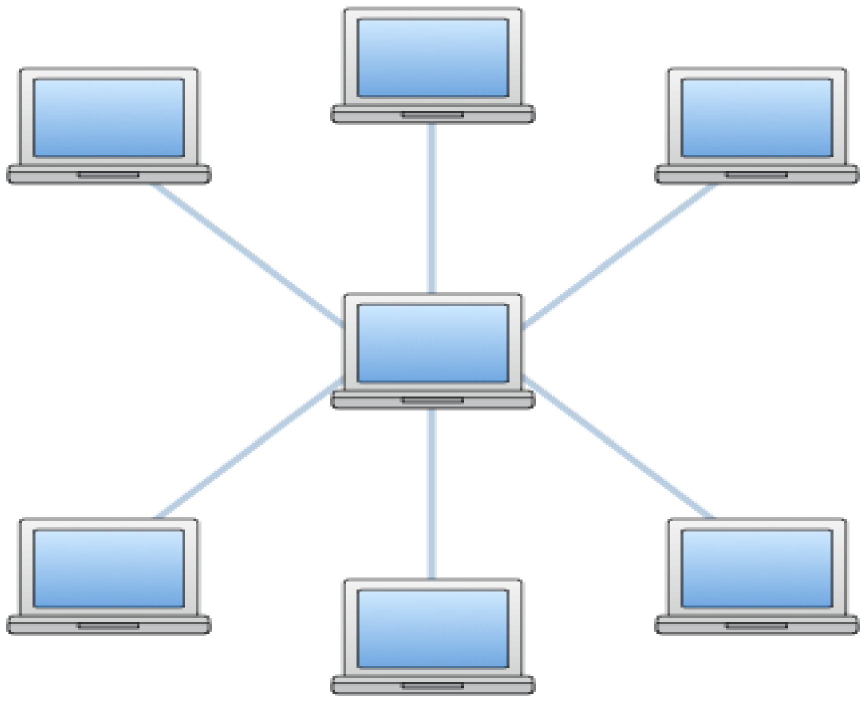
Star Topology

- A star topology is a topology for a Local Area Network (LAN) in which server is connected to each node individually.
- Server is also called the central node.
- Any exchange of data between two nodes must take place through the server.



Advantages

- Failure of one node does not affect the network
- Troubleshooting is easy as faulty node can be detected from central node immediately

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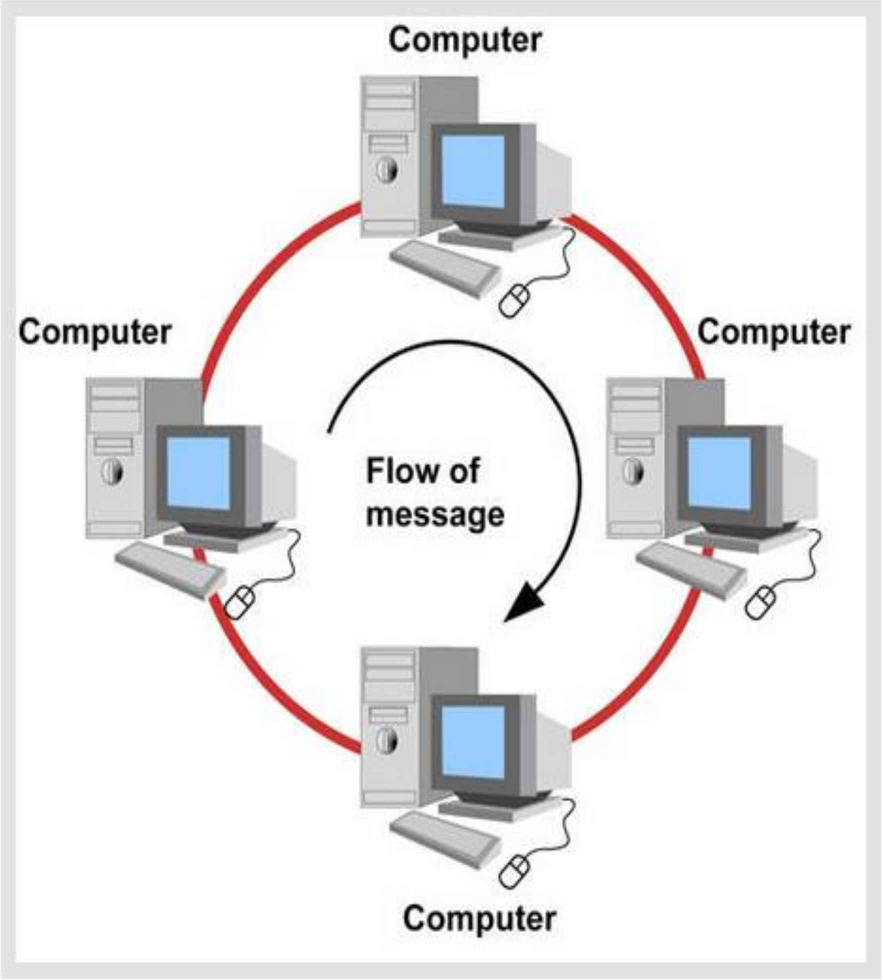
Ring Topology

- In ring topology each terminal is connected to exactly two nodes, giving the network a circular shape.
- Data travels in only one pre-determined direction.



Advantages:

- Very high transmission speeds possible
- Small cable segments are needed to connect two nodes

Disadvantages :

- Failure of single node brings down the whole network
- Troubleshooting is difficult as many nodes may have to be inspected before faulty one is identified
- Difficult to remove one or more nodes while keeping the rest of the network intact

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Tree Topology

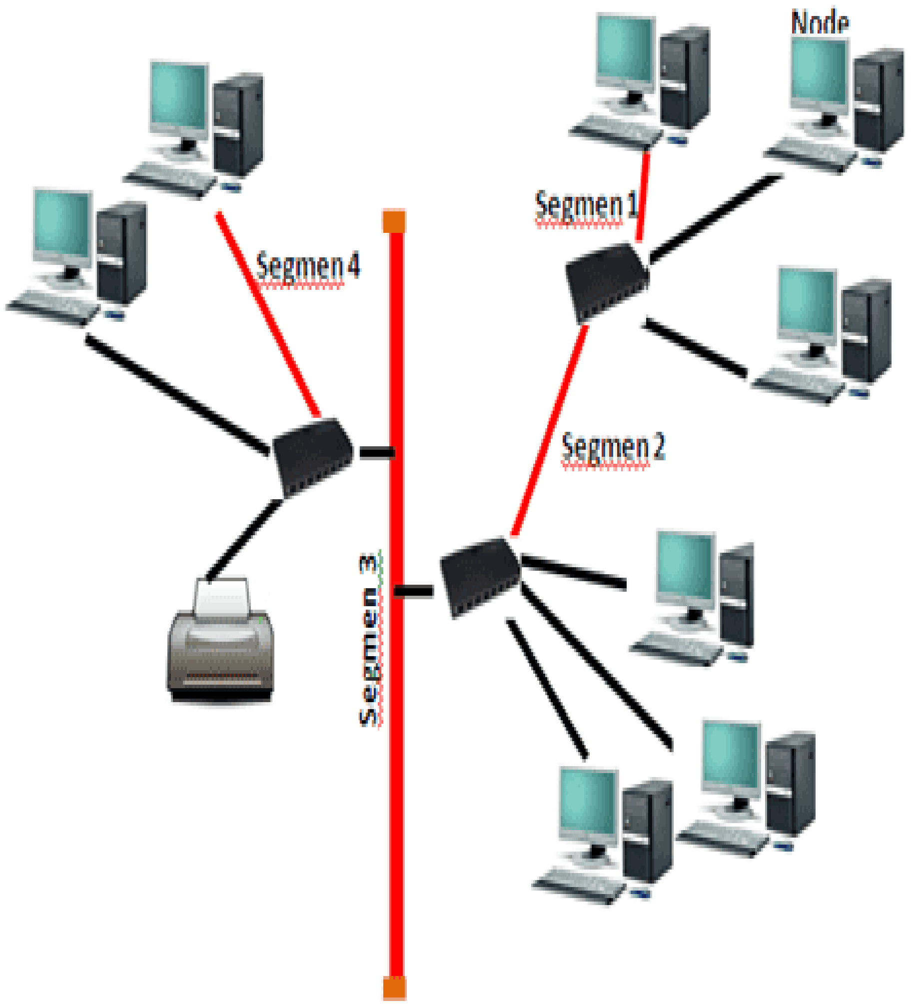
- Tree topology has a group of star networks connected to a linear bus backbone cable.
- It incorporates features of both star and bus topologies.
- Tree topology is also called hierarchical topology.

Advantages

- Existing network can be easily expanded
- Easier installation and maintenance
- Well suited for temporary networks

Disadvantages

- Failure of backbone cable brings down entire network
- Insecure network
- Maintenance difficult for large networks



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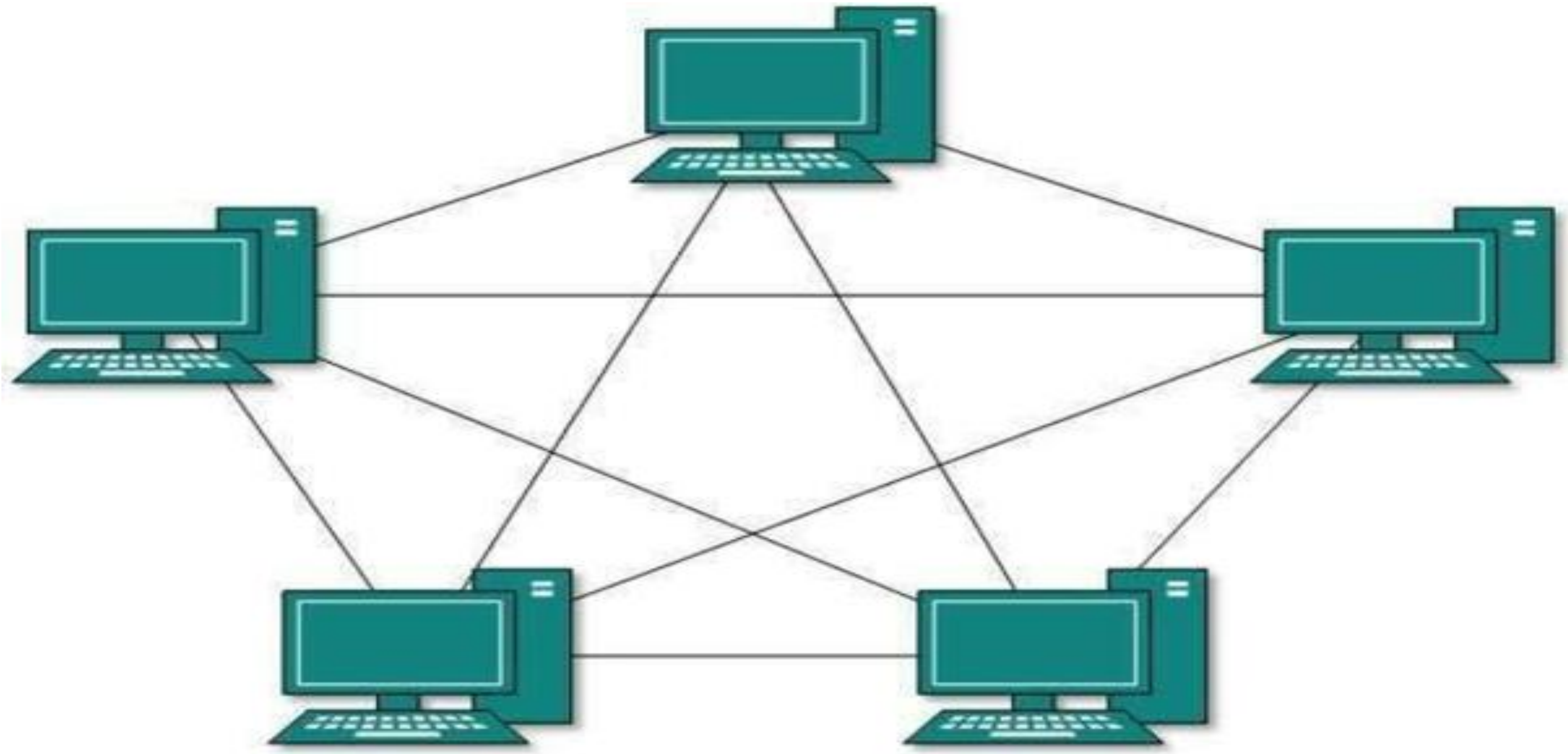
Mesh Topology

- In this type of topology, a host is connected to one or multiple hosts.
- This topology has hosts in point-to-point connection with every other host
- It provides the most reliable network structure among all network topologies.



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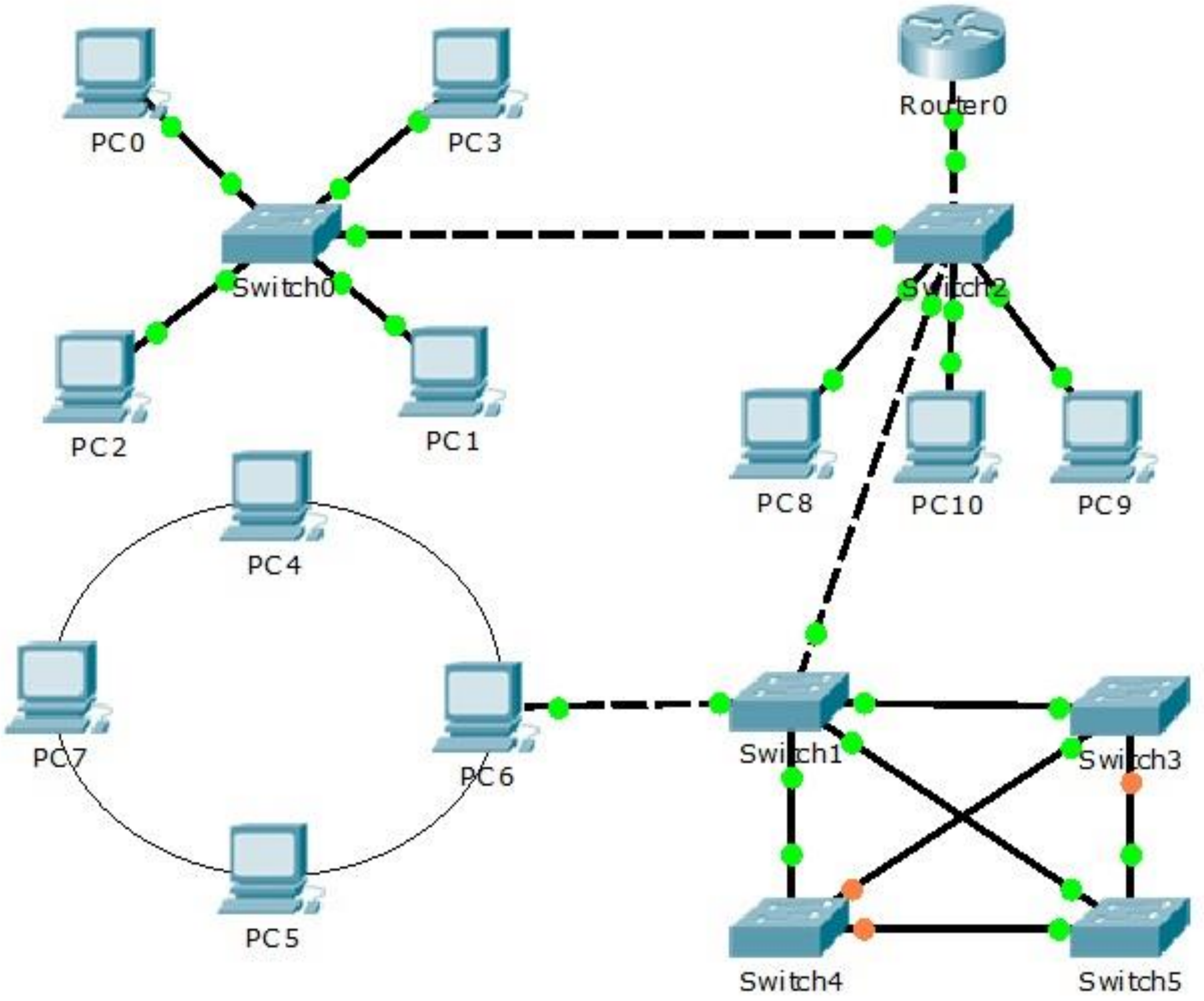
Hybrid Topology

It is the combination of two or more different topologies.



Advantages:

- It is extremely flexible.
- It is very reliable.
- Error detecting and troubleshooting are easy.
- Handles a large volume of traffic.
- It is used to create large networks.

Disadvantages:

- It is a type of network expensive.
- The design of a hybrid network is very complex.
- There is a change in the hardware to connect one topology with another topology.

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Network Hardware

Network Hardware can classify either by ***transmission technology*** or by ***scale***

- There are two types of transmission technology that are in widespread use: **broadcast** links and **point-to-point** links.
- Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called **unicasting**.
- *In contrast, on a broadcast network, the communication channel is shared by all the machines on the network; packets sent by any machine are received by all the others. An address field within each packet specifies the intended recipient. Upon receiving a packet, a machine checks the address field. If the packet is intended for the receiving machine, that machine processes the packet; if the packet is intended for some other machine, it is just ignored.*
- Broadcast systems usually allow the possibility of addressing a packet to *all* destinations by using a special code in the address field. This mode of operation is called **broadcasting**.
- Some broadcast systems also support transmission to a subset of the machines, which known as **multicasting**.

- An alternative criterion for classifying networks is by ***scale***.
- Distance is important as a classification metric because different technologies are used at different scales.

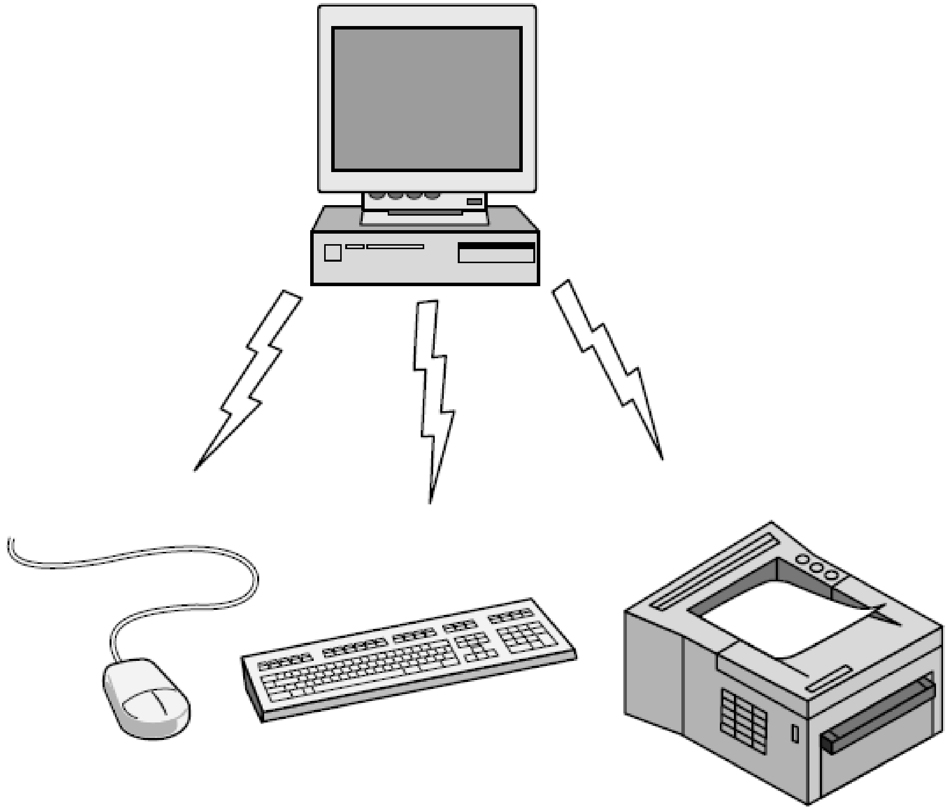
Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	Local area network
1 km	Campus	Local area network
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	Wide area network
10,000 km	Planet	The Internet

Classification of interconnected processors by scale.

Personal Area Networks

- **PANs** (**Personal Area Networks**) let devices communicate over the range of a person. A common example is a wireless network that connects a computer with its peripherals.
- Almost every computer has an attached monitor, keyboard, mouse, and printer. Without using wireless, this connection must be done with cables.
- Also short-range wireless network called **Bluetooth** to connect these components without wires.
  - Bluetooth networks use the master-slave paradigm

Bluetooth PAN configuration

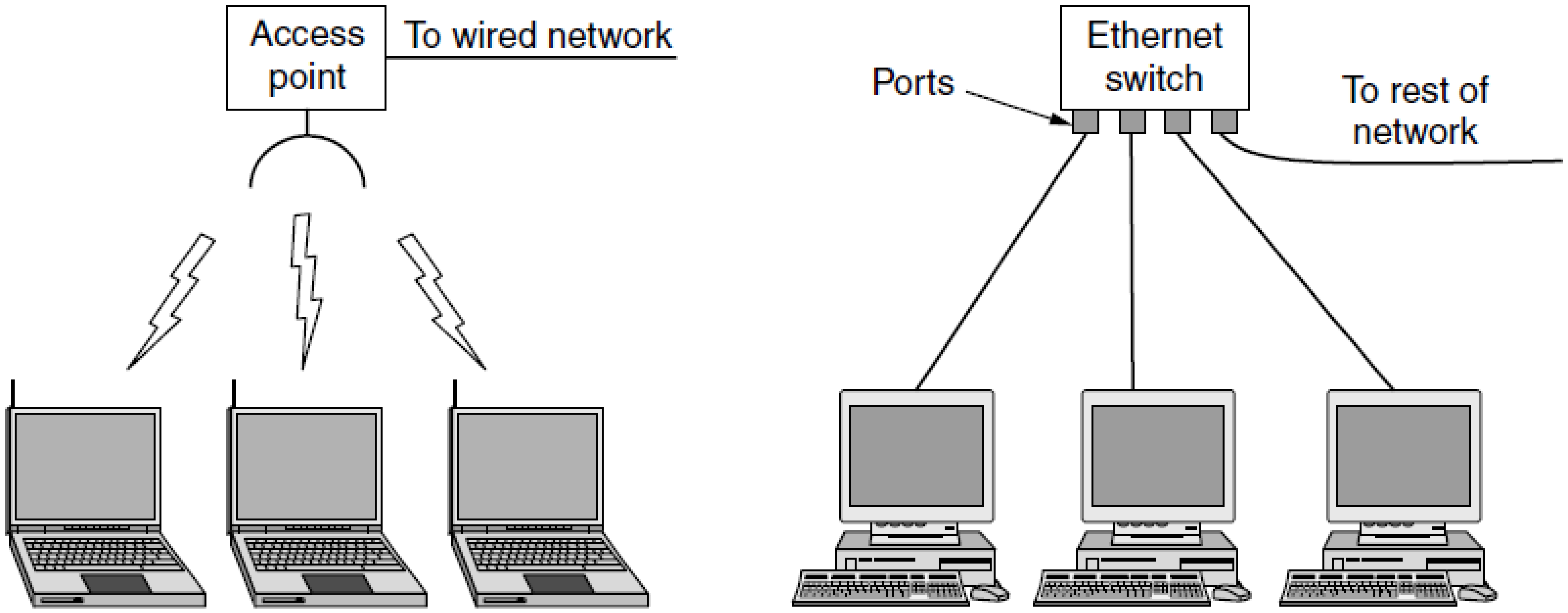


Local Area Networks

- A LAN is a privately owned network that operates within and nearby a single building like a home, office or factory.
- When LANs are used by companies, they are called **enterprise networks**.
- Wireless LANs are very popular these days where it is too much trouble to install cables.
- In these systems, every computer has a ***radio modem*** and an ***antenna*** that it uses to communicate with other computers.
- This device, called an **AP** (**Access Point**), **wireless router**, or **base station**, relays packets between the wireless computers and also between them and the Internet.
- There is a standard for wireless LANs called **IEEE 802.11**, popularly known as **WiFi.**
- It runs at speeds anywhere from 11 to hundreds of Mbps.
- Wired LANs use a range of different transmission technologies like copper wires, optical fiber etc., and LANs are restricted in size
- The topology of many wired LANs is built from point-to-point links. IEEE 802.3, popularly called **Ethernet**

Features

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher security.



Wireless and wired LANs. (a) 802.11. (b) Switched Ethernet.

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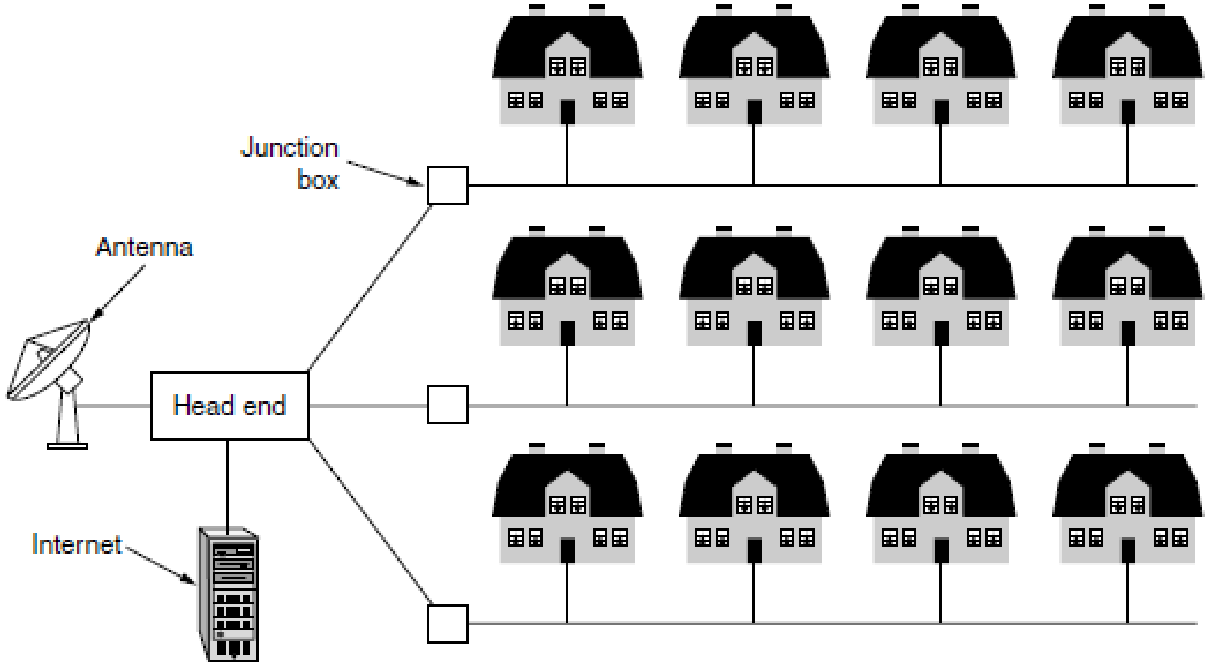
Metropolitan Area Networks

A **MAN** (**Metropolitan Area Network**) covers a city.

- Design to extend over a large area.
- Connecting number of LAN's to form larger network, so that resources can be shared.
- Networks can be up to 5 to 50 km.
- Owned by organization or individual.
- Data transfer rate is low compare to LAN.

Example: Organization with different branches located in the city.



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Uses Of Metropolitan Area Network:

- MAN is used in communication between the banks in a city.
- It can be used in an Airline Reservation.
- It can be used in a college within a city.
- It can also be used for communication in the military.

Wide Area Networks

- A **WAN** (**Wide Area Network**) spans a large geographical area, often a country or continent
- The network that connects these hosts is then called the **communication subnet**, or just **subnet** for short.
- The job of the subnet is to carry messages from host to host
- In most WANs, the subnet consists of two distinct components: transmission lines and switching elements. **Transmission lines** move bits between machines.
- **Switching elements**, or just **switches**, are specialized computers that connect two or more transmission lines.
  - When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them.
  - These switching computers have been called by the name **router** (Unfortunately, some people pronounce it ‘‘rooter’’)

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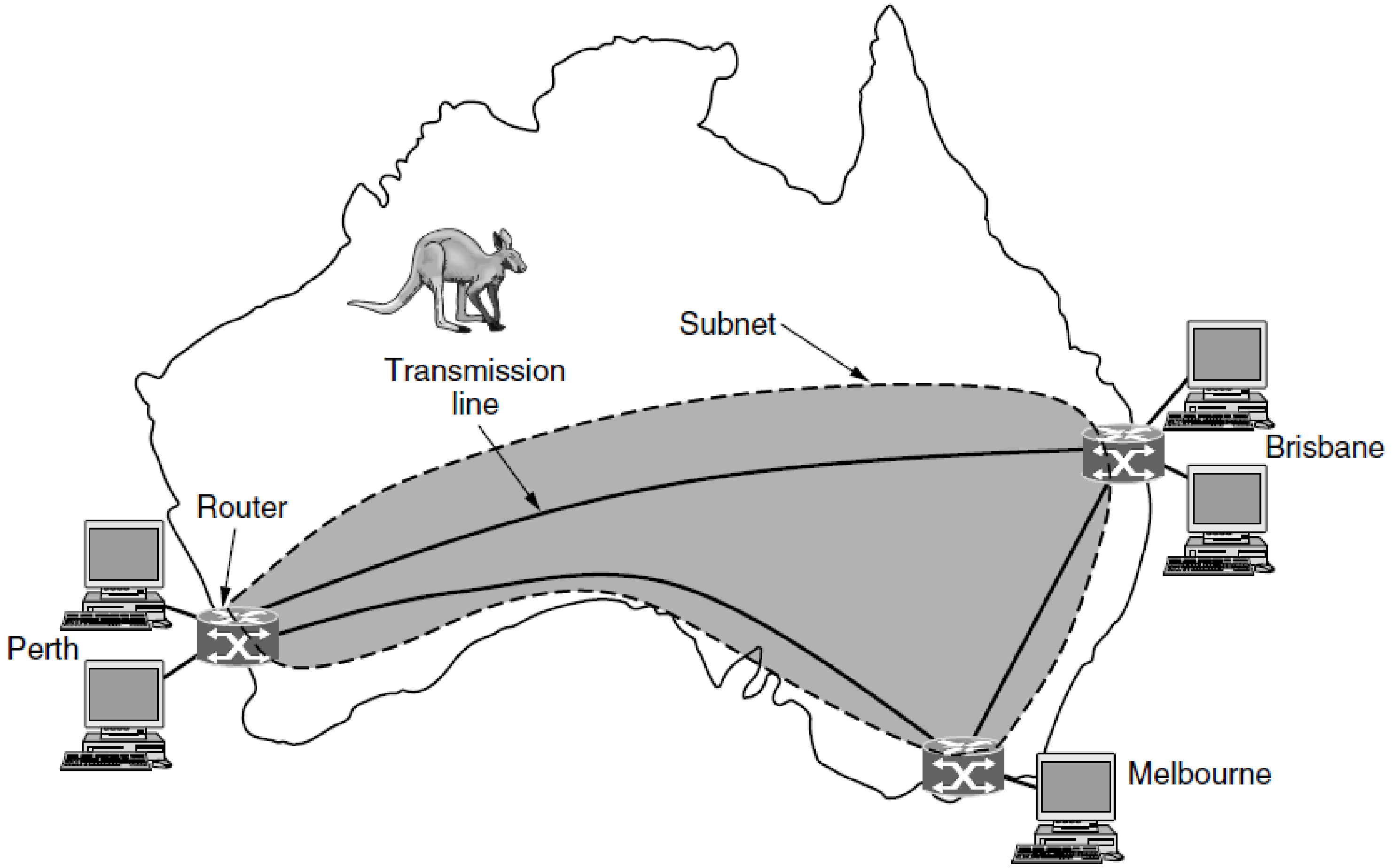
Originally subnet is the collection of routers and communication lines that moves packets from the source host to the destination host.

- In a WAN, the hosts and subnet are owned and operated by different people.
- The WAN in Fig. is a network that connects offices in Perth, Melbourne, and Brisbane. Each of these offices contains computers intended for running user (i.e., application) programs.



WAN that connects three branch offices in Australia.

Features of WAN

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.

NETWORK SOFTWARE

- The first computer networks were designed with the hardware as the main concern and the software as an afterthought.
- This strategy no longer works. Network software is now highly structured.

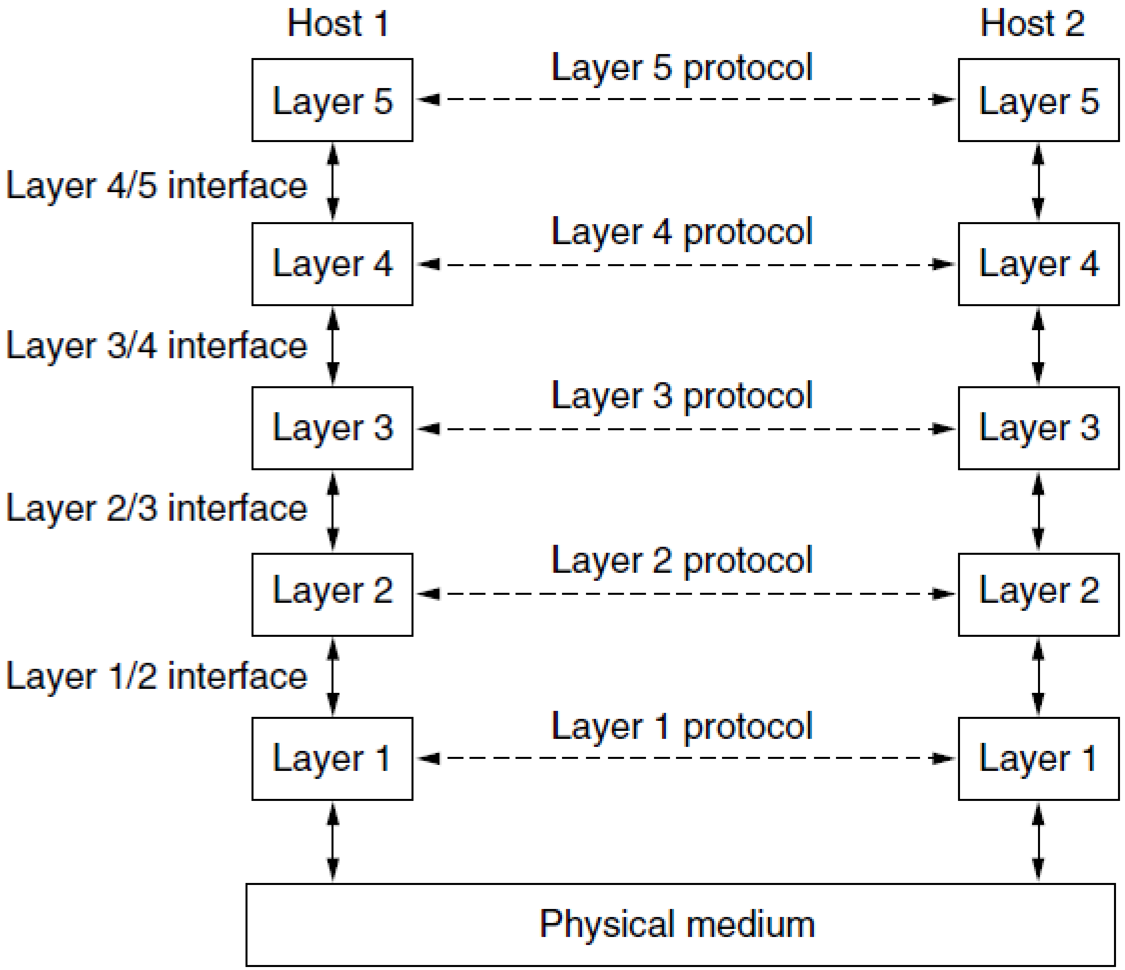
Functions of Network Software

- Helps to set up and install computer networks
- Enables users to have access to network resources in a seamless manner
- Allows administrations to add or remove users from the network
- Helps to define locations of data storage and allows users to access that data
- Helps administrators and security system to protect the network from data breaches, unauthorized access and attacks on a network
- Enables network virtualizations

Protocol Hierarchies

- To reduce their design complexity, most networks are organized as a stack of **layers** or **levels**, each one built upon the one below it.
- The purpose of each layer is to offer certain services to the higher layers
- When layer *n* on one machine carries on a conversation with layer *n* on another machine, the rules and conventions used in this conversation are collectively known as the layer *n* **protocol**.
- Basically, a **protocol** is an agreement between the communicating parties on how communication is to proceed.
- A five-layer network is illustrated in the following Fig. The entities comprising the corresponding layers on different machines are called **peers**.

Layers, protocols, and interfaces.



- In reality, no data are directly transferred from layer *n* on one machine to layer *n* on another machine.
- Below layer 1 is the **physical medium** through which actual communication occurs.
- Between each pair of adjacent layers is an **interface**. The interface defines which primitive operations and services the lower layer makes available to the upper one.

Design Issues for the Layers

- **Reliability:** is the design issue of making a network that operates correctly even though it is made up of a collection of components that are themselves unreliable.
- **Scalability:** Networks are continuously evolving. The sizes are continually increasing leading to congestion. Hence, the design should be done so that the networks are scalable and can accommodate such additions and alterations.
- **Addressing:** At a particular time, innumerable messages are being transferred between large numbers of computers. So, a naming or addressing system should exist so that each layer can identify the sender and receivers of each message.
- **Error Control:** Unreliable channels introduce a number of errors in the data streams that are communicated. So, the layers need to agree upon common error detection and error correction methods
- **Flow Control:** If the rate at which data is produced by the sender is higher than the rate at which data is received by the receiver, a proper flow control mechanism needs to be implemented.

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- **Routing:** There may be multiple paths from the source to the destination. Routing involves choosing an optimal path among all possible paths
- **Congestion:** overloading of the network because too many computers want to send too much traffic
- **Quality of service:** provide service to applications that want high throughput
- **Security:** Major factor of data communication is to defend it against threats like eavesdropping and surreptitious alteration of messages by ***confidentiality*** and ***authentication***

Connection-Oriented Versus Connectionless Service

Connection-Oriented Service

A connection-oriented service is a network service that was designed and developed after the telephone system. A connection-oriented service is used to create an end to end connection between the sender and the receiver before transmitting the data over the same or different networks. In connection-oriented service, packets are transmitted to the receiver in the same order the sender has sent them. It uses a handshake method that creates a connection between the user and sender for transmitting the data over the network. Hence it is also known as a reliable network service.

Connectionless Service

A connection is similar to a **postal system**, in which each letter takes along different route paths from the source to the destination address. Connectionless service is used in the network system to transfer data from one end to another end without creating any connection. So it does not require establishing a connection before sending the data from the sender to the receiver. It is not a reliable network service because it does not guarantee the transfer of data packets to the receiver, and data packets can be received in any order to the receiver.

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

Six different types of service.

Service Primitives

- A service is formally specified by a set of **primitives** (operations) available to user processes to access the service.
- These primitives tell the service to perform some action or report on an action taken by a peer entity.
- The set of primitives available depends on the nature of the service being provided.
- The primitives for connection-oriented service are different from those of cconnectionless service.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Six service primitives that provide a simple connection-oriented service.

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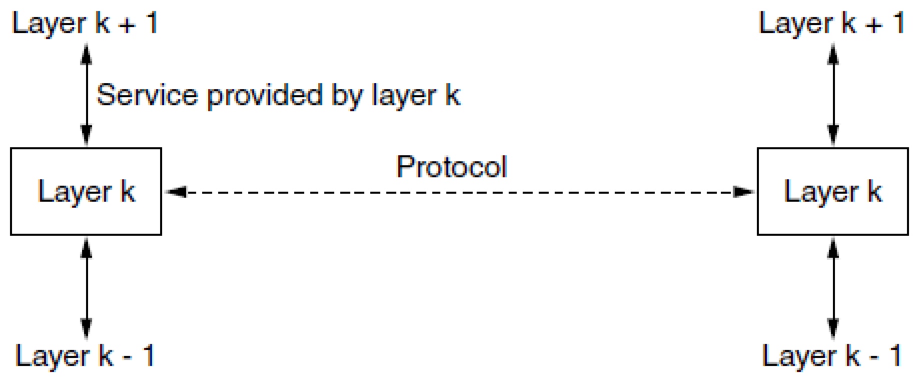
The Relationship of Services to Protocols

Services and protocols are distinct concepts.

- A *service* is a set of primitives (operations) that a layer provides to the layer above it.
  - A service relates to an interface between two layers, with the lower layer being the service provider and the upper layer being the service user.
- A *protocol*, in contrast, is a set of rules governing the format and meaning of the packets, or messages that are exchanged by the peer entities within a layer.



The relationship between a service and a protocol

service defines operations that can be performed on an object but does not specify how these operations are implemented. In contrast, a ***protocol*** relates to the *implementation* of the service

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REFERENCE MODELS

- The two important network architectures:

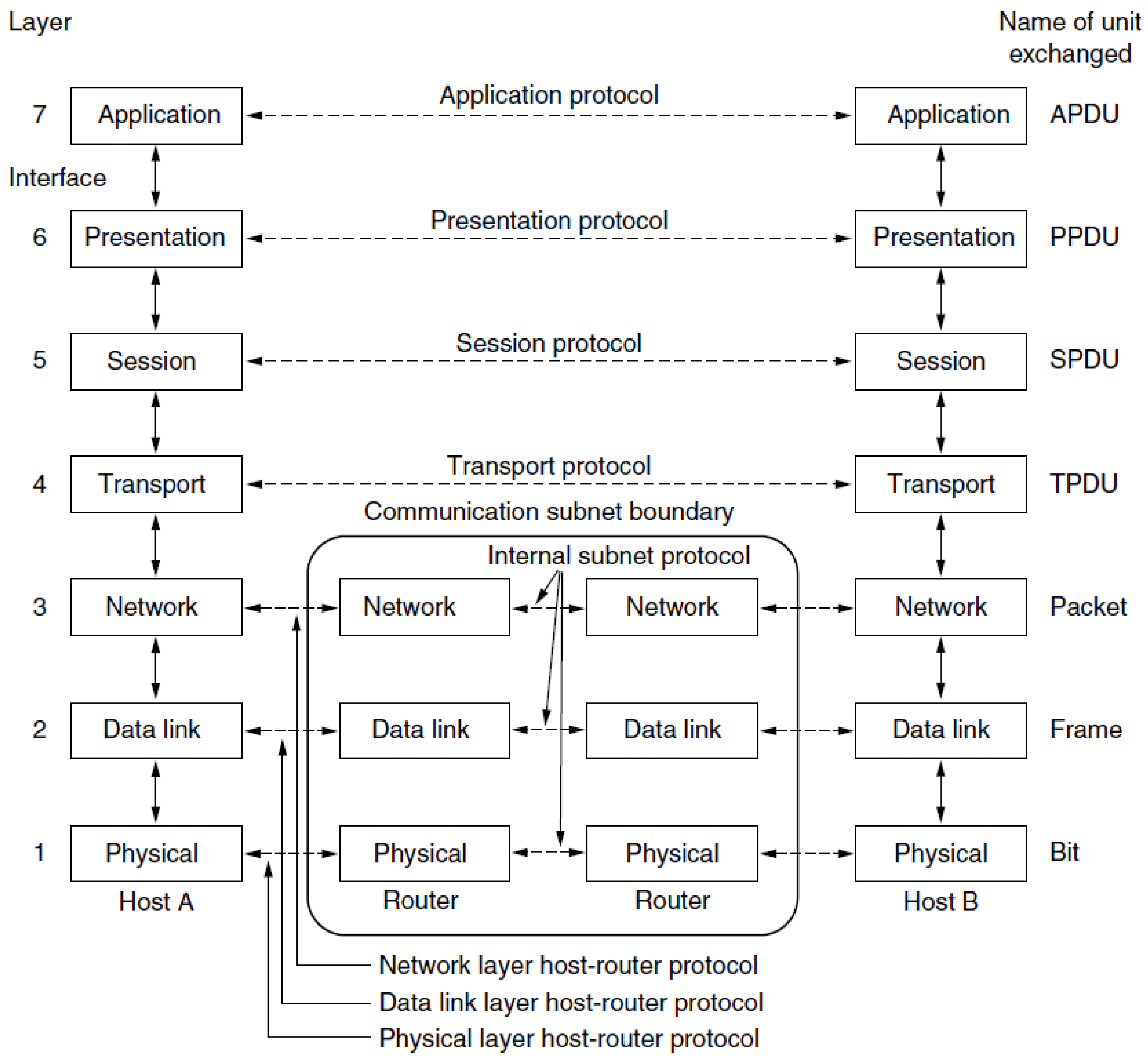
The OSI reference model and The TCP/IP reference model.

The OSI Reference Model

- This model is based on a proposal developed by the International Standards Organization (ISO) as a first step toward international standardization of the protocols used in the various layers
- The model is called the ISO **OSI** (**Open Systems Interconnection**) Reference Model because it deals with connecting open systems—that is, systems that are open for communication with other systems. We will just call it the **OSI model** for short.
- The OSI model has seven layers.

OSI Reference Model principles:

- 1. A layer should be created where a different abstraction is needed.
- 2. Each layer should perform a well-defined function.
- 3. The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
- 4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
- 5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.


Layer
Name of unit exchanged
7
Application
Application protocol
Application
APDU
Interface
6
Presentation
Presentation protocol
Presentation
PPDU
5
Session
Session protocol
Session
SPDU
4
Transport
Transport protocol
Transport
TPDU
Communication subnet boundary
Internal subnet protocol
3
Network
Network
Network
Network
Packet
2
Data link
Data link
Data link
Data link
Frame
1
Physical
Physical
Physical
Physical
Bit
Host A
Router
Router
Host B
Network layer host-router protocol
Data link layer host-router protocol
Physical layer host-router protocol

The Physical Layer

- The **physical layer** is concerned with transmitting raw bits over a communication channel.

The Data Link Layer

- The main task of the **data link layer** is to transform a raw transmission facility into a line that appears free of undetected transmission errors.
- It accomplishes this task by having the sender break up the input data into **data frames** (typically a few hundred or a few thousand bytes) and transmit the frames sequentially.
- If the service is reliable, the receiver confirms correct receipt of each frame by sending back an **acknowledgement frame**.
- Another issue is how to keep a fast transmitter from drowning a slow receiver in data. Some traffic regulation mechanism like Flow control is needed
- Broadcast networks have an additional issue to control access to the shared channel. A special sublayer of the data link layer, the **medium access control** sublayer, deals with this problem.

The Network Layer

- The **network layer** controls the operation of the subnet.
- A key design issue is determining how packets are ***routed*** from source to destination.
- Handling ***congestion*** is also a responsibility of the network layer, in conjunction with higher layers that adapt the load
- More generally, the ***quality of service*** provided (delay, transit time, jitter, etc.) is also a network layer issue.

The Transport Layer

- The basic function of the **transport layer** is to accept data from above it, split it up into smaller units call ***segments***
- The transport layer also determines what type of service to provide to the session layer
- This layer can be termed as an end-to-end layer as it provides a point-to-point connection between source and destination to deliver the data reliably.

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The Session Layer

- The session layer allows users on different machines to establish **sessions** between them.
- Sessions offer various services, including
- **dialog control:** keeping track of whose turn it is to transmit
- **token management:** preventing two parties from attempting the same critical operation simultaneously
- **Synchronization:** checkpointing long transmissions to allow them to pick up from where they left off in the event of a crash and subsequent recovery

The Presentation Layer

- The **presentation layer** is concerned with the syntax and semantics of the information transmitted.

The Application Layer

- The **application layer** contains a variety of protocols that are commonly needed by users. One widely used application protocol is **HTTP** (**HyperTextTransfer Protocol**), which is the basis for the World Wide Web.
- Other application protocols are used for file transfer (**FTP**), electronic mail(**SMTP**), and so on.

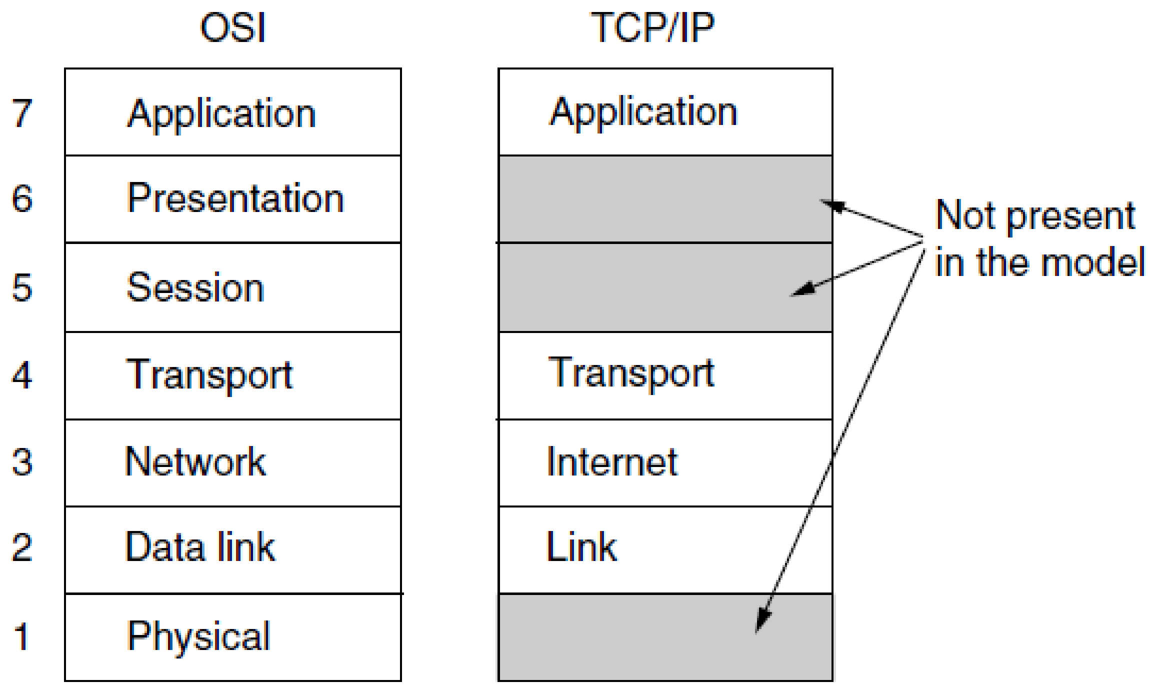
The TCP/IP Reference Model:

- The ARPANET was a research network sponsored by the DoD (U.S. Department of Defense).
- It eventually connected hundreds of universities and government installations, using leased telephone lines.
- When satellite and radio networks were added later, the existing protocols had trouble interworking with them, so a new reference architecture was needed.
- Thus, from nearly the beginning, the ability to connect multiple networks in a seamless way was one of the major design goals.
- This architecture later became known as the **TCP/IP Reference Model**.
- The DoD wanted connections to remain intact as long as the source and destination machines were functioning, even if some of the machines or transmission lines in between were suddenly put out of operation.
- Ranging from transferring files to real-time speech transmission, a flexible architecture was needed.
- All these requirements led to the choice of a packet-switching network based on a connectionless layer that runs across different networks.

The TCP/IP Reference Model:

- TCP/IP Reference Model is a four-layered suite of communication protocols. It was developed by the DoD (Department of Defence) in the 1960s
- TCP stands for Transmission Control Protocol and IP stands for Internet Protocol.
- TCP/IP Protocol suite have 4 layers

1. **The Link Layer**
2. **The Internet Layer**
3. **The Transport Layer**
4. **The Application Layer**



The TCP/IP reference model.

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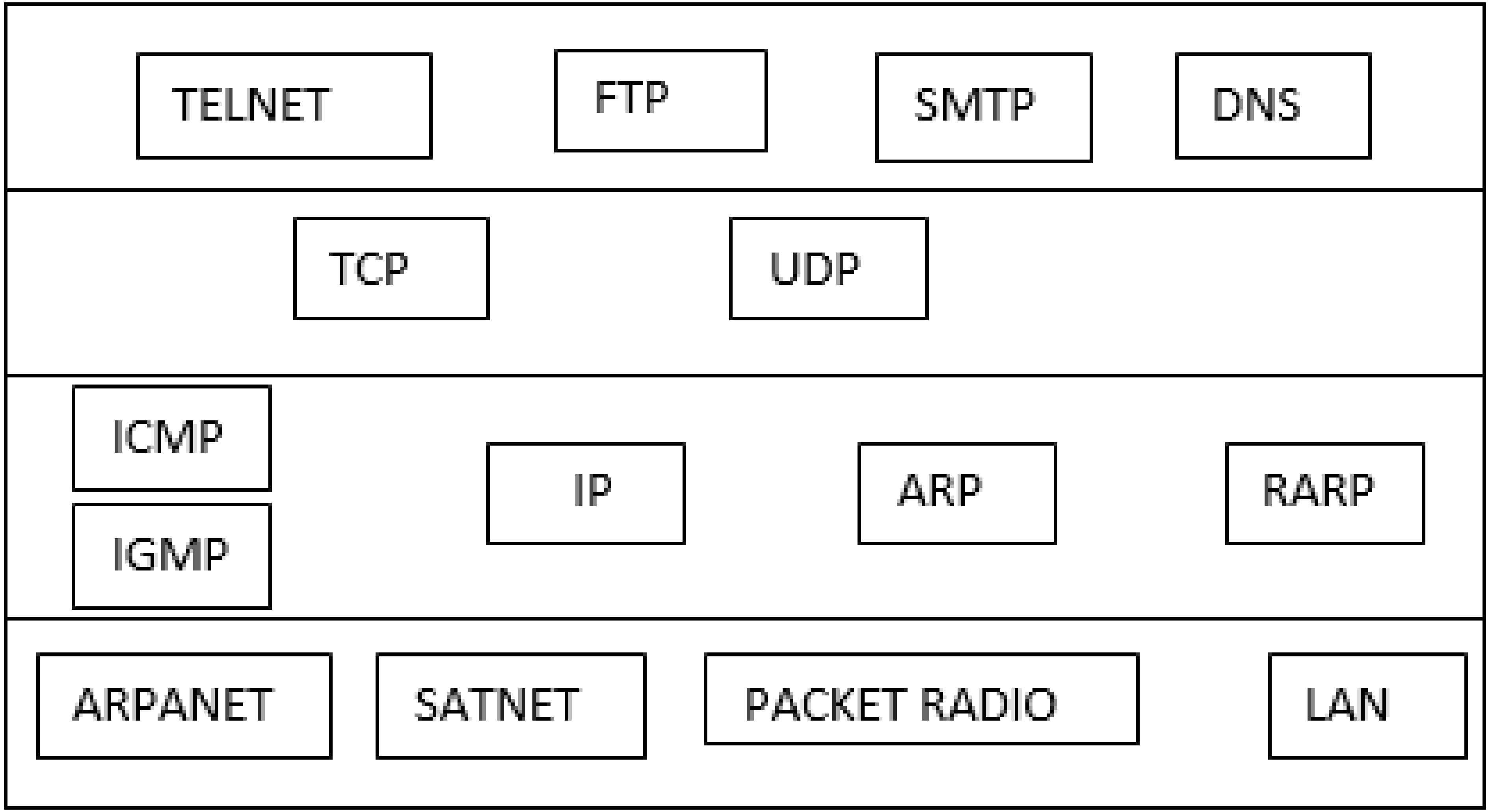
TELNET
FTP
SMTP
DNS
TCP
UDP
ICMP
IGMP
IP
ARP
RARP
ARPANET
SATNET
PACKET RADIO
LAN
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The Link Layer

- It is also called a network interface layer or link layer. It can be considered as the combination of physical layer and data link layer of the OSI model.
- It defines how bits are to be encoded into optical or electrical pulses.
- It accepts IP packets from the network layer and encapsulates them into frames.
- It is not really a layer at all, in the normal sense of the term, but rather an interface between hosts and transmission links.

Protocols used in this layer are:

ARPANET:-

- Advanced Research Project Agency Software
- It is world first packet switching Software & core Software came to compose the global Internet.

SATNET:-

- SATNET, also known as satellite network that formed as an initial segment of the Internet.
- It was implemented by BBN(Body-to-Body Network) Technologies under the direction of the Advanced Research Projects Agency.

Packet radio

- Packet radio is a digital radio communications mode used to send packets of data.
- Packet radio uses packet switching to transmit datagrams.
- Packet radio can be used to transmit data long distances.

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The Internet Layer

The Internet layer is responsible for logical transmission of data packets over the internet.

Main functions of the internet layer are −

- It transmits data packets to the link layer.
- It routes each of the data packets independently from the source to the destination, using the optimal route.
- It reassembles the out-of-order packets when they reach the destination.
- It handles the error in transmission of data packets and fragmentation of data packets.

The protocols used in this layer are −

- **Internet Protocol, IP –** It is a connectionless and unreliable protocol that provides a best effort delivery service. It transports data packets called datagrams that travel over different routes across multiple nodes.
- **Address Resolution Protocol, ARP –**This protocol maps the logical address or the Internet address of a host to its physical address, as printed in the network interface card.
- **Reverse Address Resolution Protocol, RARP –** This is to find the Internet address of a host when its physical address is known.
- **Internet Control Message Protocol, ICMP –** It monitors sending the queries as well as the error messages.
- **Internet Group Message Protocol, IGMP –**It allows the transmission of a message to a group of recipients simultaneously.

The Transport Layer

- The transport layer is responsible for error-free, end-to-end delivery of data from the source host to the destination host.

In Transport layer there are two protocols namely, TCP and UDP.

- ***Transmission Control Protocol, TCP*** − It is a reliable connection-oriented protocol that transmits data from the source to the destination machine without any error. A connectio
- **User Datagram Protocol, UDP** − It is a message-oriented protocol that provides a simple unreliable, connectionless, unacknowledged service. It is suitable for applications that do not require TCP's sequencing, error control or flow control. It is used for transmitting a small amount of data where the speed of delivery is more important than the accuracy of delivery.
- Stream Control Transmission Protocol, SCTP − It combines the features of both TCP and UDP. It is message oriented like the UDP, which providing the reliable, connection-oriented service like TCP. It is used for telephony over the Internet.

The Application Layer

- The application layer is the highest abstraction layer of the TCP/IP model that provides the interfaces and protocols needed by the users. It combines the functionalities of the session layer, the presentation layer and the application layer of the OSI model.

This layer uses a number of protocols, the main among which are as follows −

- ***Hyper Text Transfer Protocol, HTTP*** − It is the underlying protocol for world wide web. It defines how hypermedia messages are formatted and transmitted.
- ***File Transfer Protocol, FTP*** − It is a client-server based protocol for transfer of files between client and server over the network.
- ***Simple Mail Transfer Protocol, SMTP*** − It lays down the rules and semantics for sending and receiving electronic mails (e-mails).
- ***Domain Name System, DNS*** − It is a naming system for devices in networks. It provides services for translating domain names to IP addresses.
- ***TELNET*** − It provides bi-directional text-oriented services for remote login to the hosts over the network.
- ***Simple Network Management Protocol, SNMP*** − It is for managing, monitoring the network and for organizing information about the networked devices.

EXAMPLE NETWORKS

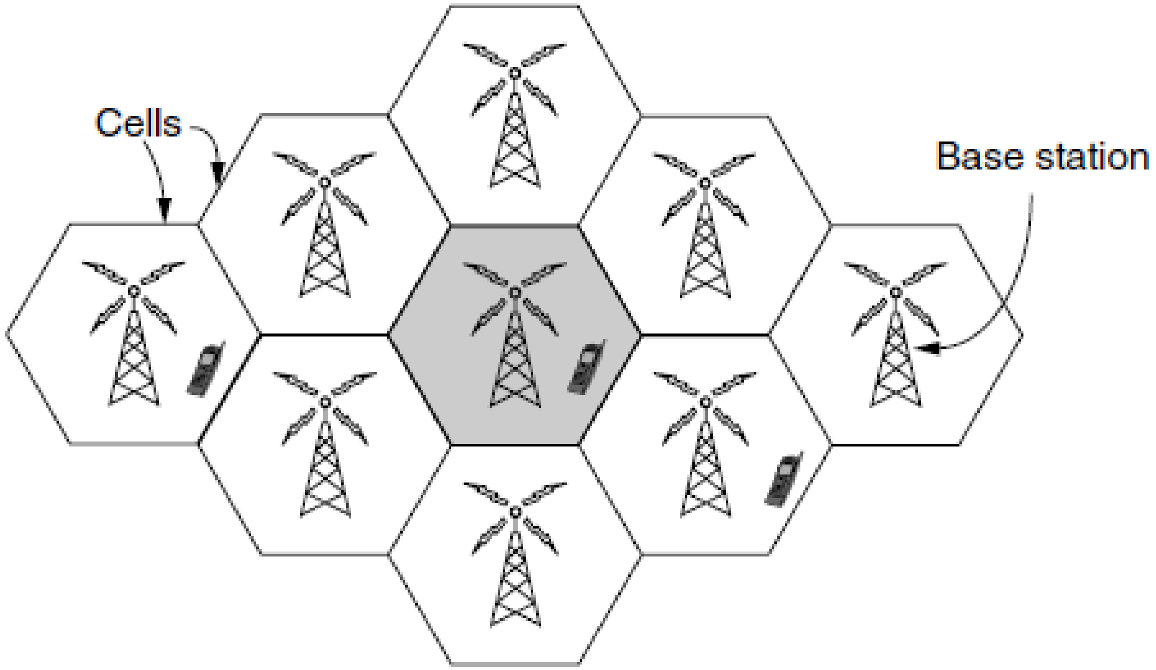
- The subject of computer networking covers many different kinds of networks, large and small, well known and less well known. They have different goals, scales, and technologies. They are:
- Internet
- Third-Generation Mobile Phone Networks
- Wireless LANs (IEEE 802.11)
- RFID and Sensor Networks

The Internet

- In simplest terms, the Internet is a global network comprised of smaller networks that are interconnected using **standardized** communication protocols. The Internet standards describe a framework known as the Internet protocol suite. This model divides methods into a *layered system of protocols.*
- The Internet provides a variety of information and communication facilities; contains forums, databases, email, hypertext, etc. It consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies.

Third-Generation Mobile Phone Networks

- People love to talk on the phone even more than they like to surf the Internet, and this has made the mobile phone network the most successful network in the world.
- It has more than four billion subscribers worldwide roughly 60% of the world's population and more than the number of Internet hosts and fixed telephone lines combined
- First-generation mobile phone systems transmitted voice calls as continuously varying (analog) signals rather than sequences of (digital) bits, i.e., **AMPS** (**Advanced Mobile Phone System**) in 1982.
- Second-generation mobile phone systems switched to transmitting voice calls in digital form to increase capacity, improve security, and offer text messaging i.e., **GSM** (**Global System for Mobile communications**) in 1991. [most widely used mobile phone system in the world, is a 2G system]
- The third generation, or 3G, systems were initially deployed in 2001 and offer both digital voice and broadband digital data services. **UMTS** (**Universal Mobile Telecommunications System**), also called **WCDMA** (**Wideband Code Division Multiple Access**), is the main 3G system that is being rapidly deployed worldwide.



Cellular design of mobile phone networks.

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Wireless LANs (IEEE 802.11)

- Wireless LANs (WLANs) are wireless computer networks that use high-frequency radio waves instead of cables for connecting the devices within a limited area forming LAN (Local Area Network). Users connected by wireless LANs can move around within this limited area such as home, school, campus, office building, railway platform, etc.
- Most WLANs are based upon the standard IEEE 802.11 standard or WiFi.

Components of WLANs:

- Wireless Access Point (WAP or AP)
- Client

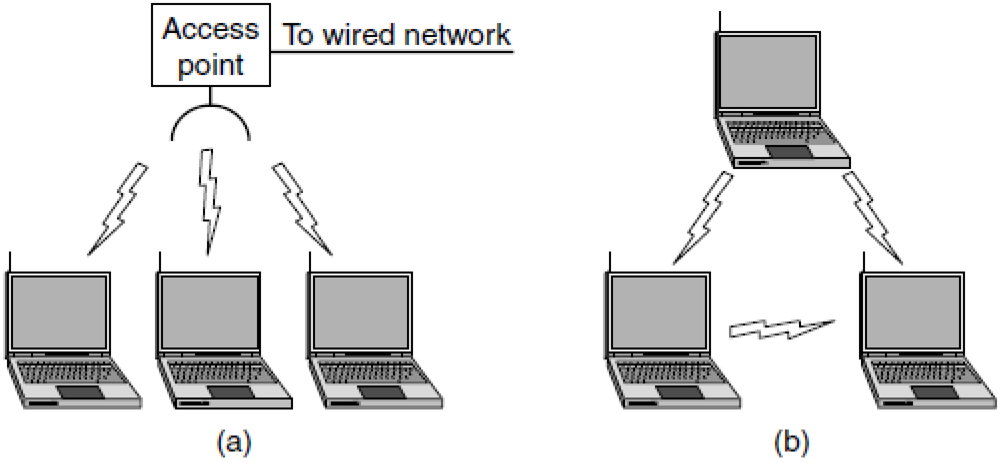
Types of WLANS

Wireless network with an access point:

Mobile devices or clients connect to an access point (AP) that in turn connects via a bridge to the LAN or Internet.

The client transmits frames to other clients via the AP.

Ad hoc network**:** Clients transmit frames directly to each other in a peer-to-peer fashion.



(a) Wireless network with an access point. (b) Ad hoc network.

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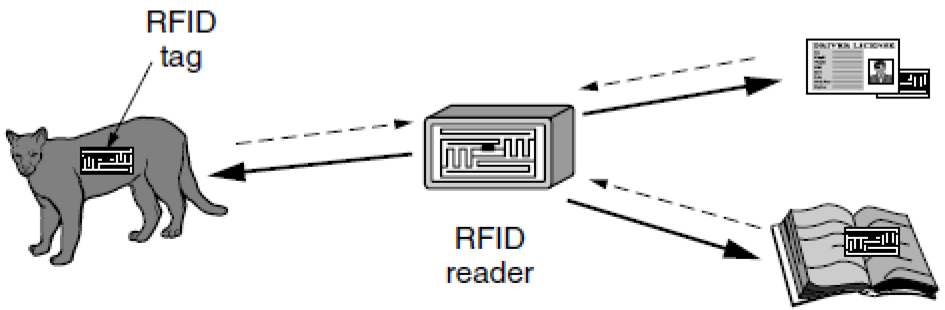
RFID and Sensor Networks

- The networks we have studied so far are made up of computing devices that are easy to recognize, from computers to mobile phones.
- With **Radio Frequency IDentification** (**RFID**), everyday objects can also be part of a computer network.
- An RFID tag looks like a postage stamp-sized sticker that can be affixed to (or embedded in) an object so that it can be tracked.
- The tag consists of a small microchip with a unique identifier and an antenna that receives radio transmissions.
- RFID readers installed at tracking points find tags when they come into range and interrogate them for their information as shown in Fig.



- Applications include checking identities, managing the supply chain, timing races, and replacing barcodes.

Internet Based Applications

- Internet applications are server-based applications. Following are a few Internet Applications −
- World Wide Web (WWW)
- Electronic mail (e-mail)
- File Transfer Protocol (FTP)
- Telnet (i.e., log-in to the computer located remotely)
- Internet Relay Chat (IRC) (Real time video chatting)

Network Devices

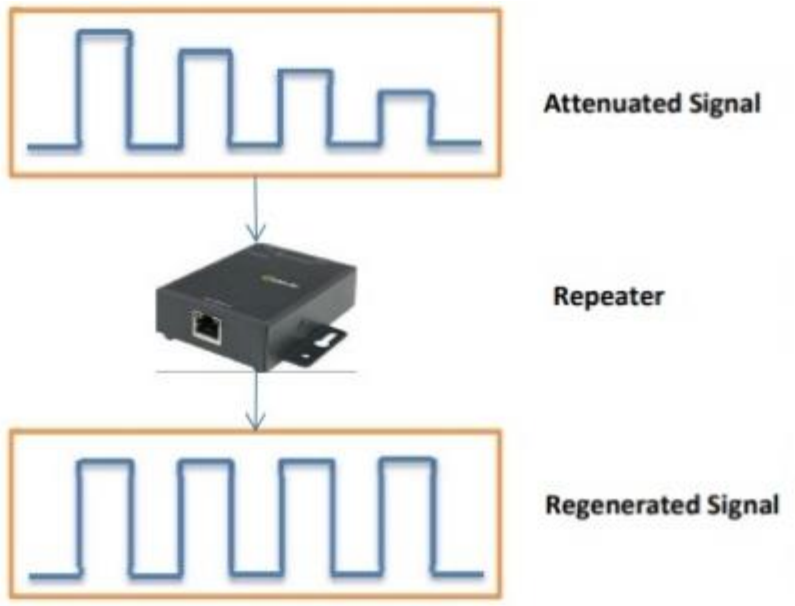
Hardware devices that are used to connect computers, printers, fax machines and other electronic devices to a network are called **network devices**.

These devices transfer data in a fast, secure and correct way over same or different networks.

Network devices may be inter-network or intra-network.

Repeaters

- Repeaters are network devices operating at physical layer of the OSI model that amplify or regenerate an incoming signal before retransmitting it. They are incorporated in networks to expand its coverage area. They are also known as *signal boosters*.
- When an electrical signal is transmitted via a channel, it gets attenuated **(loss of transmission signal strength)** depending upon the nature of the channel or the technology. This poses a limitation upon the length of the LAN or coverage area of cellular networks. This problem is alleviated by installing repeaters at certain intervals.



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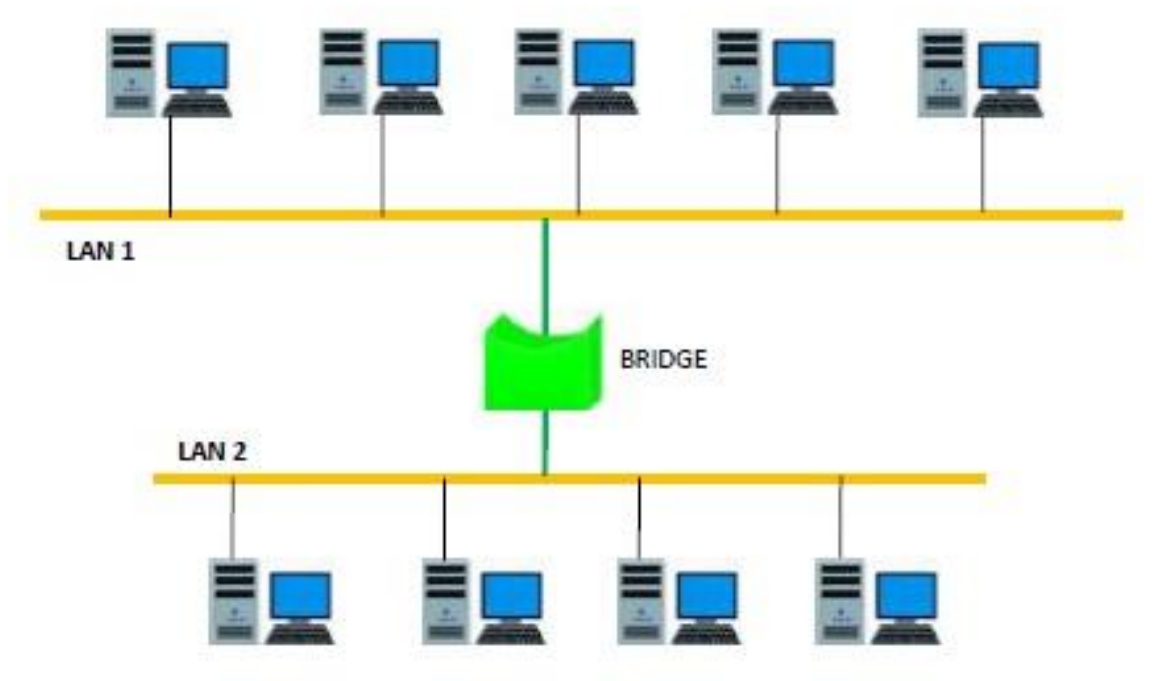
Bridges

- A bridge is a network device that connects multiple LANs (local area networks) together to form a larger LAN. The process of aggregating networks is called network bridging. A bridge connects the different components so that they appear as parts of a single network. Bridges operate at the data link layer of the OSI model and hence also referred as Layer 2 switches.



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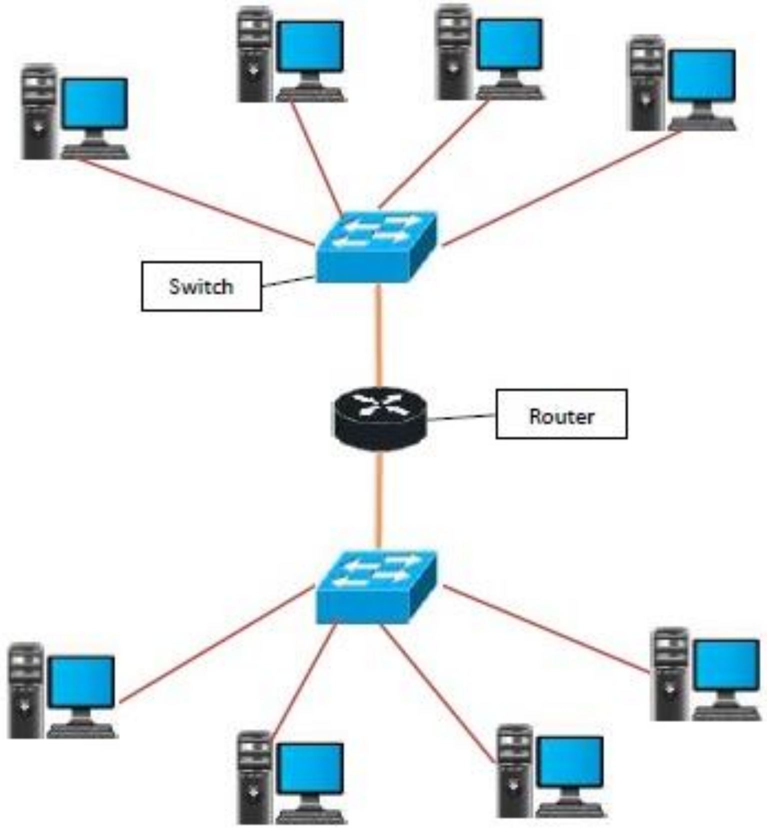
Routers

A **router** is a **network layer** hardware device that transmits data from one LAN to another if both networks support the same set of protocols.

So a **router** is typically connected to at least two LANs and the **internet service provider** (ISP).

It receives its data in the form of **packets**, which are **data frames** with their **destination address** added.

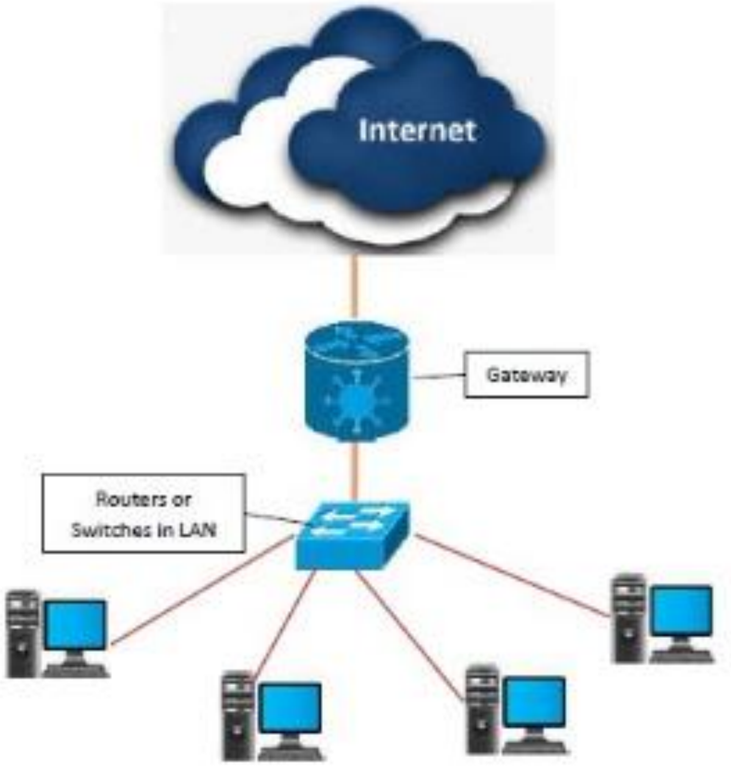
Router also strengthens the signals before transmitting them. That is why sometimes it is also called **repeater**.



Router Connecting Two Networks

Gateways

- **Gateway** is a network device used to connect two or more dissimilar networks. A gateway usually is a computer with multiple **NICs(**Network Interface Card**)** connected to different networks. A gateway can also be configured completely using software. As networks connect to a different network through gateways, these gateways are usually hosts or end points of the network.
- **Gateway** uses **packet switching** technique to transmit data from one network to another. In this way it is similar to a **router**, the only difference being router can transmit data only over networks that use same protocols.



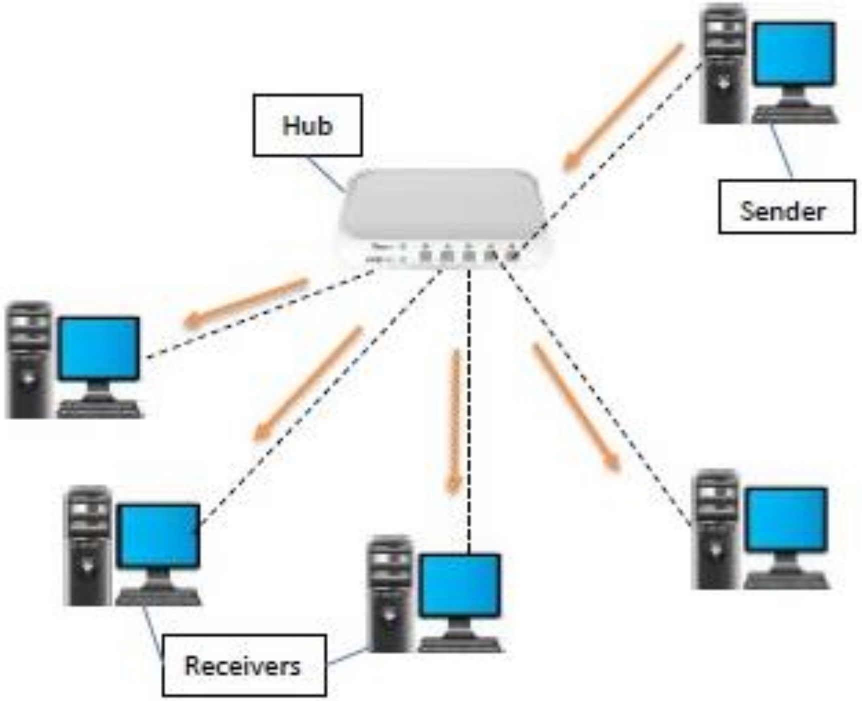
Gateway between a LAN and Internet

Multiprotocol Routers

- Multiprotocol Routers uses Multiprotocol Label Switching (MPLS) routing technique that augments speed and control of the network traffic by directing data from one node to the next node based on short path labels. Instead of being routed using long network addresses, the data packets are routed through path labels that identify virtual paths between the nodes rather than endpoints. MPLS speeds up traffic flows by avoiding complex lookups in the routing table at each node as in conventional routing algorithms.

Hubs

- Hubs are networking devices operating at a physical layer of the OSI model that are used to connect multiple devices in a network. They are generally used to connect computers in a LAN.
- A hub has many ports in it. A computer which intends to be connected to the network is plugged in to one of these ports. When a data frame arrives at a port, it is broadcast to every other port, without considering whether it is destined for a particular destination device or not.



Broadcasting by a Hub

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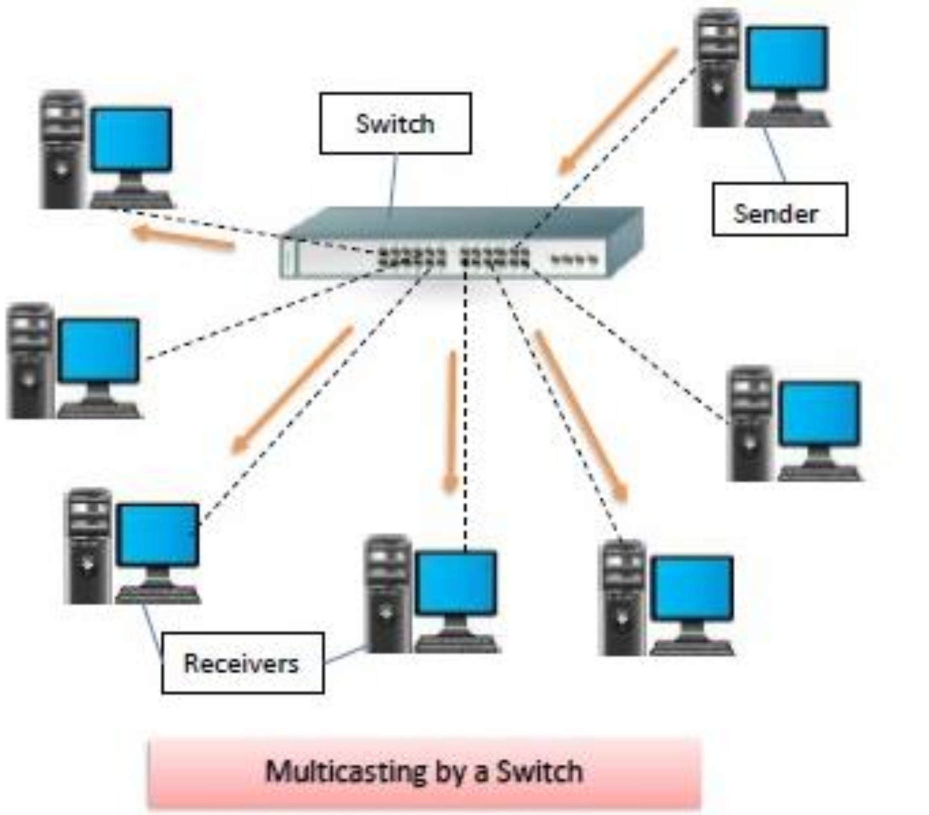
Switches

- Switches are networking devices operating at layer 2 or a data link layer of the OSI model. They connect devices in a network and use packet switching to send, receive or forward data packets or data frames over the network.
- A switch has many ports, to which computers are plugged in. When a data frame arrives at any port of a network switch, it examines the destination address, performs necessary checks and sends the frame to the corresponding device(s).It supports unicast, multicast as well as broadcast communications.



Modems

- Modem is a device that enables a computer to send or receive data over telephone or cable lines.
- The main function of the modem is to convert digital signal into analog and vice versa. Modem is a combination of two devices − **modulator** and **demodulator**.
  The **modulator** converts digital data into analog data when the data is being sent by the computer. The **demodulator** converts analog data signals into digital data when it is being received by the computer.

Digital Data Waveform

Analog Data Waveform

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CSU/DSU (Channel Service Unit/Data Service Unit)

- A CSU/DSU (Channel Service Unit/Data Service Unit) is a hardware device that converts a digital data frame from the communications technology used on a local area network (LAN) into a frame appropriate to a wide-area network (WAN) and vice versa.

To connect to a local area network, the DSU communicates with the Channel Service Unit.

- The data encoded in the digital circuit is converted into synchronous serial data by the digital service unit.
- Wherever the digital communication line is utilized, the digital service unit also electrically separates it from the networking equipment.
- The Channel Service Unit and the Digital Service Unit must be owned or manufactured by the same company or manufacturer for the digital service unit to function correctly.

NIC

- NIC or network interface card is a network adapter that is used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique id that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and the router or modem. NIC card is a layer 2 device which means that it works on both the physical and data link layers of the network model.

Wireless Access Points

- A Wireless Access Point (WAP) is a networking device that allows connecting the devices with the wired network. A Wireless Access Point (WAP) is used to create the WLAN (Wireless Local Area Network)
- A wireless AP connects the wired networks to the wireless client. It eases access to the network for mobile users which increases productivity and reduces the infrastructure cost.

Transceivers

- IrDA communication involves a transmitter unit for transmitting the data over IR and a receiver for receiving the data.
- Infrared Light Emitting Diode (LED) is the IR source for transmitter and at the receiving end a photodiode acts as the receiver.
- Both transmitter and receiver unit will be present in each device supporting IrDA communication for bidirectional data transfer. Such IR units are known as ‘Transceiver’.

Firewalls

- A firewall can be defined as a special type of network security device or a software program that monitors and filters incoming and outgoing network traffic based on a defined set of security rules. It acts as a barrier between internal private networks and external sources (such as the public Internet).
- The primary purpose of a firewall is to allow non-threatening traffic and prevent malicious or unwanted data traffic for protecting the computer from viruses and attacks.

Proxies/Proxy server

- The proxy server is a computer on the internet that accepts the incoming requests from the client and forwards those requests to the destination server.
- The proxy server allows us to access any websites with a different IP address. It separates the client system and web server from the global network.
- It plays an intermediary role between users and targeted websites or servers. It collects and provides information related to user requests. The most important point about a proxy server is that it does not encrypt traffic.

That's all about

UNIT 1

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Unit 2

Data Link Layer

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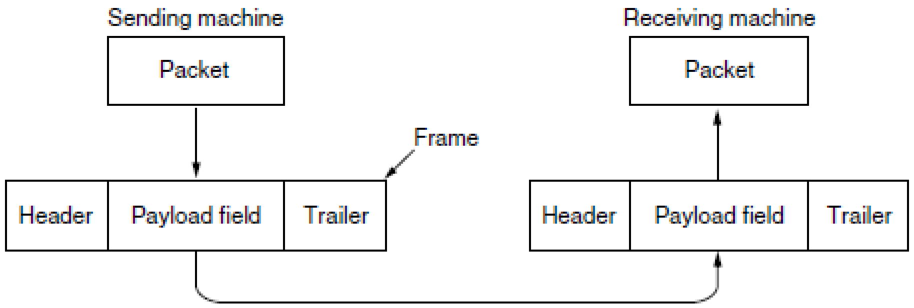
The data link layer uses the services of the physical layer to send and receive bits over communication channels. It has a number of functions, including:

1. Providing a well-defined service interface to the network layer.
2. Dealing with transmission errors.
3. Regulating the flow of data so that slow receivers are not swamped by fast senders.

- To accomplish these goals, the data link layer takes the packets it gets from the network layer and encapsulates them into **frames** for transmission.



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Relationship between packets and frames

<u>Design issues of Data layer</u> :

- Providing services to the network layer
- Framing
- Error Control
- Flow Control

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Services Provided to the Network Layer

- The function of the data link layer is to provide services to the network layer.
- The principal service is transferring data from the network layer on the source machine to the network layer on the destination machine.
- On the source machine is an entity, call it a process, in the network layer that hands some bits to the data link layer for transmission to the destination.
- The job of the data link layer is to transmit the bits to the destination machine so they can be handed over to the network layer there.
- The data link layer can be designed to offer various services like
  1. Unacknowledged connectionless service.
  2. Acknowledged connectionless service.
  3. Acknowledged connection-oriented service.

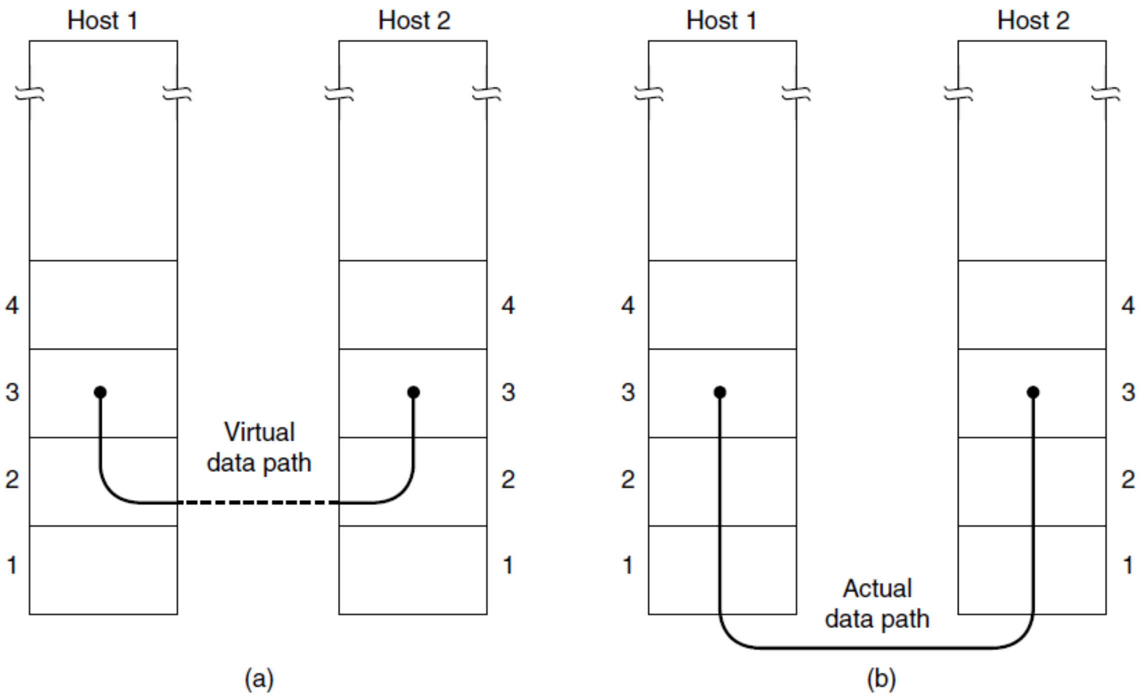


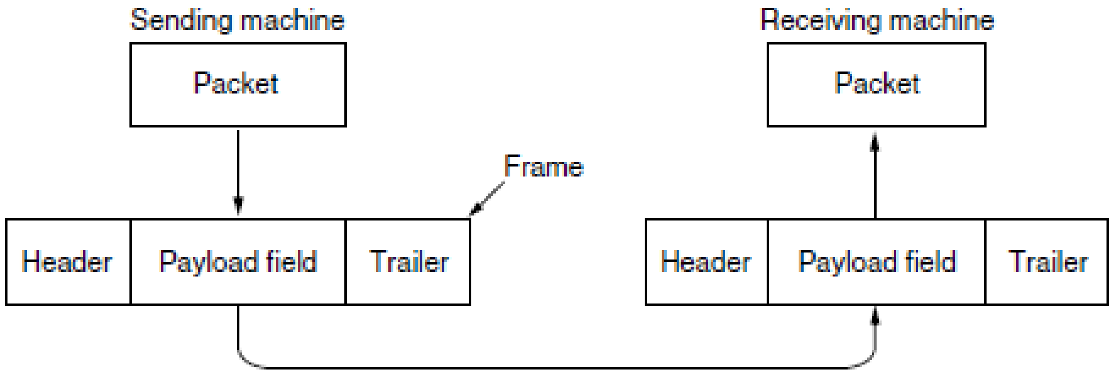
Figure : (a) Virtual communication. (b) Actual communication.

Framing

- The data link layer encapsulates each data packet from the network layer into frames that are then transmitted.

A frame has three parts, namely –

- Frame Header
- Payload field that contains the data packet from network layer
- Trailer



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- The usual approach is for the data link layer to break up the bit stream into discrete frames, compute a short token called a checksum for each frame, and include the checksum in the frame when it is transmitted.
- When a frame arrives at the destination, the checksum is recomputed. If the newly computed checksum is different from the one contained in the frame, the data link layer knows that an error has occurred and takes steps to deal with it (e.g., discarding the bad frame and possibly also sending back an error report).
- Breaking up the bit stream into frames is in four methods:

1. Byte count.

2. Flag bytes with byte stuffing.

3. Flag bits with bit stuffing.

4. Physical layer coding violations.

1. Byte count.

- This method is rarely used and is generally required to count total number of bytes that are present in frame. This is be done by using field in header. Byte count method ensures data link layer at the receiver or destination about total number of bytes that follow, and about where the frame ends.
- There is disadvantage also of using this method i.e., if anyhow character count is disturbed or distorted by an error occurring during transmission, then destination or receiver might lose synchronization. The destination or receiver might also be not able to locate or identify beginning of next frame.

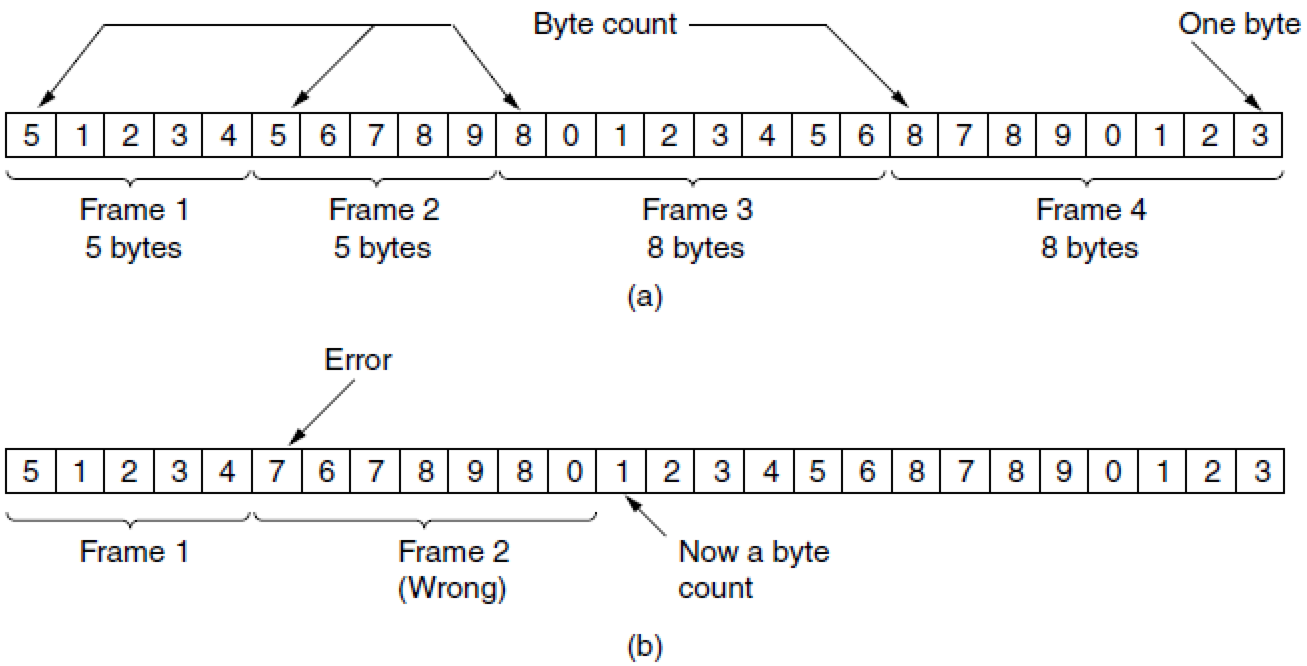

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Figure: A byte stream. (a) Without errors. (b) With one error.

2. Flag bytes with byte stuffing.

- The second framing method gets around the problem of resynchronization after an error by having each frame start and end with special bytes.
- Often the same byte, called a **flag byte**, is used as both the starting and ending delimiter.
- Two consecutive flag bytes indicate the end of one frame and the start of the next.
- Thus, if the receiver ever loses synchronization it can just search for two flag bytes to find the end of the current frame and the start of the next frame
- However, there is a still a problem we have to solve. It may happen that the flag byte occurs in the data, especially when binary data such as photographs or songs are being transmitted.
- One way to solve this problem is to have the sender’s data link layer insert a special escape byte (ESC) just before each ‘‘accidental’’ flag byte in the data.
- The data link layer on the receiving end removes the escape bytes before giving the data to the network layer. This technique is called **byte stuffing**.
- Of course, the next question is: what happens if an escape byte occurs in the middle of the data? The answer is that it, too, is stuffed with an escape byte.

Figure: A frame delimited by flag bytes.

FLAG	Header	Payload field	Trailer	FLAG

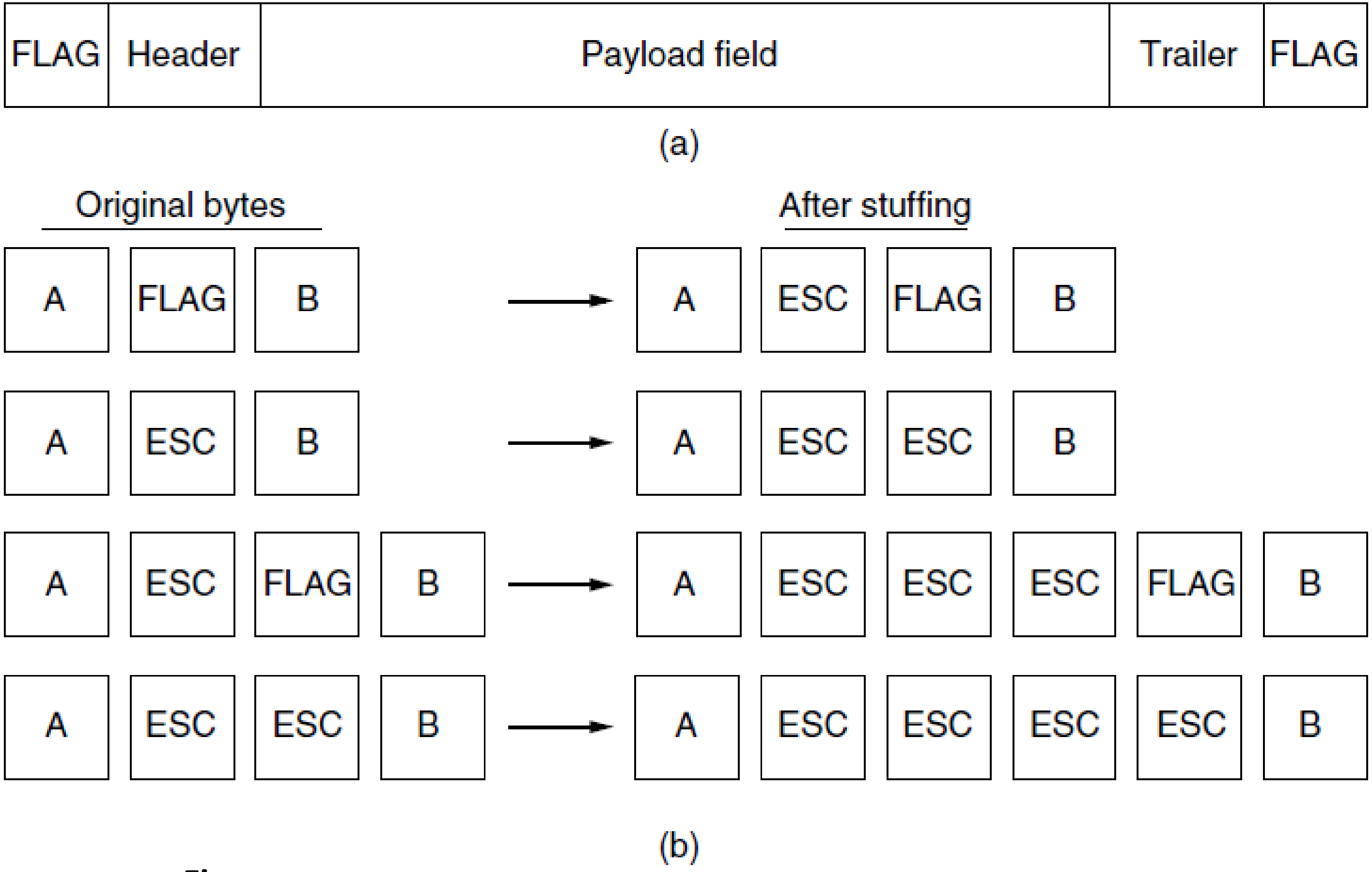

Figure:
(a) A frame delimited by flag bytes.
(b) Four examples of byte sequences before and after byte stuffing.

3. Flag bits with bit stuffing.

- Framing can be also be done at the bit level, so frames can contain an arbitrary number of bits made up of units of any size. It was developed for the once very popular **HDLC** (**Highlevel Data Link Control**) protocol.
- Each frame begins and ends with a special bit pattern, 01111110 or 0x7E in hexadecimal.
- This pattern is a flag byte. Whenever the sender's data link layer encounters five consecutive 1s in the data, it automatically stuffs a 0 bit into the outgoing bit stream.
- Bit stuffing is very essential part of transmission process in network and communication protocol. It is also required in USB.

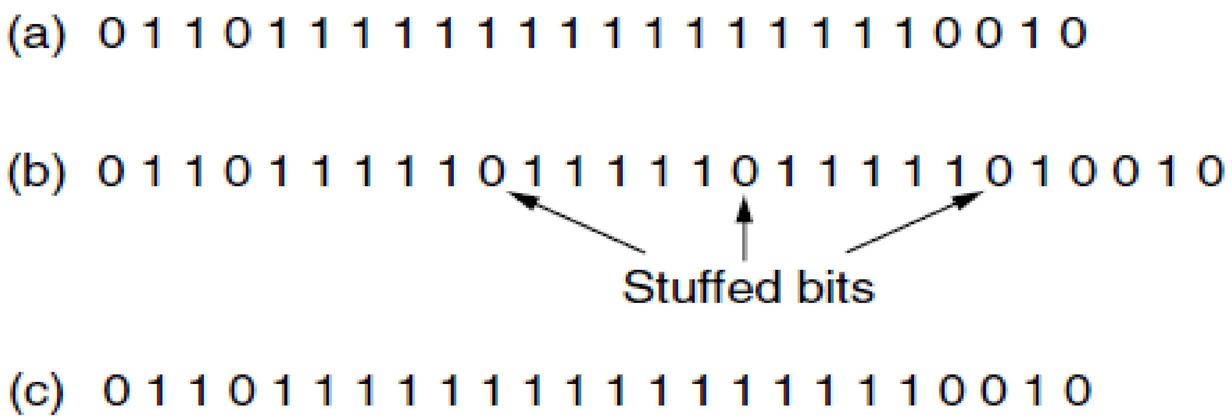

Figure 3-5. Bit stuffing. (a) The original data.
(b) The data as they appear on the line.
(c) The data as they are stored in the receiver's memory after destuffing.

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4. Physical layer coding violations.

- Encoding violation is method that is used only for network in which encoding on physical medium includes some sort of redundancy i.e., use of more than one graphical or visual structure to simply encode or represent one variable of data.

Error Control

The issues it caters to with respect to error control are −

- Dealing with transmission errors
- Sending acknowledgement frames in reliable connections
- Retransmitting lost frames
- Identifying duplicate frames and deleting them
- Controlling access to shared channels in case of broadcasting

Flow Control

- The data link layer regulates flow control so that a fast sender does not drown a slow receiver. When the sender sends frames at very high speeds, a slow receiver may not be able to handle it. There will be frame losses even if the transmission is error-free.

The two common approaches for flow control are −

- *Feedback based flow control* - the receiver sends back information to the sender giving it permission to send more data, or at least telling the sender how the receiver is doing.
- *Rate based flow control* - the protocol has a built-in mechanism that limits the rate at which senders may transmit data, without using feedback from the receiver.

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ERROR DETECTION AND CORRECTION

Error

- A condition when the receiver's information does not match with the sender's information. During transmission, digital signals suffer from noise that can introduce errors in the binary bits travelling from sender to receiver. That means a 0 bit may change to 1 or a 1 bit may change to 0.
- Network designers have developed two basic strategies for dealing with errors.
- **Error Detecting** and **Correcting Codes**.

- **Error Detecting** and **Correcting Codes**.
- we use error-detecting codes which are additional data added to a given digital message to help us detect if any error has occurred during transmission of the message.
- Basic approach used for error detection is the use of redundancy bits, where additional bits are added to facilitate detection of errors.

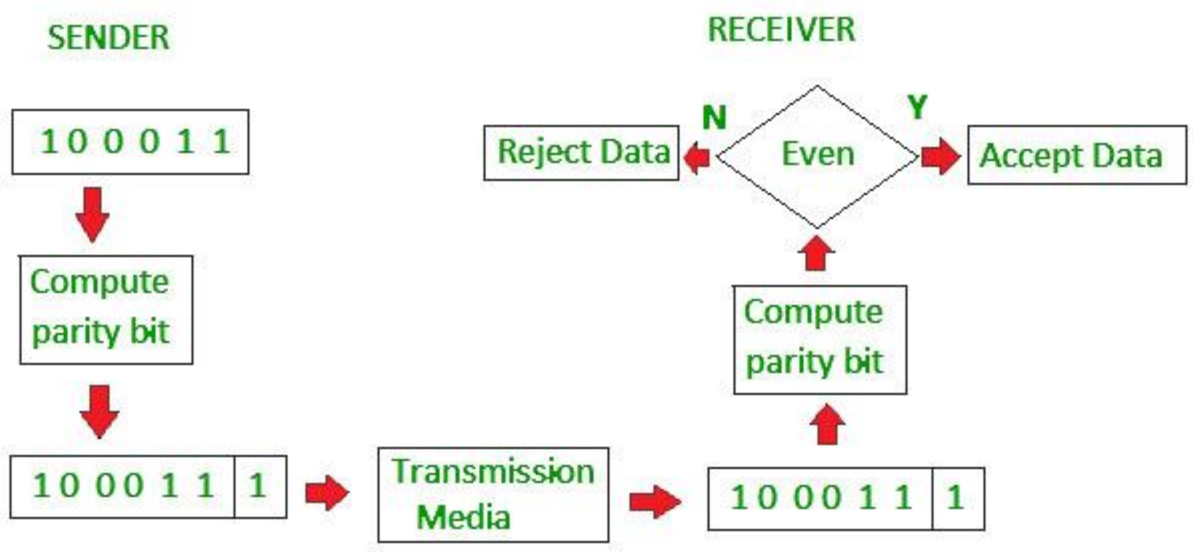
Some popular techniques for error detection are:

1. Simple Parity check
2. Two-dimensional Parity check
3. Checksum
4. Cyclic redundancy check

1. Simple Parity check

Blocks of data from the source are subjected to a check bit or parity bit generator form, where a parity of :

- 1 is added to the block if it contains odd number of 1’s, and
- 0 is added if it contains even number of 1’s
- This scheme makes the total number of 1’s even, that is why it is called even parity checking.



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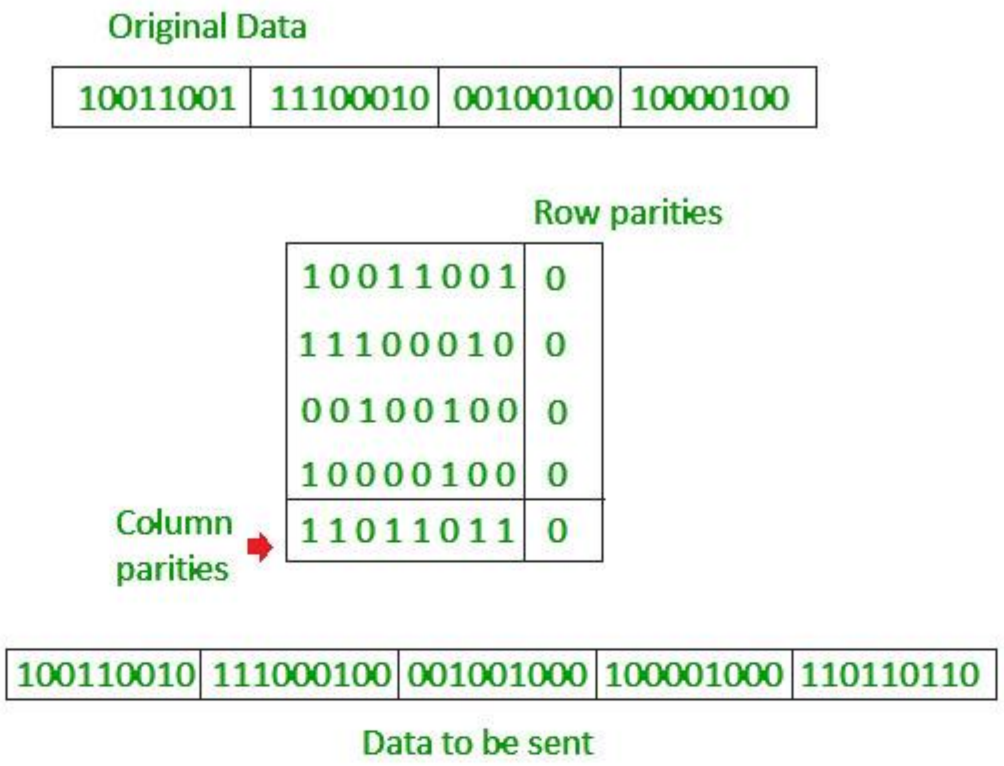
2. Two-dimensional Parity check

- Parity check bits are calculated for each row, which is equivalent to a simple parity check bit.
- Parity check bits are also calculated for all columns, then both are sent along with the data. At the receiving end these are compared with the parity bits calculated on the received data.



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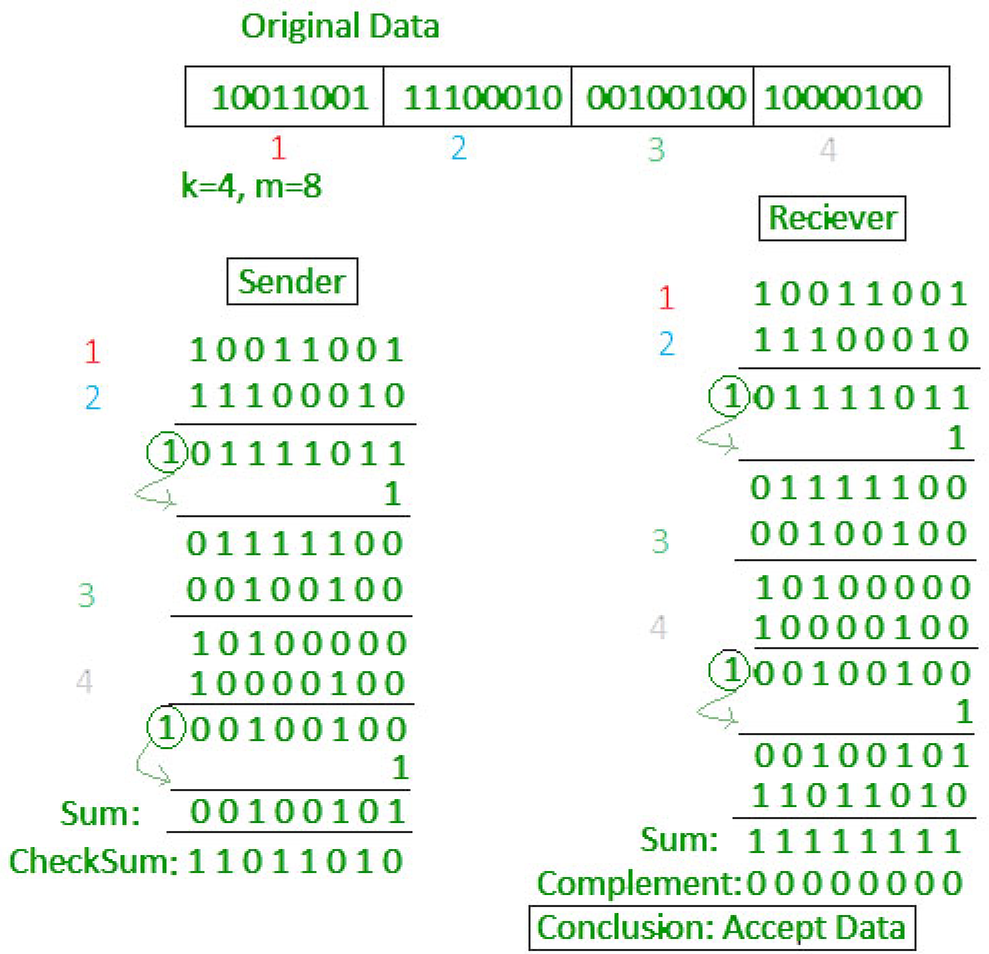
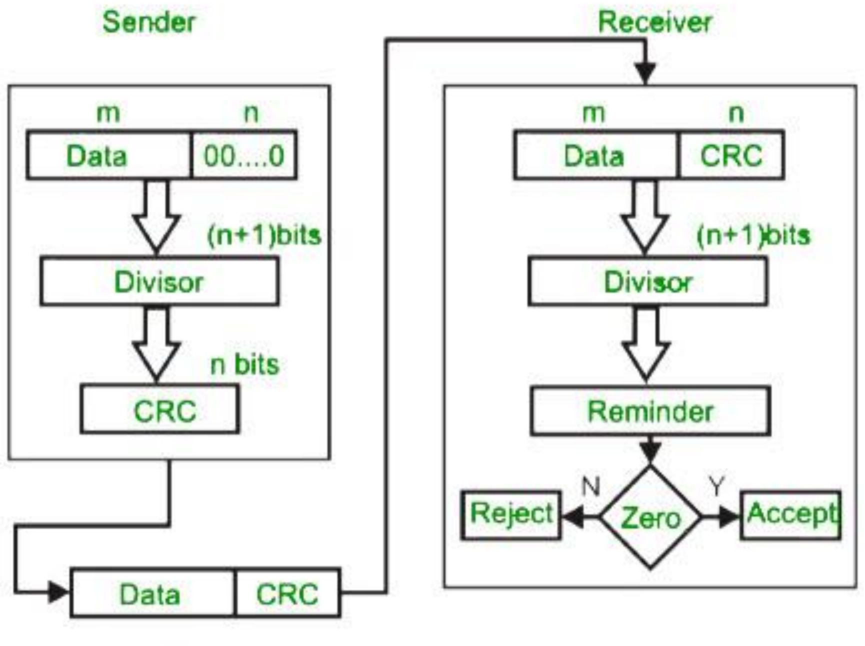
3. Checksum

- In checksum error detection scheme, the data is divided into k segments each of m bits.
- In the sender's end the segments are added using 1's complement arithmetic to get the sum.
- The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.
- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum.
- The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.



4. Cyclic redundancy check (CRC)

- Unlike checksum scheme, which is based on addition, CRC is based on binary division.
- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
- At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.



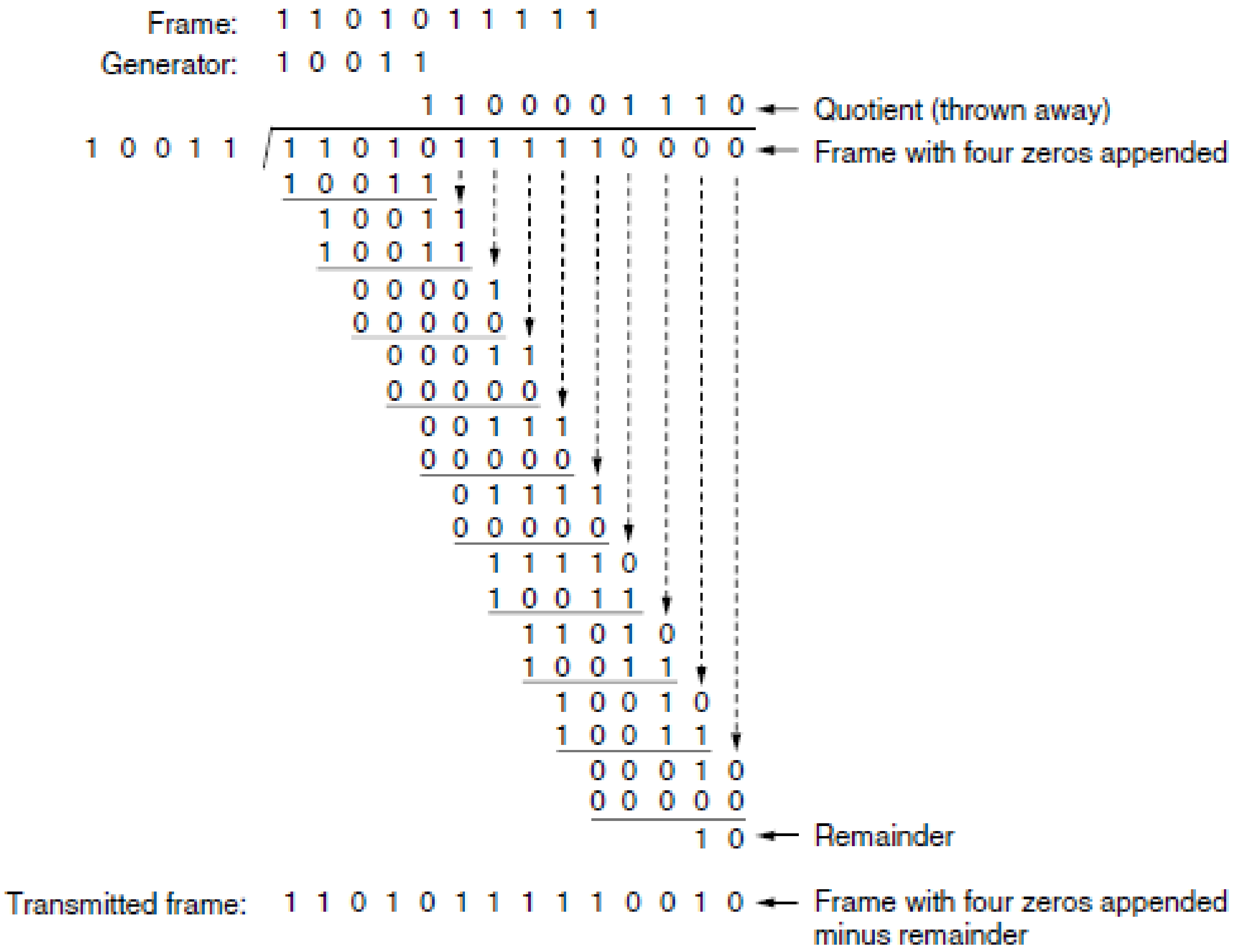


Figure: Example calculation of the CRC.

original message

1010000

@ means X-OR

Generator polynomial

x^3+1

$1.x^3+0.x^2+0.x^1+1.x^0$

CRC generator

$\boxed{1001}$ 4-bit

If CRC generator is of n bit then append (n-1) zeros in the end of original message

Sender

```
1001 | 1010000000
      @1001
       0011000000
        @1001
         0101000
          @1001
          0011000
           @1001
            01010
           @1001
             0011
```

Message to be transmitted

```
1010000000
      +011
1010000011
```

Receiver

```
1001 | 1010000011
      @1001
       0011000011
        @1001
         01010011
          @1001
           0011011
            @1001
             01001
            @1001
              0000
```

Zero means data is accepted

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Example

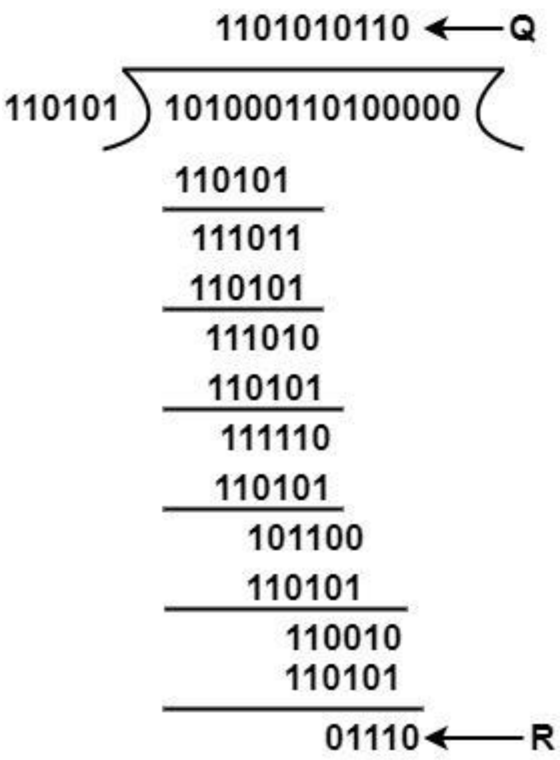
Message D = 1010001101 (10 bits)

Predetermined P = 110101 (6 bits)

Hence, n = 15 K = 10 and (n – k) = 5

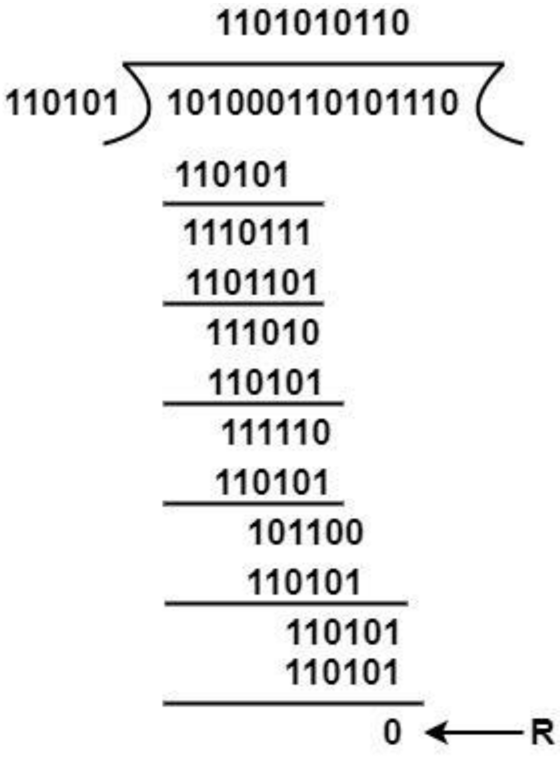
The message is generated through 2^5:accommodating 101000110100000

The product is divided by P.



The remainder is inserted to 2^5D to provide T = 101000110101110 that is sent.

Suppose that there are no errors, and the receiver gets T perfect. The received frame is divided by P.



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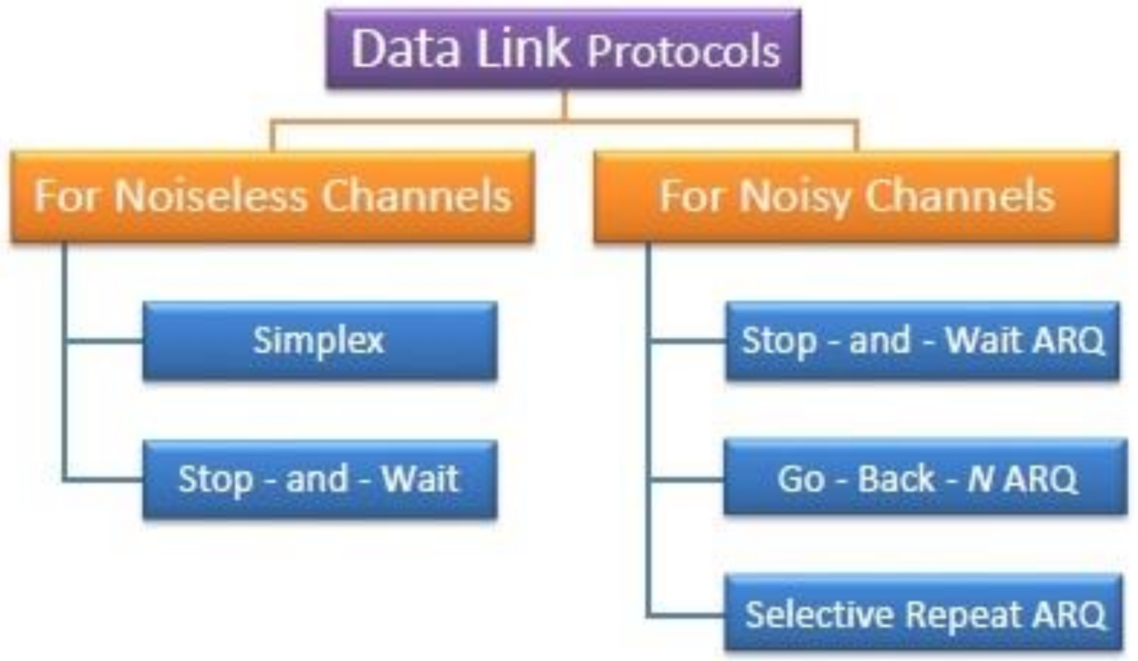
Because of no remainder, there are no errors.

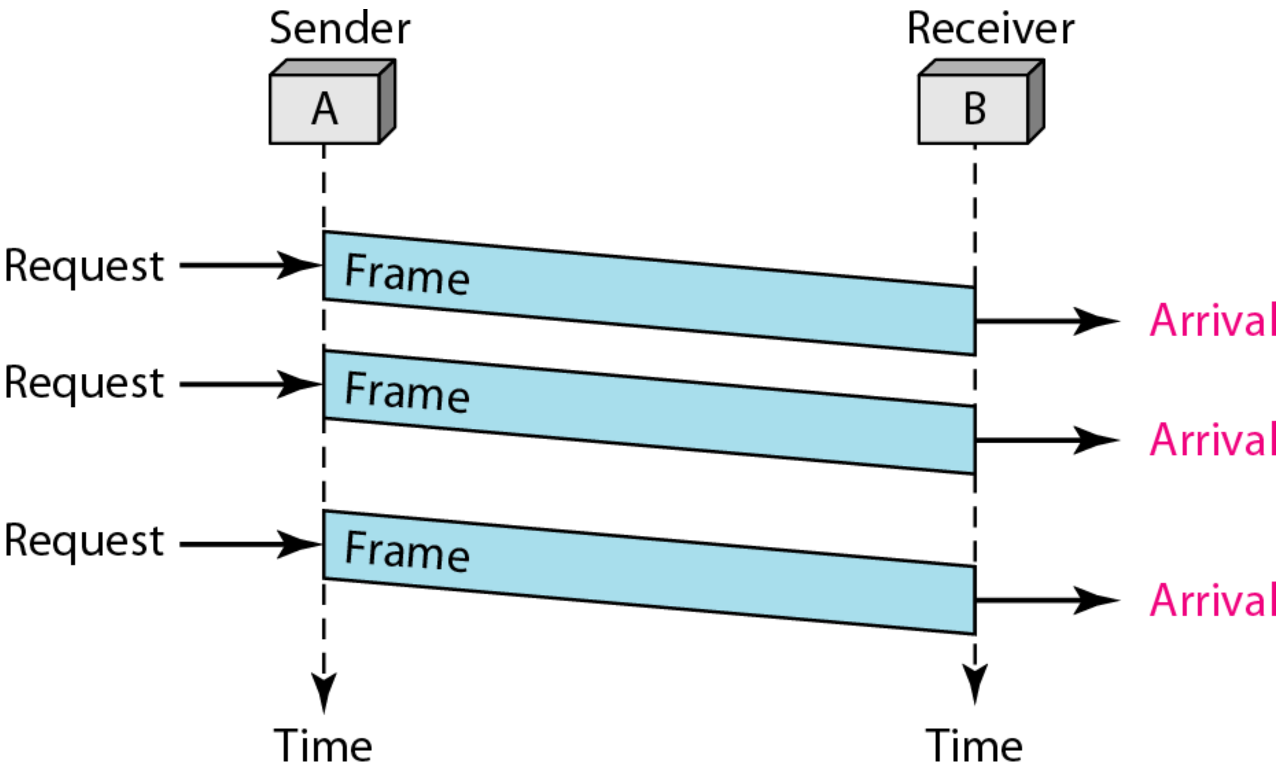
Elementary Data Link Protocols

- Data link protocols can be broadly divided into two categories, depending on whether the transmission channel is noiseless or noisy.



Simplex Protocol

- The Simplex protocol is hypothetical protocol designed for unidirectional data transmission over an ideal channel, i.e. a channel through which transmission can never go wrong.
- The sender simply sends all its data available onto the channel as soon as they are available its buffer. The receiver is assumed to process all incoming data instantly. It is hypothetical since it does not handle flow control or error control.



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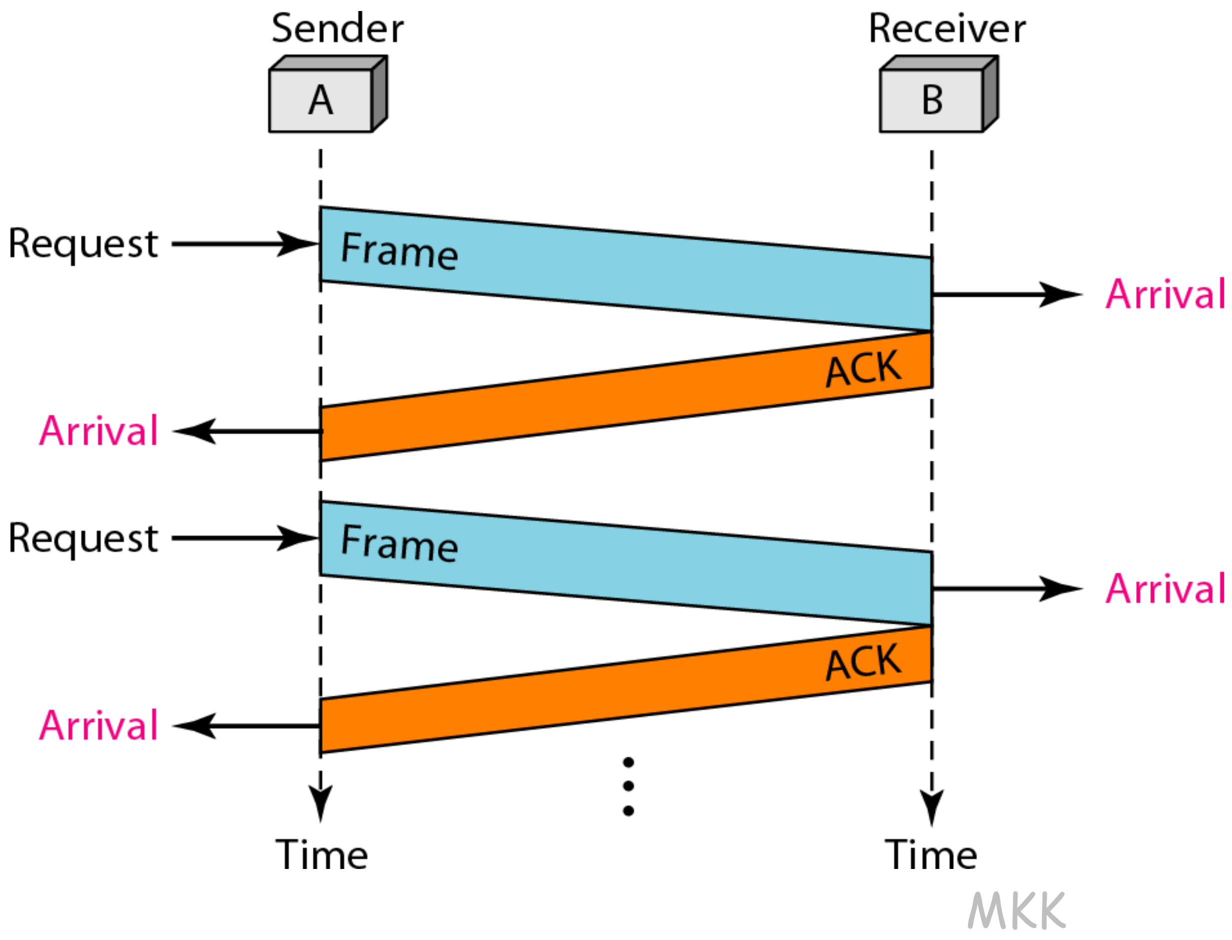
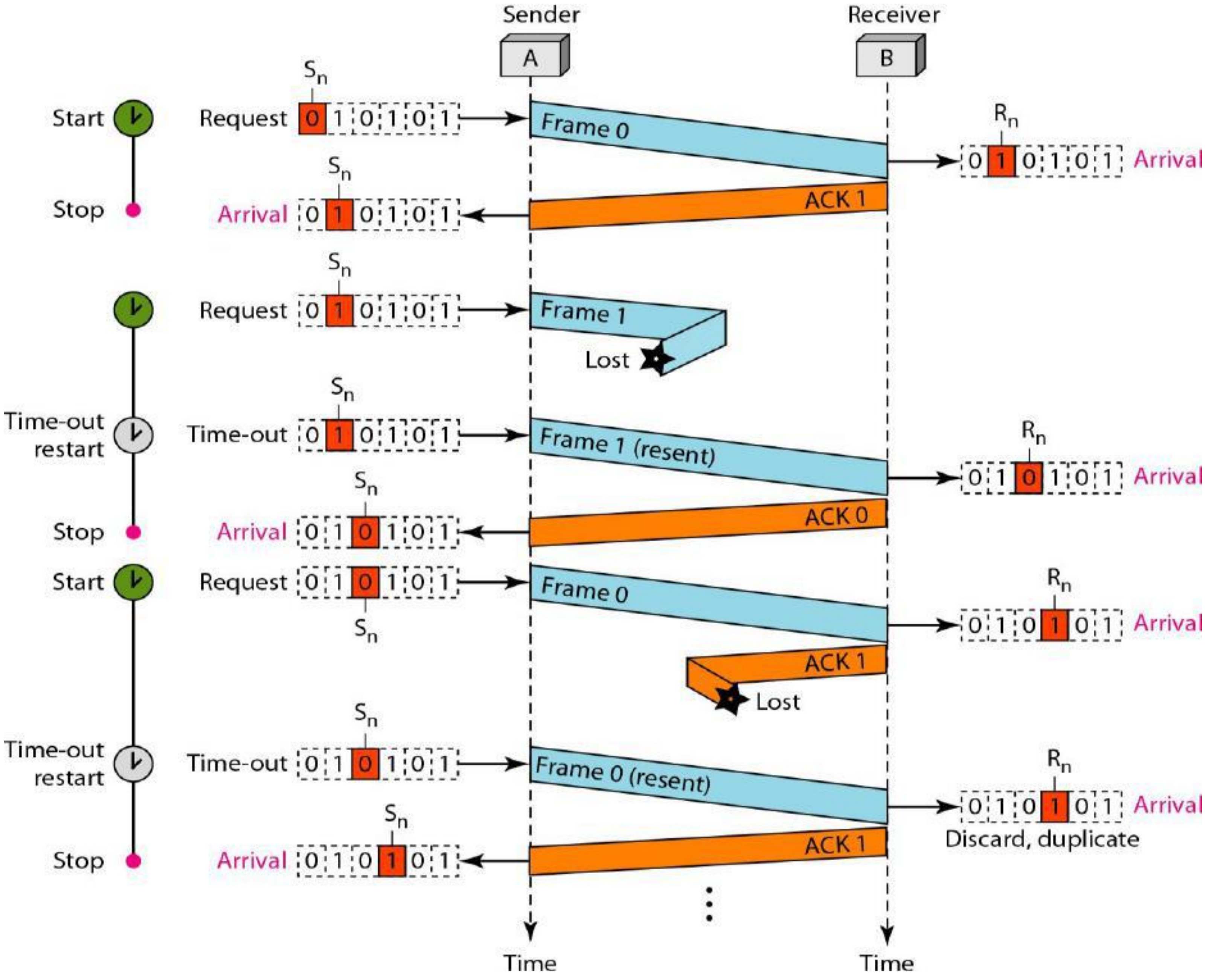
Stop – and – Wait Protocol

- Stop – and – Wait protocol is for noiseless channel too. It provides unidirectional data transmission without any error control facilities. However, it provides for flow control.
- The sender sends one frame and waits for feedback from the receiver. When the ACK arrives, the sender sends the next frame It is Stop-and-Wait Protocol because the sender sends one frame, stops until it receives confirmation from the receiver (okay to go ahead), and then sends the next frame.
- Only ACK frames (simple tokens of acknowledgment) travel from the other direction.



Stop – and – Wait ARQ

- Stop – and – wait Automatic Repeat Request (Stop – and – Wait ARQ) is a variation of the above protocol with added error control mechanisms, appropriate for ***noisy channels***.
- The sender keeps a copy of the sent frame. It then waits for a finite time to receive a positive acknowledgement from receiver.
- If the timer expires or a negative acknowledgement is received, the frame is retransmitted.
- If a positive acknowledgement is received then the next frame is sent.



Sender
Receiver
A
B
Start
Stop
Time-out
restart
Request
Arrival
Time-out
Frame 0
ACK 1
Frame 1
Lost
Frame 1 (resent)
ACK 0
Frame 0 (resent)
Discard, duplicate
Time
Time

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Sliding window protocols

Go – Back – N ARQ

- Go – Back – N ARQ provides for sending multiple frames before receiving the acknowledgement for the first frame. It uses the concept of sliding window, and so is also called sliding window protocol. The frames are sequentially numbered and a finite number of frames are sent. If the acknowledgement of a frame is not received within the time period, all frames starting from that frame are retransmitted.
- In the Go-Back-N Protocol, the sequence numbers are modulo 2m, where m is the size of the sequence number field in bits.
- The sequence numbers range from 0 to $2^m - 1$.
- For example, if m is 4, the only sequence numbers are 0 through 15 inclusive.

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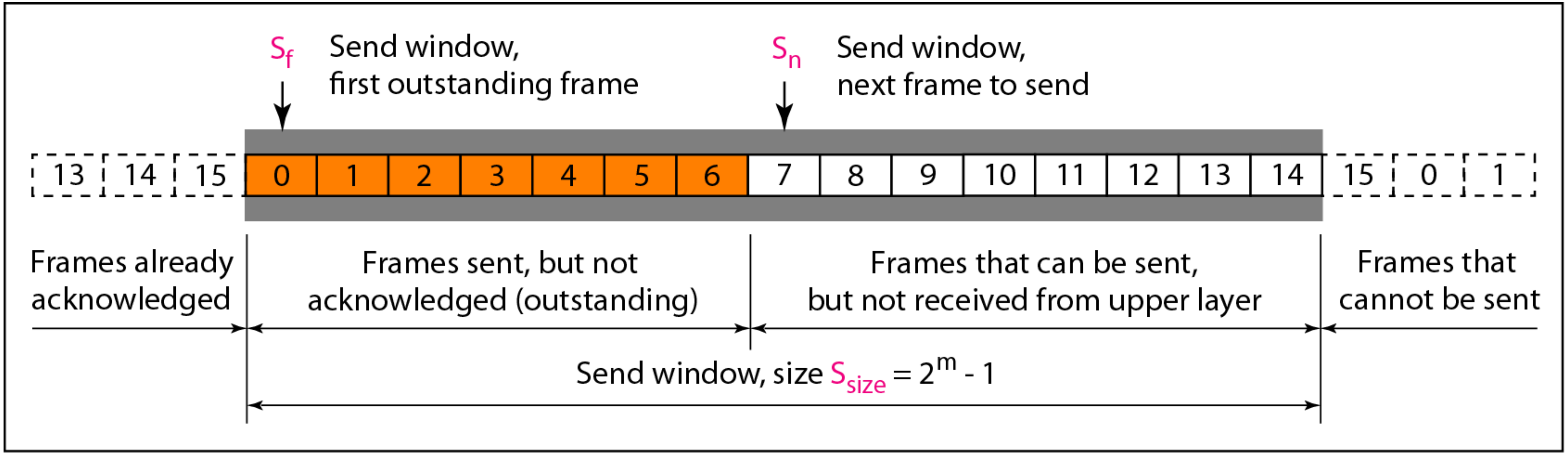
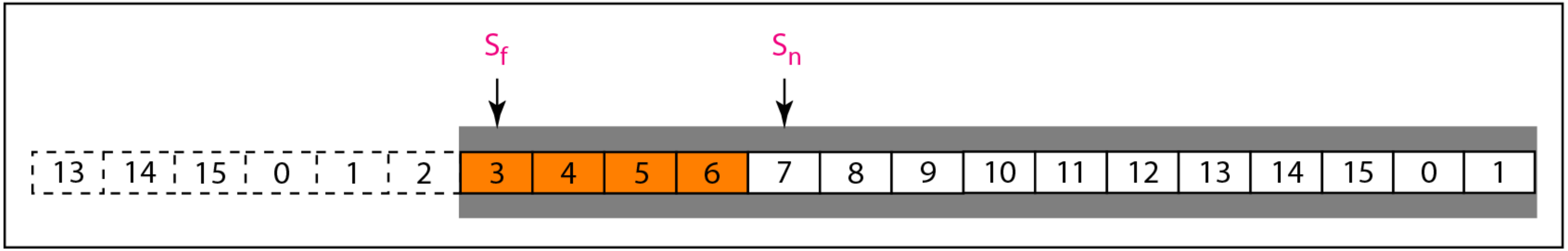


a. Send window before sliding



b. Send window after sliding

- The sender window at any time divides the possible sequence numbers into four regions.
- The first region, from the far left to the left wall of the window, defines the sequence numbers belonging to frames that are already acknowledged. The sender does not worry about these frames and keeps no copies of them.
- The second region, colored in Figure (a), defines the range of sequence numbers belonging to the frames that are sent and have an unknown status. The sender needs to wait to find out if these frames have been received or were lost. We call these outstanding frames.
- The third range, white in the figure, defines the range of sequence numbers for frames that can be sent; however, the corresponding data packets have not yet been received from the network layer.
- Finally, the fourth region defines sequence numbers that cannot be used until the window slides

- The send window is an abstract concept defining an imaginary box of size 2m − 1 with three variables: Sf, Sn, and Ssize.
- The variable Sf defines the sequence number of the first (oldest) outstanding frame.
- The variable Sn holds the sequence number that will be assigned to the next frame to be sent.
- Finally, the variable Ssize defines the size of the window.

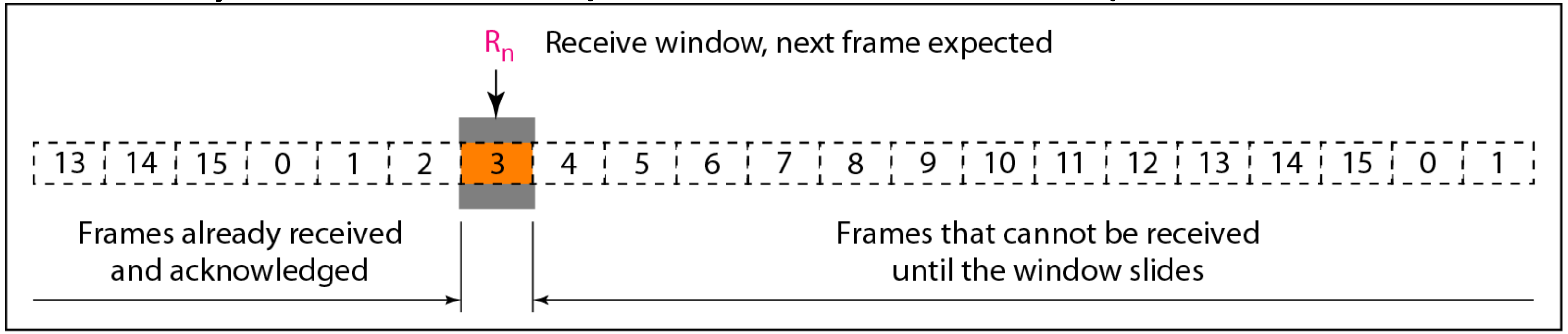
Figure (b) shows how a send window can slide one or more slots to the right when an acknowledgment arrives from the other end. The acknowledgments in this protocol are cumulative, meaning that more than one frame can be acknowledged by an ACK frame.

- In Figure, frames 0, I, and 2 are acknowledged, so the window has slide to the right three slots.
- Note that the value of Sf is 3 because frame 3 is now the first outstanding frame.
- The send window can slide one or more slots when a valid acknowledgment arrives.

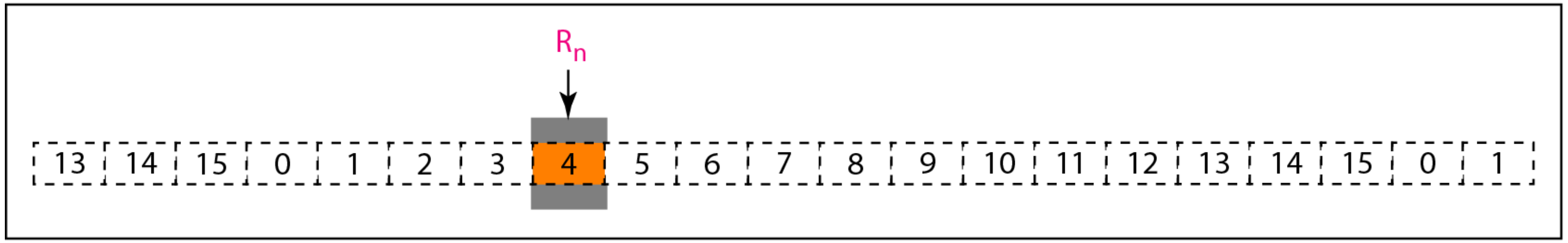
Receiver window: variable Rn (receive window, next frame expected) .

- The sequence numbers to the left of the window belong to the frames already received and acknowledged; the sequence numbers to the right of this window define the frames that cannot be received. Any received frame with a sequence number in these two regions is discarded. Only a frame with a sequence number matching the value of Rn is accepted and acknowledged.
- The receive window also slides, but only one slot at a time. When a correct frame is received (and a frame is received only one at a time), the window slides.(see below figure for receiving window)

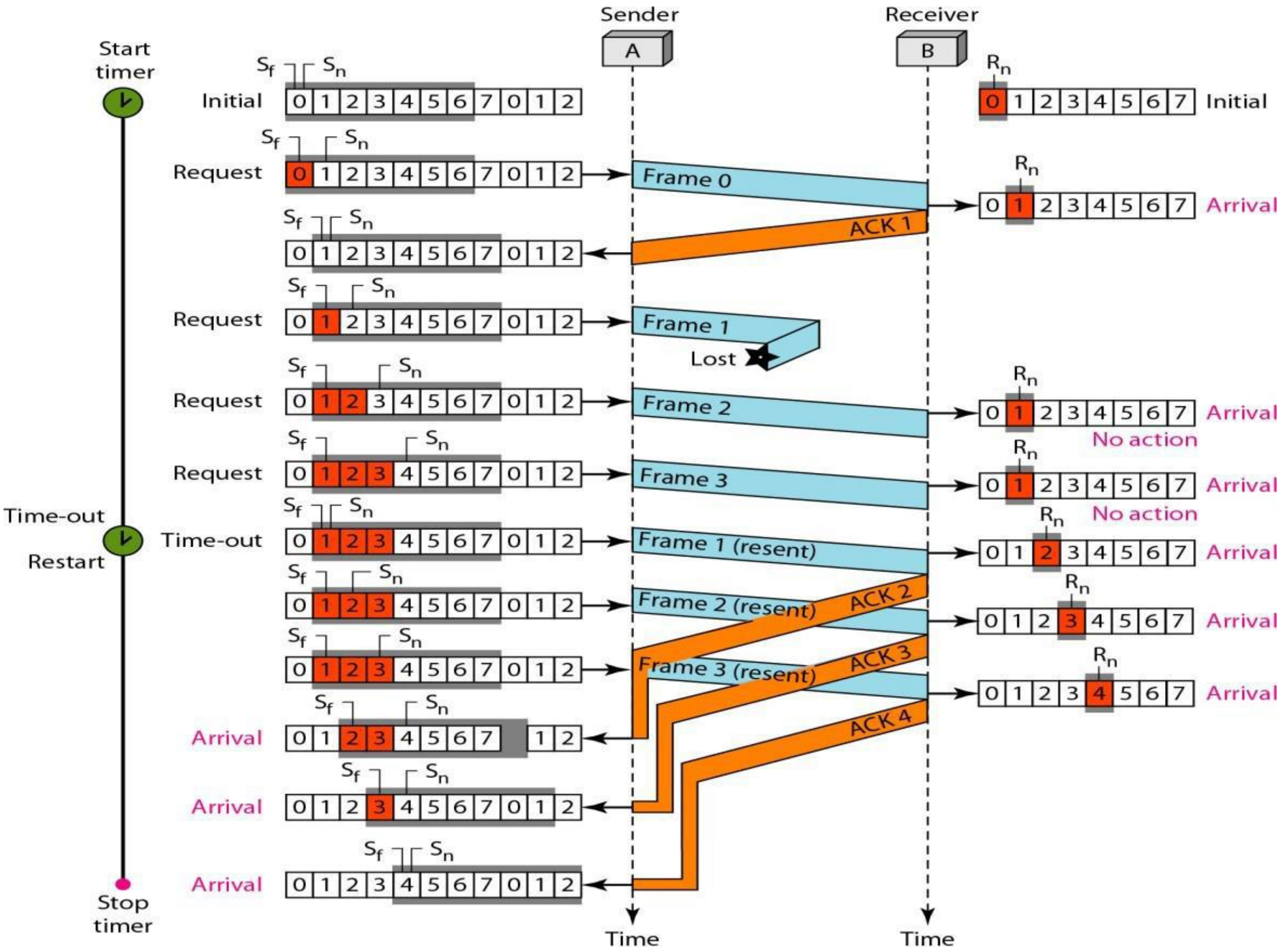
Fig: Receiver window
(a)before sliding
(b)After sliding



a. Receive window

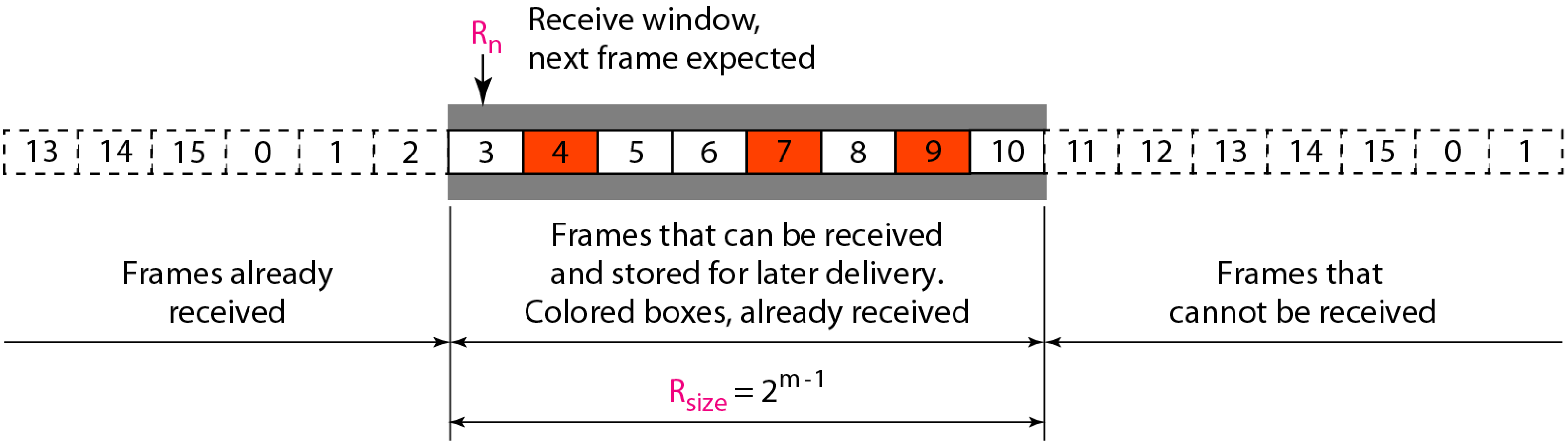


b. Window after sliding


Sender
Receiver
A
B
Start timer
Initial
Request
Request
Request
Request
Time-out
Restart
Time-out
Arrival
Arrival
Arrival
Stop timer
Frame 0
ACK 1
Frame 1
Lost
Frame 2
Frame 3
Frame 1 (resent)
Frame 2 (resent)
Frame 3 (resent)
ACK 2
ACK 3
ACK 4
Initial
Arrival
Arrival
No action
Arrival
No action
Arrival
Arrival
Arrival
Time
Time
S_f
S_n
R_n

Selective Repeat ARQ

- This protocol also provides for sending multiple frames before receiving the acknowledgement for the first frame. However, here only the erroneous or lost frames are retransmitted, while the good frames are received and buffered.



In Selective Repeat ARQ, the size of the sender and receiver window must be at most one-half of 2^m

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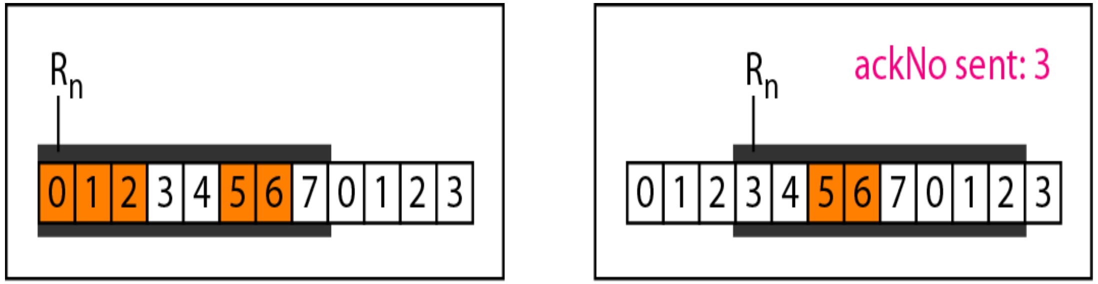
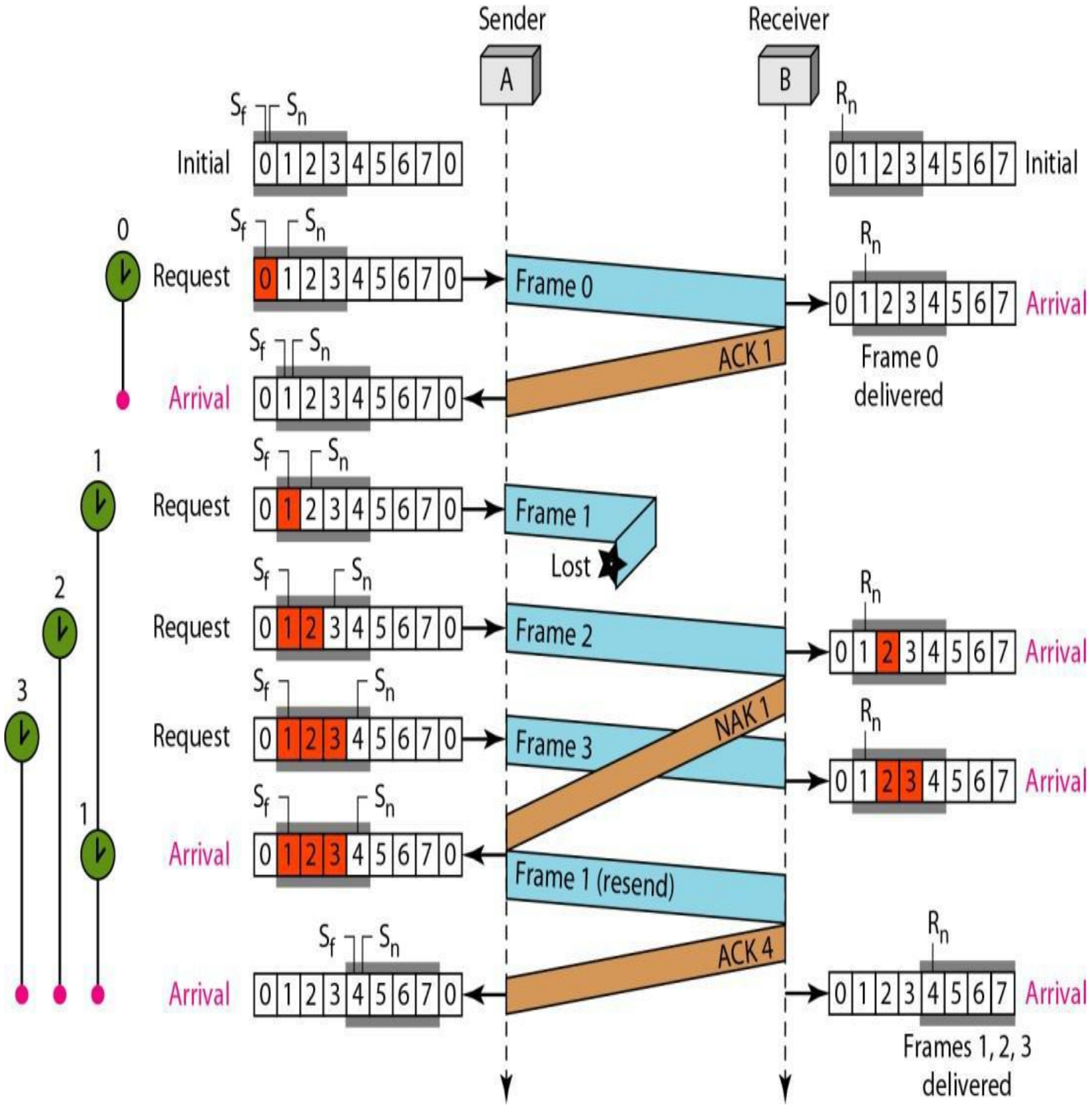
Delivery of Data in Selective Repeat ARQ:



a. Before delivery

b. After delivery

Flow Diagram



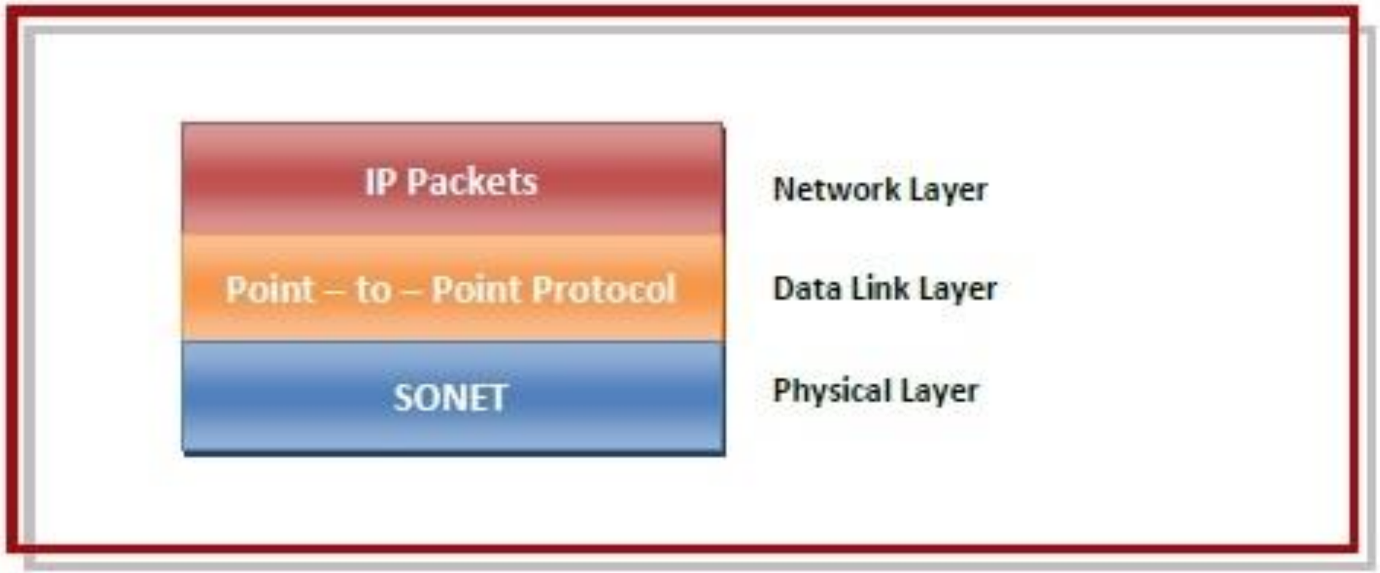
EXAMPLE DATA LINK PROTOCOLS

- Here we will examine the data link protocols found on point-to-point lines in the Internet in two common situations.
- The first situation is when packets are sent over SONET optical fiber links in wide-area networks. These links are widely used, for example, to connect routers in the different locations of an ISP’s network.
- The second situation is for ADSL links running on the local loop of the telephone network at the edge of the Internet. These links connect millions of individuals and businesses to the Internet.

A standard protocol called **PPP** (**Point-to-Point Protocol**) is used to send packets over these links.

Packet over SONET

- Synchronous optical networking (SONET) is a physical layer protocol for transmitting multiple digital bit streams over optical fiber links that form the backbone of the communication networks.
- Packet-over-SONET (POS) is a standard that maps IP packets into SONET frames.



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Features provides by PPP in POS

- **Framing** − It encapsulates the datagram in a frame so that it can be transmitted over the specified physical layer. It delineates the beginning and end of the frames and provides for error detection.
- **Link Control Protocol (LCP)** − It is responsible for establishing, configuring, testing, maintaining and terminating links for transmission. It also imparts negotiation for set up of options and use of features by the two endpoints of the links.
- **Network Control Protocols (NCPs)** − These protocols are used for negotiating the parameters and facilities for the network layer. For every higher-layer protocol supported by PPP, one NCP is there.

Application of POS

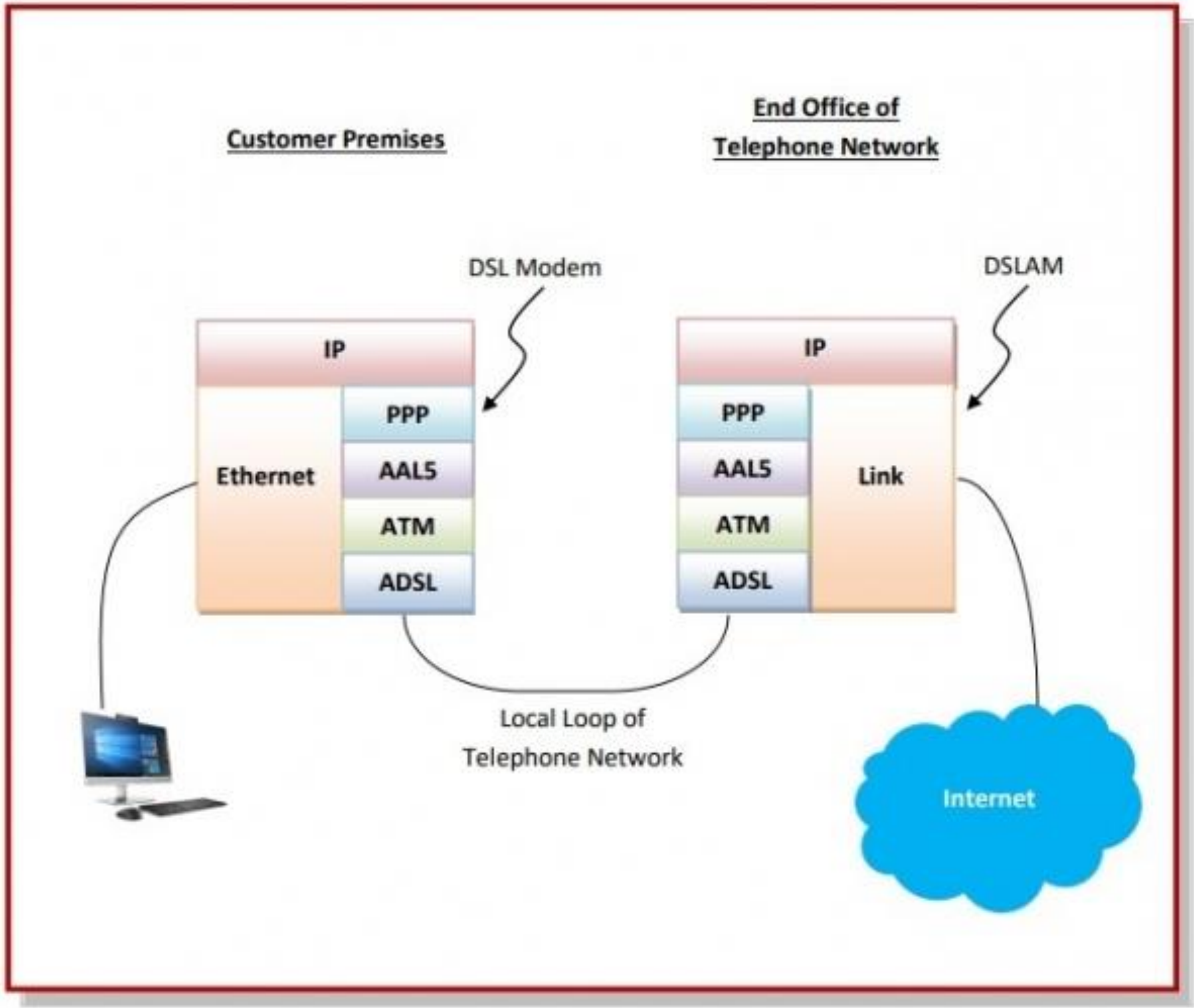
- For sending a large amount of network traffic over the Internet.
- For transmitting IP packets over Wide Area Networks (WANs).

ADSL (Asymmetric Digital Subscriber Loop)

- Asymmetric Digital Subscriber Line (ADSL) is a type of broadband communications technology that transmits digital data at a high bandwidth over existing phone lines to homes and businesses.
- ADSL protocol stack depicts the set of protocols and devices that are used along with ADSL.
- In order to access ADSL, a Digital Subscriber Line modem (DSL modem) is installed at the customer site. The DSL modem sends data bits over the local loop of the telephone network. The local loop is a two − wire connection between the subscriber’s house and the end office of the telephone company. The data bits are accepted at the end office by a device called Digital Subscriber Line Access Multiplexer (DSLAM).

Working Principle

- In the customer site, IP packets are sent to the DSL modem using a link layer like Ethernet. The DSL modem transmits the data through the local loop of the telephone network to the DSLAM at the end office. The DSLAM extracts the IP packets and sends them over the Internet.
- The protocols start at the physical layer with ADSL. Above it are the protocols ATM (Asynchronous Transfer Mode) and AAL5 (ATM Adaptation Layer 5). At the top of the stack, just below the IP is PPP (Point − to − Point) Protocol.


Customer Premises
End Office of
Telephone Network
DSL Modem
DSLAM
IP
PPP
AAL5
ATM
ADSL
Ethernet
IP
PPP
AAL5
ATM
ADSL
Link
Local Loop of
Telephone Network
Internet

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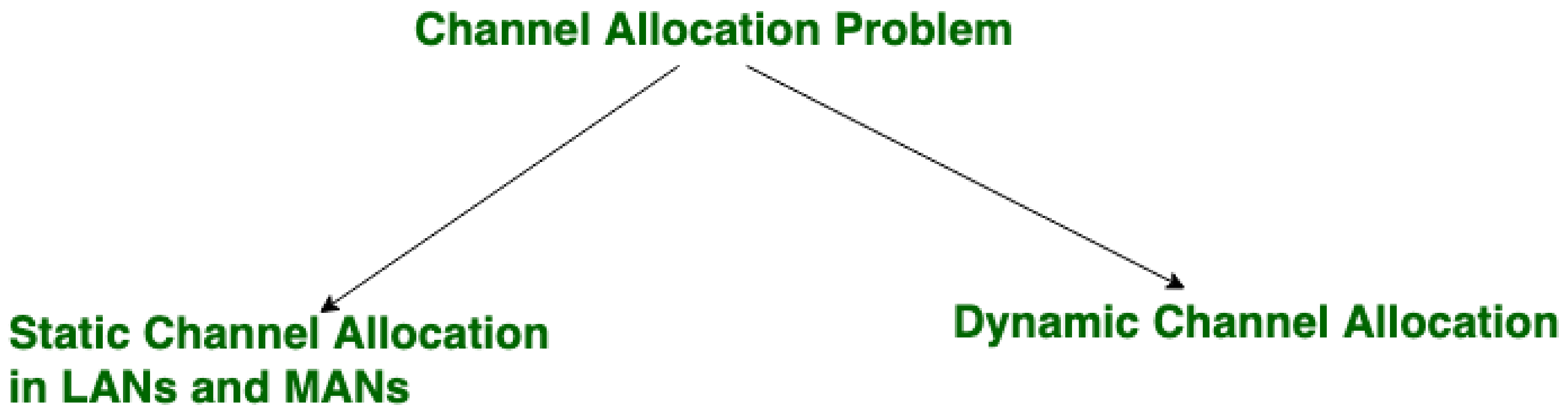
The Medium Access Control

- Network links can be divided into two categories: those using *point-to-point* connections and those using *broadcast* channels.
- In any broadcast network, the key issue is how to determine who gets to use the channel when there is competition for it
  - To make this point, consider a conference call in which six people, on six different telephones, are all connected so that each one can hear and talk to all the others. It is very likely that when one of them stops speaking, two or more will start talking at once, leading to chaos. In a face-to-face meeting, chaos is avoided by external means. For example, at a meeting, people raise their hands to request permission to speak.
- The protocols used to determine who goes next on a multiaccess channel belong to a sublayer of the data link layer called the **MAC** (**Medium Access Control**) sublayer.

THE CHANNEL ALLOCATION PROBLEM

Channel allocation is a process in which a single channel is divided and allotted to multiple users in order to carry user specific tasks. The user's quantity may vary every time the process takes place. If there are N number of users and channel is divided into N equal-sized sub channels, Each user is assigned one portion. If the number of users are small and don't vary at times, then (FDM)Frequency Division Multiplexing can be used as it is a simple and efficient channel bandwidth allocating technique.

- Channel allocation problem can be solved by two schemes: Static Channel Allocation in LANs and MANs, and Dynamic Channel Allocation.



1. Static Channel Allocation in LANs and MANs:

It is the classical or traditional approach of allocating a single channel among multiple competing users through ***Frequency Division Multiplexing (FDM)***. If there are N users, the bandwidth is divided into N equal sized portions each user being assigned one portion. since each user has a private frequency band, there is no interface between users.

$$T = \frac{1}{\mu C - \lambda}$$

T(FDM) = N*T(1/μ (C/N)- λ /N)

T = mean time delay,
C = capacity of channel,
λ = arrival rate of frames,
1/μ = bits/frame,
N = number of sub channels,

T(FDM) = Frequency Division Multiplexing Time

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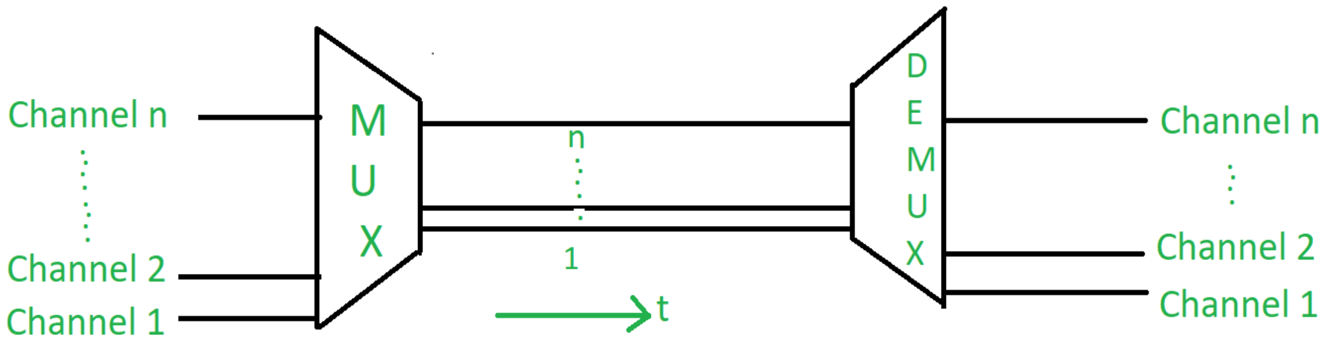
Multiplexing is used in cases where the signals are of lower bandwidth and the transmitting media is having higher bandwidth. In this case, the possibility of sending a number of signals is more. In this, the signals are combined into one and are sent over a link that has greater bandwidth of media than the communicating nodes.

1. Frequency Division Multiplexing (FDM):

- In this, a number of signals are transmitted at the same time, and each source transfers its signals in the allotted frequency range. There is a suitable frequency gap between the 2 adjacent signals to avoid over-lapping. Since the signals are transmitted in the allotted frequencies so this decreases the probability of collision. The frequency spectrum is divided into several logical channels, in which every user feels that they possess a particular bandwidth. A number of signals are sent simultaneously at the same time allocating separate frequency bands or channels to each signal. It is used in radio and TV transmission. Therefore to avoid interference between two successive channels **Guard bands** are used.



Application of FDM:

1. In the first generation of mobile phones, FDM was used.
2. The use of FDM in television broadcasting
3. FDM is used to broadcast FM and AM radio frequencies.

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2. Dynamic Channel Allocation

In dynamic channel allocation scheme, frequency bands are not permanently assigned to the users. Instead channels are allotted to users dynamically as needed, from a central pool. The allocation is done considering a number of parameters/assumptions.

Possible assumptions include:

1. *Independent Traffic:* The model consists of N independent stations (e.g., computers, telephones), each with a program or user that generates frames for transmission.
2. *Single Channel:* A single channel is available for all communication. All stations can transmit on it and all can receive from it.
3. *Observable Collisions:* If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a collision.
4. *Continuous or Slotted Time:* Time may be assumed continuous, in this case frame transmission can begin at any instant. Alternatively, time may be slotted or divided into discrete intervals (called slots).
5. *Carrier Sense or No Carrier Sense*: With the carrier sense assumption, stations can sense if the channel is in use before trying to use it. No station will attempt to use the channel while it is sensed as busy. If there is no carrier sense, stations cannot sense the channel before trying to use it. They just go ahead and transmit.

Carrier Sense Multiple Access (CSMA) Protocols

Carrier Sense Multiple Access (CSMA) is a network protocol for carrier transmission that operates in the Medium Access Control (MAC) layer. It senses or listens whether the shared channel for transmission is busy or not, and transmits if the channel is not busy. Using CMSA protocols, more than one users or nodes send and receive data through a shared medium that may be a single cable or optical fiber connecting multiple nodes, or a portion of the wireless spectrum.

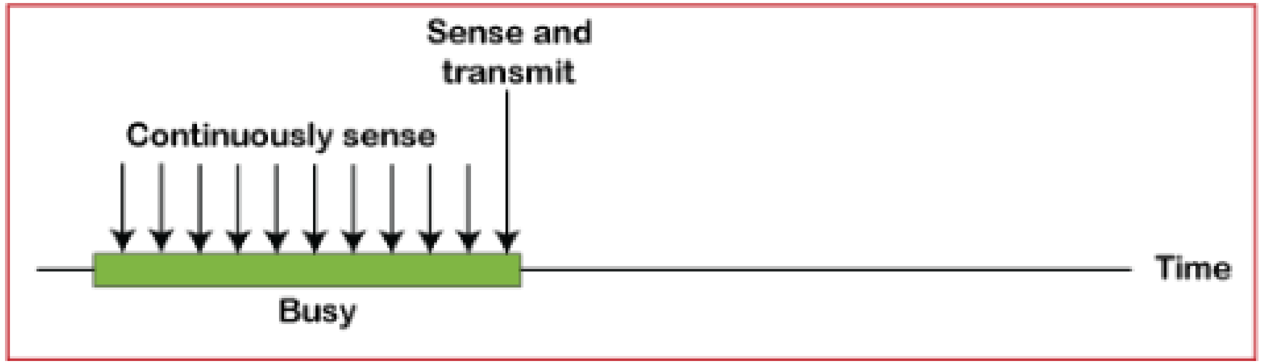
Working Principle

- When a station has frames to transmit, it attempts to detect presence of the carrier signal from the other nodes connected to the shared channel. If a carrier signal is detected, it implies that a transmission is in progress. The station waits till the ongoing transmission executes to completion, and then initiates its own transmission. Generally, transmissions by the node are received by all other nodes connected to the channel.
- Since, the nodes detect for a transmission before sending their own frames, collision of frames is reduced. However, if two nodes detect an idle channel at the same time, they may simultaneously initiate transmission. This would cause the frames to garble resulting in a collision.

The versions of CSMA access modes are−

- 1-Persistent
- Non-Persistent
- P-Persistent
- O- Persistent

a) **1-Persistent:** In the 1-Persistent mode of CSMA that defines each node, first sense the shared channel and if the channel is idle, it immediately sends the data. Else it must wait and keep track of the status of the channel to be idle and broadcast the frame unconditionally as soon as the channel is idle.



a. 1-persistent

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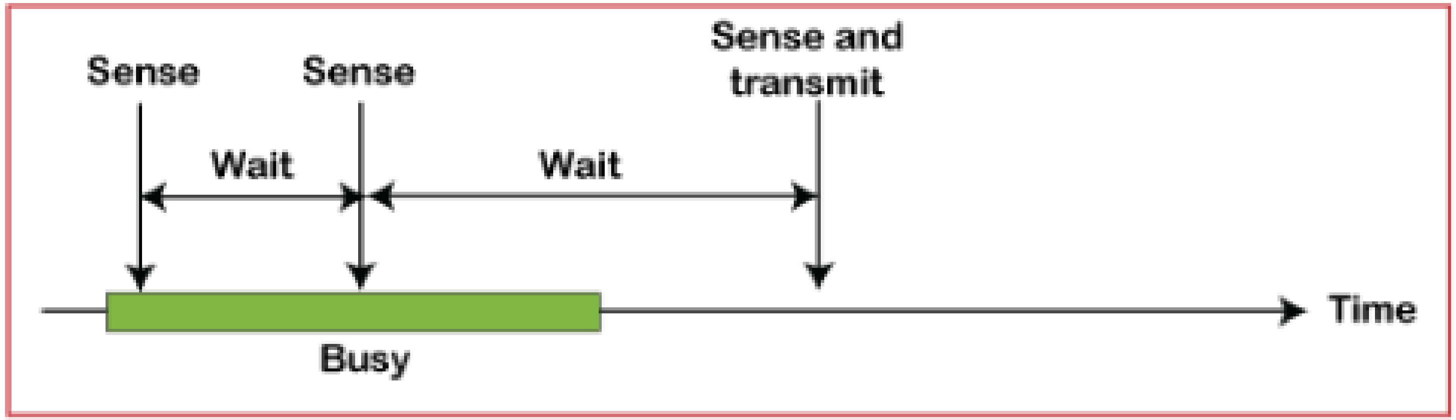
b) **Non-Persistent:** It is the access mode of CSMA that defines before transmitting the data, each node must sense the channel, and if the channel is inactive, it immediately sends the data. Otherwise, the station must wait for a random time (not continuously), and when the channel is found to be idle, it transmits the frames.



b. Nonpersistent

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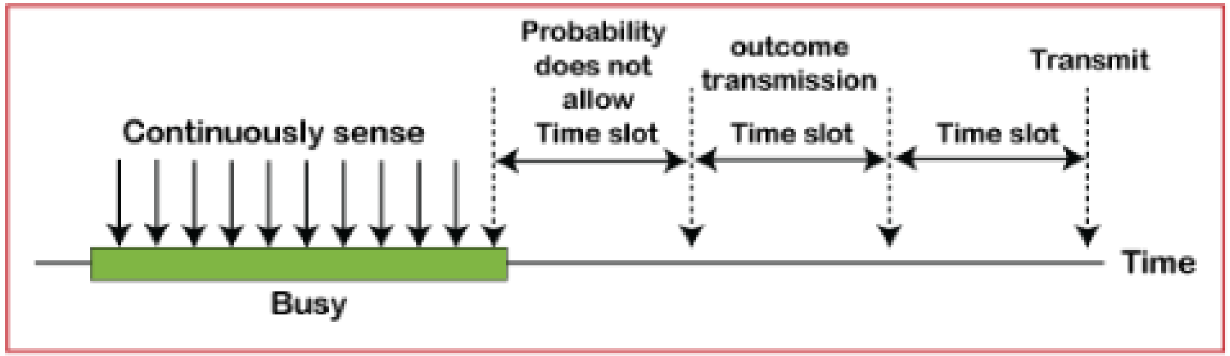
c) **P-Persistent:** It is the combination of 1-Persistent and Non-persistent modes. The P-Persistent mode defines that each node senses the channel, and if the channel is inactive, it sends a frame with a **P** probability. If the data is not transmitted, it waits for a (**q = 1-p probability**) random time and resumes the frame with the next time slot.



c. p-persistent

d) **O- Persistent:** It is an O-persistent method that defines the superiority of the station before the transmission of the frame on the shared channel. If it is found that the channel is inactive, each station waits for its turn to retransmit the data.

Collision-Free Protocols

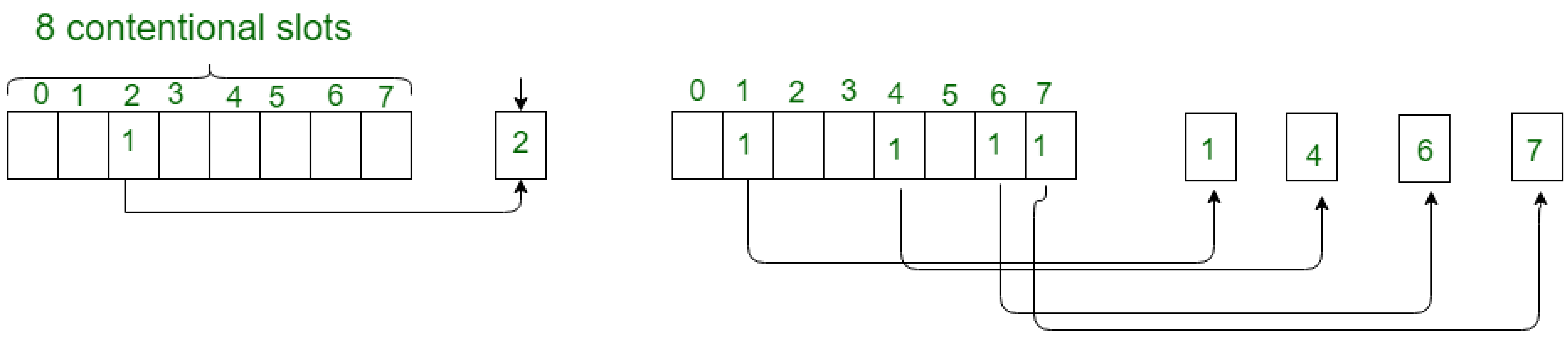
Collision – free protocols resolves collision in the contention period and so the possibilities of collisions are eliminated.

Types of Collision – free Protocols

- Bit – map Protocol
- Binary Countdown

Bit – map Protocol:

- In bit map protocol, the contention period is divided into N slots, where N is the total number of stations sharing the channel. If a station has a frame to send, it sets the corresponding bit in the slot. So, before transmission, each station knows whether the other stations want to transmit. Collisions are avoided by mutual agreement among the contending stations on who gets the channel.



A Bit-map Protocol.

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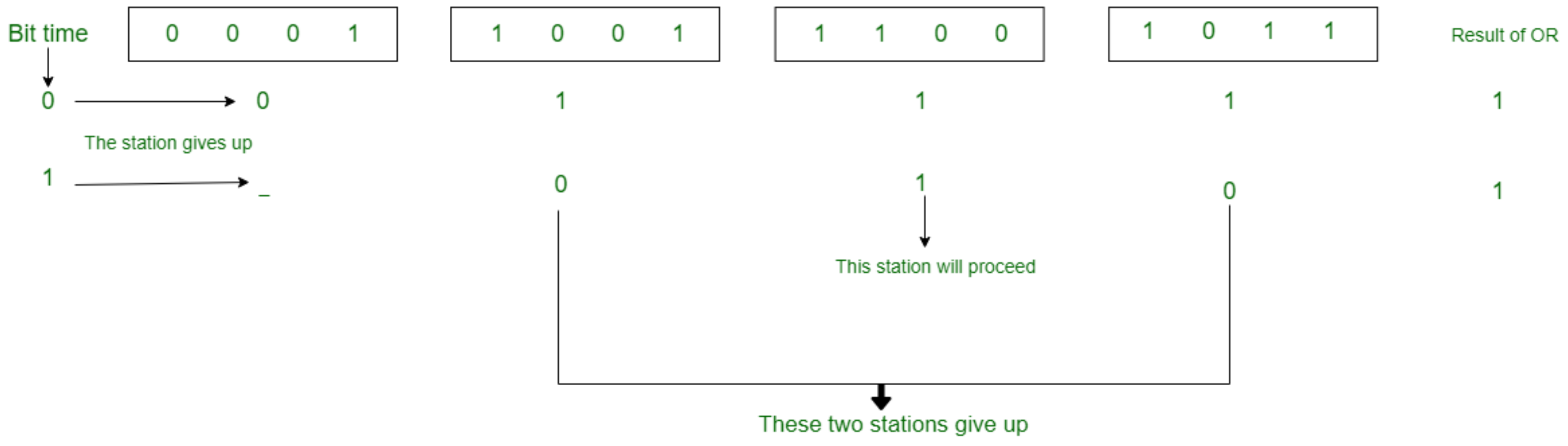
Binary Countdown

- This protocol overcomes the overhead of 1 bit per station of the bit – map protocol. Here, binary addresses of equal lengths are assigned to each station. For example, if there are 6 stations, they may be assigned the binary addresses 001, 010, 011, 100, 101 and 110. All stations wanting to communicate broadcast their addresses. The station with higher address gets the higher priority for transmitting.



Binary countdown

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The Ethernet

- Ethernet is the most widely used LAN technology, which is defined under IEEE standards 802.3. The reason behind its wide usability is Ethernet is easy to understand, implement, maintain, and allows low-cost network implementation.
- Two kinds of Ethernet exist: **classic Ethernet** and **switched Ethernet**

- Classic Ethernet is the original form and ran at rates from 3 to 10 Mbps.
- Switched Ethernet is what Ethernet has become and runs at 100, 1000, and 10,000 Mbps, in forms called fast Ethernet, gigabit Ethernet, and 10 gigabit Ethernet. In practice, only switched Ethernet is used nowadays.

These data rates are defined for operation over optical fibres and twisted-pair cables:

i) Fast Ethernet

Fast Ethernet refers to an Ethernet network that can transfer data at a rate of 100 Mbit/s.

ii) Gigabit Ethernet

Gigabit Ethernet delivers a data rate of 1,000 Mbit/s (1 Gbit/s).

iii) 10 Gigabit Ethernet

10 Gigabit Ethernet is the recent generation and delivers a data rate of 10 Gbit/s (10,000 Mbit/s). It is generally used for backbones in high-end applications requiring high data rates.

Advantages of Ethernet:

- ***Speed:*** When compared to a wireless connection, Ethernet provides significantly more speed. Because Ethernet is a one-to-one connection, this is the case. As a result, speeds of up to 10 Gigabits per second (Gbps) or even 100 Gigabits per second (Gbps) are possible.
- ***Efficiency:*** An Ethernet cable, such as Cat6, consumes less electricity, even less than a wifi connection. As a result, these Ethernet cables are thought to be the most energy-efficient.
- ***Good data transfer quality:*** Because it is resistant to noise, the information transferred is of high quality.

Baud rate = 2* Bit rate

<u>Wireless LANs</u> (IEEE 802.11)

- Wireless LANs (WLANs) are wireless computer networks that use high-frequency radio waves instead of cables for connecting the devices within a limited area forming LAN (Local Area Network). Users connected by wireless LANs can move around within this limited area such as home, school, campus, office building, railway platform, etc.
- Most WLANs are based upon the standard IEEE 802.11 standard or WiFi.

Components of WLANs:

- Wireless Access Point (WAP or AP)
- Client

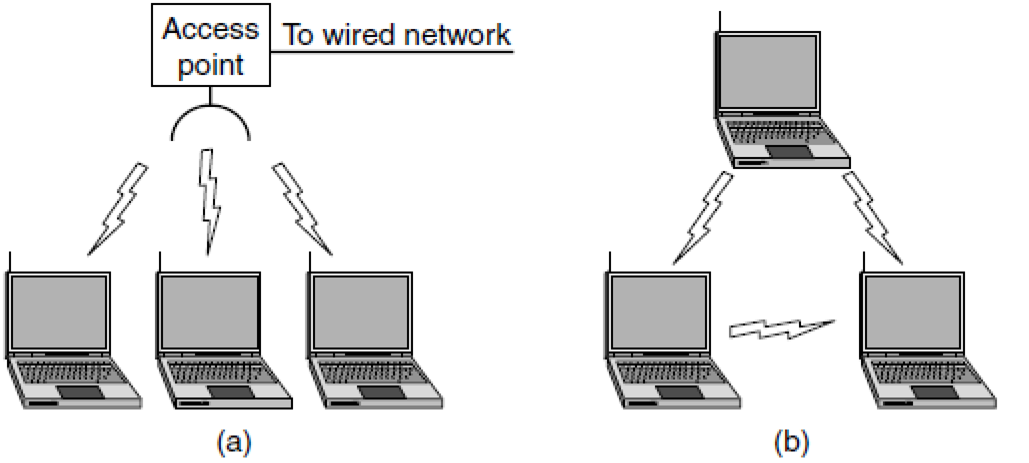
Types of WLANS

Wireless network with an access point:

Mobile devices or clients connect to an access point (AP) that in turn connects via a bridge to the LAN or Internet.

The client transmits frames to other clients via the AP.

***Ad hoc network*:** Clients transmit frames directly to each other in a peer-to-peer fashion.



(a) Wireless network with an access point. (b) Ad hoc network.

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Bluetooth

- Bluetooth is a low cost, low power, short range wireless technology for data and audio communication.
- Bluetooth was first proposed by ‘Ericsson’ in 1994.
- Bluetooth operates at 2.4GHz of the Radio Frequency spectrum and uses the Frequency Hopping Spread Spectrum (FHSS) technique for communication.
- Literally it supports a data rate of up to 1Mbps to 24Mbps (and a range of approximately 30 to 100 feet (Depending on the Bluetooth version – v1.2 supports datarate up to 1Mbps, v2.0 + EDR ‘*Enhanced Data Rate*’ supports datarate up to 3Mbps, v3.0 + HS ‘*High Speed*’ and v4.0 supports datarate up to 24Mbps)) for data communication.
- Bluetooth communication has two essential parts; a physical link part and a protocol part.
- The physical link is responsible for the physical transmission of data between devices supporting Bluetooth communication and protocol part is responsible for defining the rules of communication.
- The physical link works on the Wireless principle making use of Radio Frequency waves for communication.

- Bluetooth enabled devices essentially contain a Bluetooth wireless radio for the transmission and reception of data.
- The rules governing the Bluetooth communication is implemented in the 'Bluetooth protocol stack'.
- The Bluetooth communication IC holds the stack. Each Bluetooth device will have a 48 bit unique identification number.
- Bluetooth communication follows packet based data transfer. Bluetooth supports point-to-point (device to device) and point-to-multipoint (device to multiple device broadcasting) wireless communication.
- The point-to-point communication follows the master slave relationship. A Bluetooth device can function as either master or slave.
- When a network is formed with one Bluetooth device as master and more than one device as slaves, it is called a Piconet. A Piconet supports a maximum of seven slave devices.
- Bluetooth is the favourite choice for short range data communication in handheld embedded devices like cell phones
- Bluetooth Low Energy (BLE)/Bluetooth Smart is a latest addition to the Bluetooth technology. BLE allows devices to use much less power compared to the standard Bluetooth connections, while offering most of the connectivity of Bluetooth and maintaining a similar communication range.
- Bluetooth 4.2 specification enables IoT support through Low-power IP connectivity

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- The Bluetooth standard specifies the minimum requirements that a Bluetooth device must support for a specific usage scenario.
- The Generic Access Profile (GAP) defines the requirements for detecting a Bluetooth device and establishing a connection with it. All other specific usage profiles are based on GAP.

✓ Serial Port Profile (SPP) for serial data communication,

✓ File Transfer Profile (FTP) for file transfer between devices,

✓ Human Interface Device (HID) for supporting human interface devices like keyboard and mouse are examples for Bluetooth profiles.

- BLE implements various application specific profiles for communicating with low power Bluetooth peripherals like fitness devices, Blood Pressure and heart rate monitors etc.

The specifications for Bluetooth communication is defined and licensed by the standards body 'Bluetooth Special Interest Group (SIG)'.

That's all about

UNIT 2

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Unit 3

Network Layer

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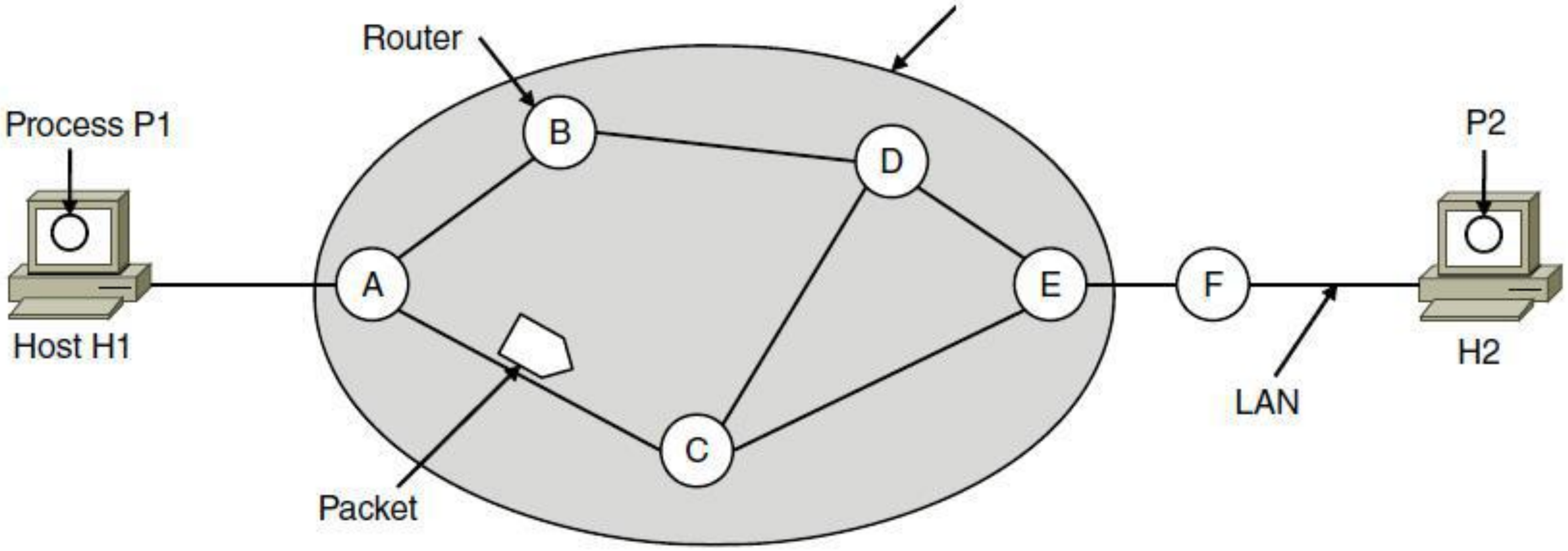
The network layer is concerned with getting packets from the source all the way to the destination. Getting to the destination may require making many hops at intermediate routers along the way. This function clearly contrasts with that of the data link layer, which has the more modest goal of just moving frames from one end of a wire to the other. Thus, the network layer is the lowest layer that deals with end-to-end transmission.

Network Layer Design Issues

1. Store-and-forward packet switching
2. Services provided to transport layer
3. Implementation of connectionless service
4. Implementation of connection-oriented service
5. Comparison of virtual-circuit and datagram networks

1. Store-and-forward packet switching

The network layer operates in an environment that uses store and forward packet switching. The node which has a packet to send, delivers it to the nearest router. The packet is stored in the router until it has fully arrived and its checksum is verified for error detection. Once, this is done, the packet is forwarded to the next router. Since, each router needs to store the entire packet before it can forward it to the next hop, the mechanism is called store − and − forward switching.



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2. Services provided to transport layer

- The network layer provides service its immediate upper layer, namely transport layer, through the network − transport layer interface. The two types of services provided are
  - Connection − Oriented Service − In this service, a path is setup between the source and the destination, and all the data packets belonging to a message are routed along this path.
  - Connectionless Service − In this service, each packet of the message is considered as an independent entity and is individually routed from the source to the destination.

The objectives of the network layer while providing these services are −

- The services should not be dependent upon the router technology.
- The router configuration details should not be of a concern to the transport layer.
- A uniform addressing plan should be made available to the transport layer, whether the network is a LAN, MAN or WAN.

3. Implementation of connectionless service

- Packet are termed as “datagrams” and corresponding subnet as “datagram subnets”. When the message size that has to be transmitted is 4 times the size of the packet, then the network layer divides into 4 packets and transmits each packet to router via. a few protocol. Each data packet has destination address and is routed independently irrespective of the packets.

4. Implementation of connection-oriented service

- To use a connection-oriented service, first we establishes a connection, use it and then release it. In connection-oriented services, the data packets are delivered to the receiver in the same order in which they have been sent by the sender.

It can be done in either two ways :

  - **Circuit Switched Connection –** A dedicated physical path or a circuit is established between the communicating nodes and then data stream is transferred.
  - **Virtual Circuit Switched Connection –** The data stream is transferred over a packet switched network, in such a way that it seems to the user that there is a dedicated path from the sender to the receiver. A virtual path is established here. While, other connections may also be using the same path.

5. Comparison of virtual-circuit and datagram networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

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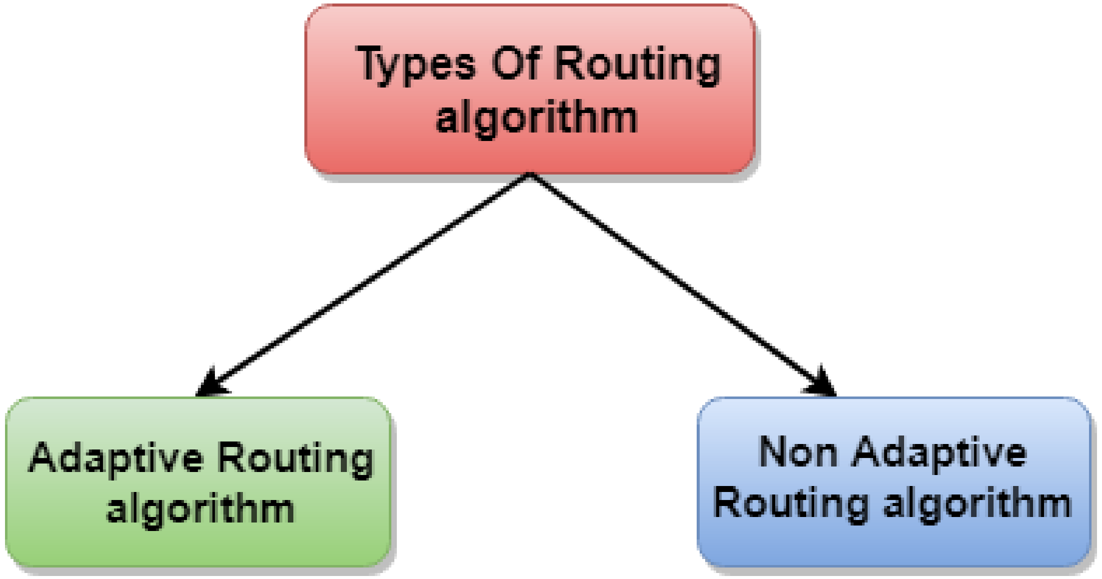
Routing Algorithms

- A routing algorithm is a procedure that lays down the route or path to transfer data packets from source to the destination. They help in directing Internet traffic efficiently. After a data packet leaves its source, it can choose among the many different paths to reach its destination. Routing algorithm mathematically computes the best path, i.e. “least – cost path” that the packet can be routed through.
- Routing algorithms can be broadly categorized into two types,

  adaptive and non adaptive routing algorithms



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- Routing algorithms can be grouped into two major classes:

  1) Non adaptive (Static Routing)

  2) Adaptive (Dynamic Routing)

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Adaptive Algorithms

- Adaptive routing algorithms, also known as dynamic routing algorithms, makes routing decisions dynamically depending on the network conditions. It constructs the routing table depending upon the network traffic and topology. They try to compute the optimized route depending upon the hop count, transit time and distance.

Non – Adaptive Routing Algorithms

- Non-adaptive Routing algorithms, also known as static routing algorithms, construct a static routing table to determine the path through which packets are to be sent. The static routing table is constructed based upon the routing information stored in the routers when the network is booted up.

The Optimality Principle

- The purpose of a routing algorithm at a router is to decide which output line an incoming packet should go. The optimal path from a particular router to another may be the least cost path, the least distance path, the least time path, the least hops path or a combination of any of the above.
- The optimality principle can be logically proved as follows −
- If a better route could be found between router J and router K, the path from router I to router K via J would be updated via this route. Thus, the optimal path from J to K will again lie on the optimal path from I to K.

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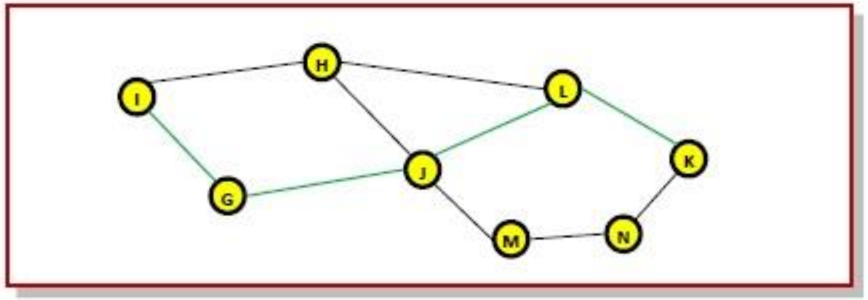
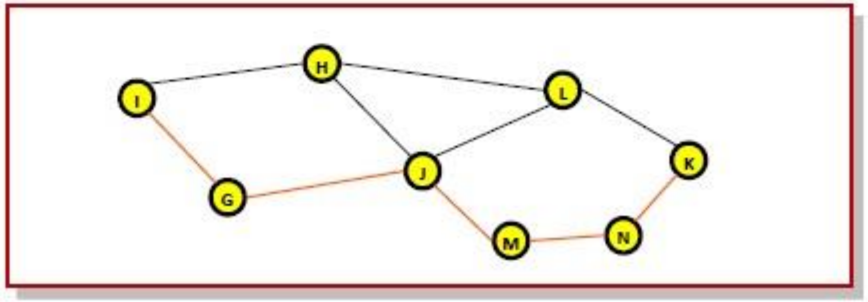
Example

- Consider a network of routers, {G, H, I, J, K, L, M, N} as shown in the figure. Let the optimal route from I to K be as shown via the green path, i.e. via the route I-G-J-L-K. According to the optimality principle, the optimal path from J to K with be along the same route, i.e. J-L-K.



- Now, suppose we find a better route from J to K is found, say along J-M-N-K. Consequently, we will also need to update the optimal route from I to K as I-G-J-M-N-K, since the previous route ceases to be optimal in this situation. This new optimal path is shown line orange lines in the following figure –
-



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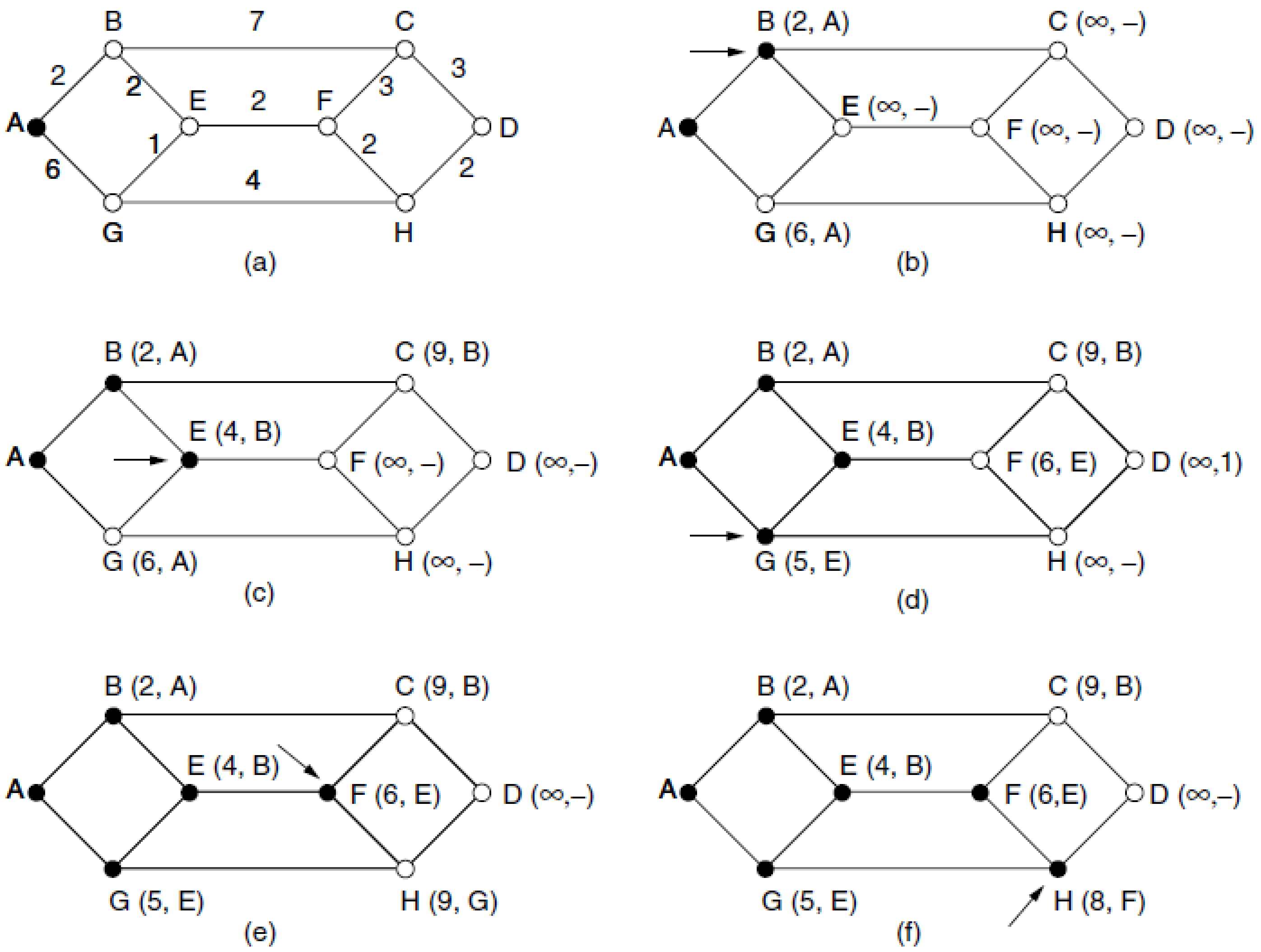
Shortest Path Algorithm

- Consider that a network comprises of N vertices (nodes or network devices) that are connected by M edges (transmission lines). Each edge is associated with a weight, representing the physical distance or the transmission delay of the transmission line. The target of shortest path algorithms is to find a route between any pair of vertices along the edges, so the sum of weights of edges is minimum. If the edges are of equal weights, the shortest path algorithm aims to find a route having minimum number of hops

Example



- The first six steps used in computing the shortest path from *A* to *D*.
- The arrows indicate the working node.

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Flooding

- When a routing algorithm is implemented, each router must make decisions based on local knowledge, not the complete picture of the network. A simple local technique is **flooding**, in which every incoming packet is sent out on every outgoing line except the one it arrived on. Flooding obviously generates vast numbers of duplicate packets, in fact, an infinite number unless some measures are taken to damp the process.
- A better technique for damming the flood is to have routers keep track of which packets have been flooded, to avoid sending them out a second time. One way to achieve this goal is to have the source router put a sequence number in each packet it receives from its hosts. Each router then needs a list per source router telling which sequence numbers originating at that source have already been seen. If an incoming packet is on the list, it is not flooded.

Distance Vector Routing (DVR)

- A **distance-vector routing (DVR)** algorithm requires that a router inform its neighbors of topology changes periodically. Historically known as the old ARPANET routing algorithm (or known as Bellman-Ford algorithm).
  - **Bellman Ford Basics –** Each router maintains a Distance Vector table containing the distance between itself and ALL possible destination nodes. Distances based on a chosen metric, are computed using information from the neighbors' distance vectors.
- Information kept by DV router - Each router has an ID
  - Associated with each link connected to a router, there is a link cost (static or dynamic).
  - Intermediate hops
- Distance Vector Table Initialization –
  - Distance to itself = 0
  - Distance to ALL other routers = infinity number.

Algorithm –

- A router transmits its distance vector to each of its neighbors in a routing packet.
- Each router receives and saves the most recently received distance vector from each of its neighbors.
- A router recalculates its distance vector when:
  - It receives a distance vector from a neighbor containing different information than before.
  - It discovers that a link to a neighbor has gone down.
- The DV calculation is based on minimizing the cost to each destination
  - **Note**
    - From time-to-time, each node sends its own distance vector estimate to neighbors.
    - When a node x receives new DV estimate from any neighbor v, it saves v's distance vector and it updates its own DV using B-F equation:

$$Dx(y) = \min \{ C(x,v) + Dv(y), Dx(y) \} \text{ for each node } y \in N$$

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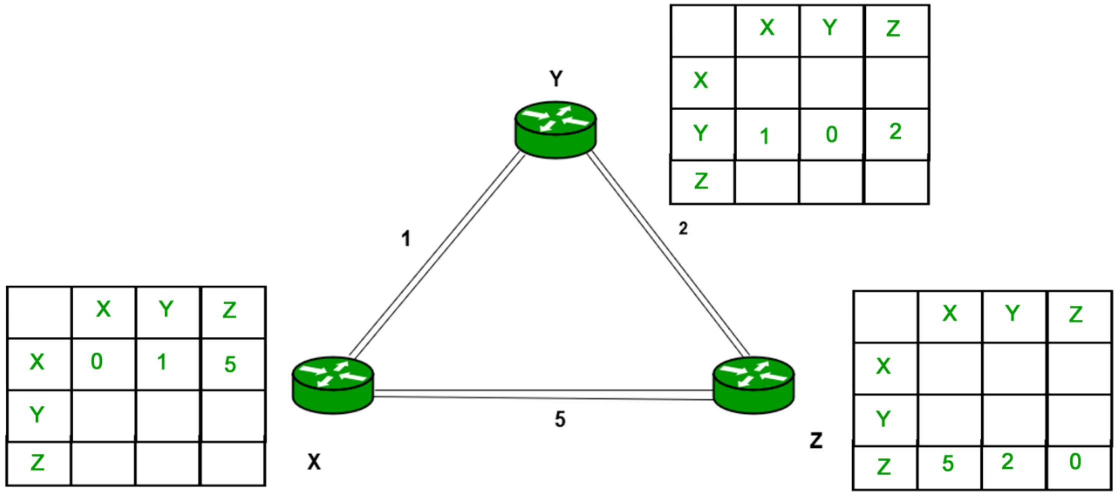
Example – Consider 3-routers X, Y and Z as shown in figure. Each router have their routing table. Every routing table will contain distance to the destination nodes.



- Consider router X , X will share it routing table to neighbors and neighbors will share it routing table to it to X and distance from node X to destination will be calculated using bellmen- ford equation.
- As we can see that distance will be less going from X to Z when Y is intermediate node(hop) so it will be update in routing table X.

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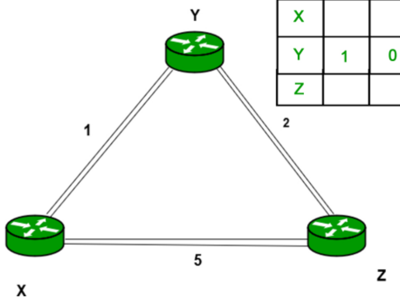
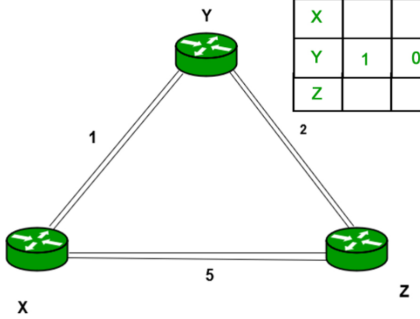


	X	Y	Z
X	0	1	~~5~~ 3
Y	1	0	2
Z			

	X	Y	Z
X			
Y	1	0	2
Z			

	X	Y	Z
X			
Y			
Z	5	2	0

Similarly for Z also –

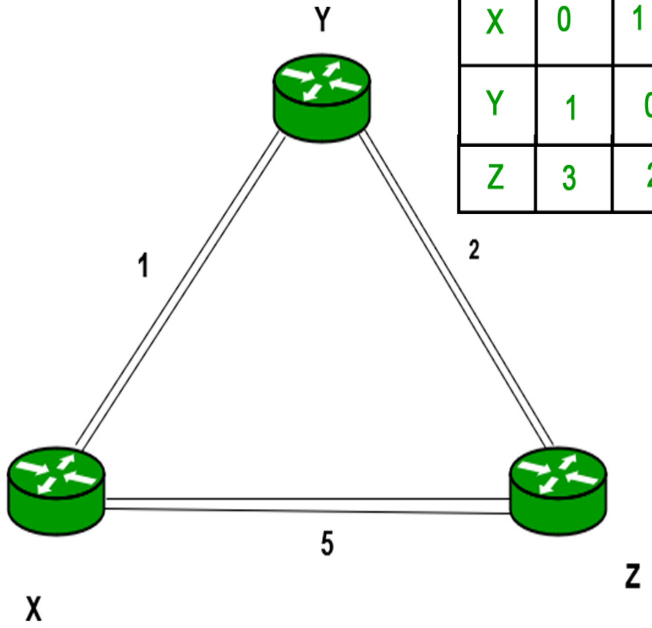


	X	Y	Z
X	0	1	~~5~~ 3
Y	1	0	2
Z	~~5~~ 3	2	0

	X	Y	Z
X			
Y	1	0	2
Z			

	X	Y	Z
X			
Y			
Z	5	2	0

Finally the routing table for all –



	X	Y	Z
X	0	1	3
Y	1	0	2
Z	3	2	0

	X	Y	Z
X	0	1	3
Y	1	0	2
Z	3	2	0

	X	Y	Z
X	0	1	3
Y	1	0	2
Z	3	2	0

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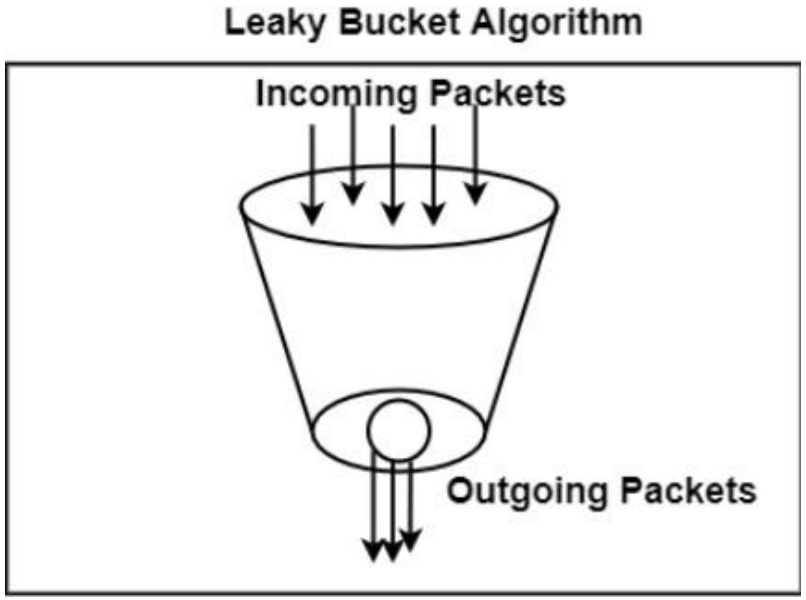
Congestion Control Algorithm

- Too many packets present in (a part of) the network causes packet delay and loss that degrades performance. This situation is called **congestion**. The network and transport layers share the responsibility for handling congestion
- There are two congestion control algorithms:

1. Leaky Bucket
2. Token Bucket Algorithm

Leaky Bucket

- The leaky bucket algorithm discovers its use in the context of network traffic shaping or rate-limiting. The algorithm allows controlling the rate at which a record is injected into a network and managing burstiness in the data rate.
- In this algorithm, a bucket with a volume of ‘b’ bytes and a hole in the bottom is considered. If the bucket is null, it means b bytes are available as storage. A packet with a size smaller than b bytes arrives at the bucket and will forward it. If the packet's size increases by more than b bytes, it will either be discarded or queued. It is also considered that the bucket leaks through the hole in its bottom at a constant rate of r bytes per second.



- The outflow is considered constant when there is any packet in the bucket and zero when it is empty. This defines that if data flows into the bucket faster than data flows out through the hole, the bucket overflows.
- The disadvantages compared with the leaky-bucket algorithm are the inefficient use of available network resources. The leak rate is a fixed parameter. In the case of the traffic, volume is deficient, the large area of network resources such as bandwidth is not being used effectively.

The leaky-bucket algorithm does not allow individual flows to burst up to port speed to effectively consume network resources when there would not be resource contention in the network.

Token Bucket Algorithm

- The leaky bucket algorithm has a rigid output design at the average rate independent of the bursty traffic. In some applications, when large bursts arrive, the output is allowed to speed up. This calls for a more flexible algorithm, preferably one that never loses information. Therefore, a token bucket algorithm finds its uses in network traffic shaping or rate-limiting.
- It is a control algorithm that indicates when traffic should be sent. This order comes based on the display of tokens in the bucket. The bucket contains tokens. Each of the tokens defines a packet of predetermined size. Tokens in the bucket are deleted for the ability to share a packet.
- When tokens are shown, a flow to transmit traffic appears in the display of tokens. No token means no flow sends its packets. Hence, a flow transfers traffic up to its peak burst rate in good tokens in the bucket.
- Thus, the token bucket algorithm adds a token to the bucket each 1 / r seconds. The volume of the bucket is ‘b’ tokens. When a token appears, and the bucket is complete, the token is discarded. If a packet of n bytes appears and n tokens are deleted from the bucket, the packet is forwarded to the network.
- When a packet of n bytes appears but fewer than n tokens are available. No tokens are removed from the bucket in such a case, and the packet is considered non-conformant. The non-conformant packets can either be dropped or queued for subsequent transmission when sufficient tokens have accumulated in the bucket.
- They can also be transmitted but marked as being non-conformant. The possibility is that they may be dropped subsequently if the network is overloaded.

The Network Layer In The Internet

- There are some principles that drove its design in the past and made it the success that it is today. They are:

1. **Make sure it works.** Do not finalize the design or standard until multiple prototypes have successfully communicated with each other.
2. **Keep it simple.** When in doubt, use the simplest solution.
3. **Make clear choices.** If there are several ways of doing the same thing, choose only one.
4. **Exploit modularity.** This principle leads directly to the idea of having protocol stacks, each of whose layers is independent of all the other ones.
5. **Expect heterogeneity.** Different types of hardware, transmission facilities, and applications will occur on any large network. To handle them, the network design must be simple, general, and flexible.
6. **Avoid static options and parameters.**
7. **Look for a good design; it need not be perfect.**
8. **Be strict when sending and tolerant when receiving.**
9. **Think about scalability.**
10. **Consider performance and cost.**

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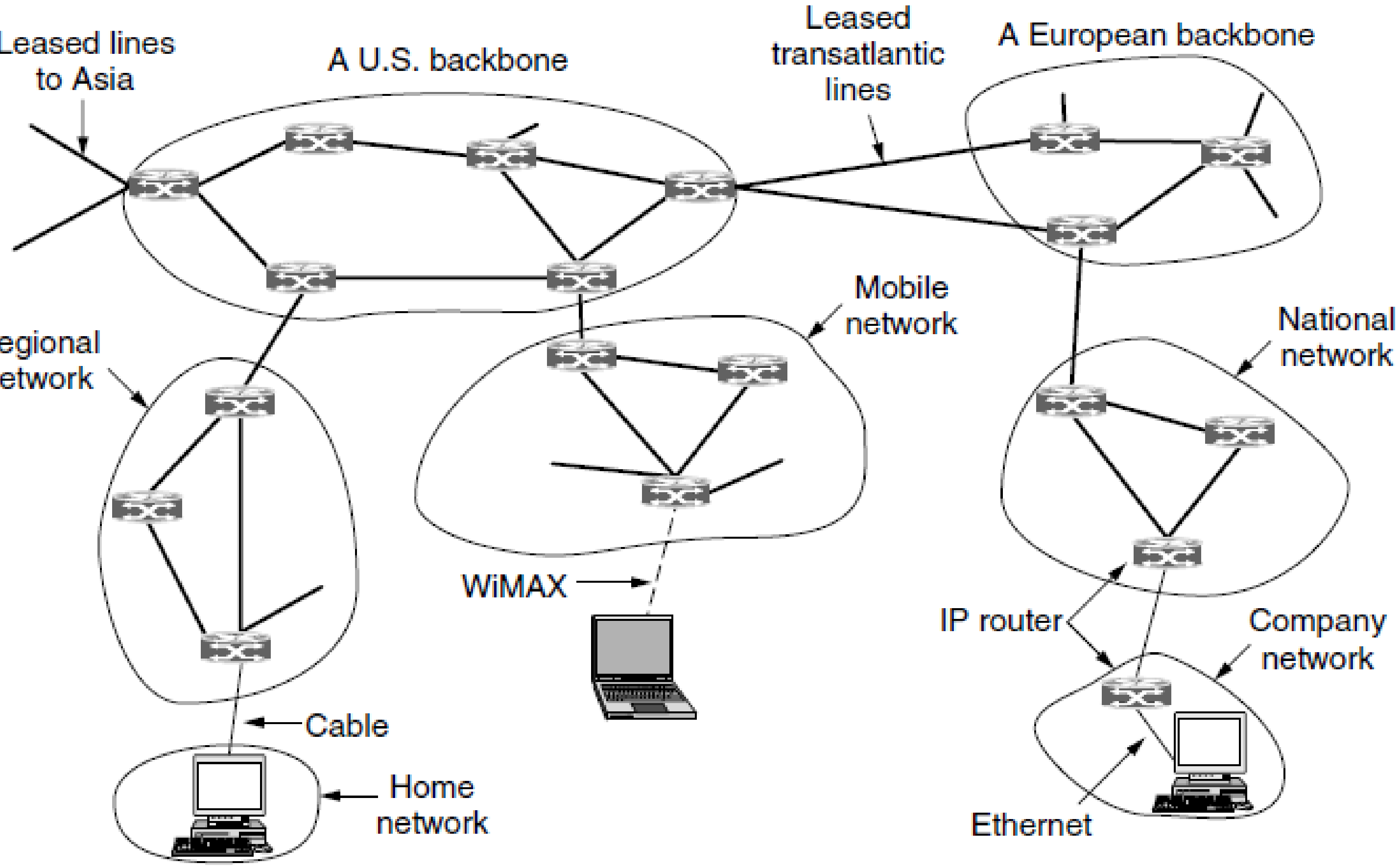

The Internet is an interconnected collection of many networks.

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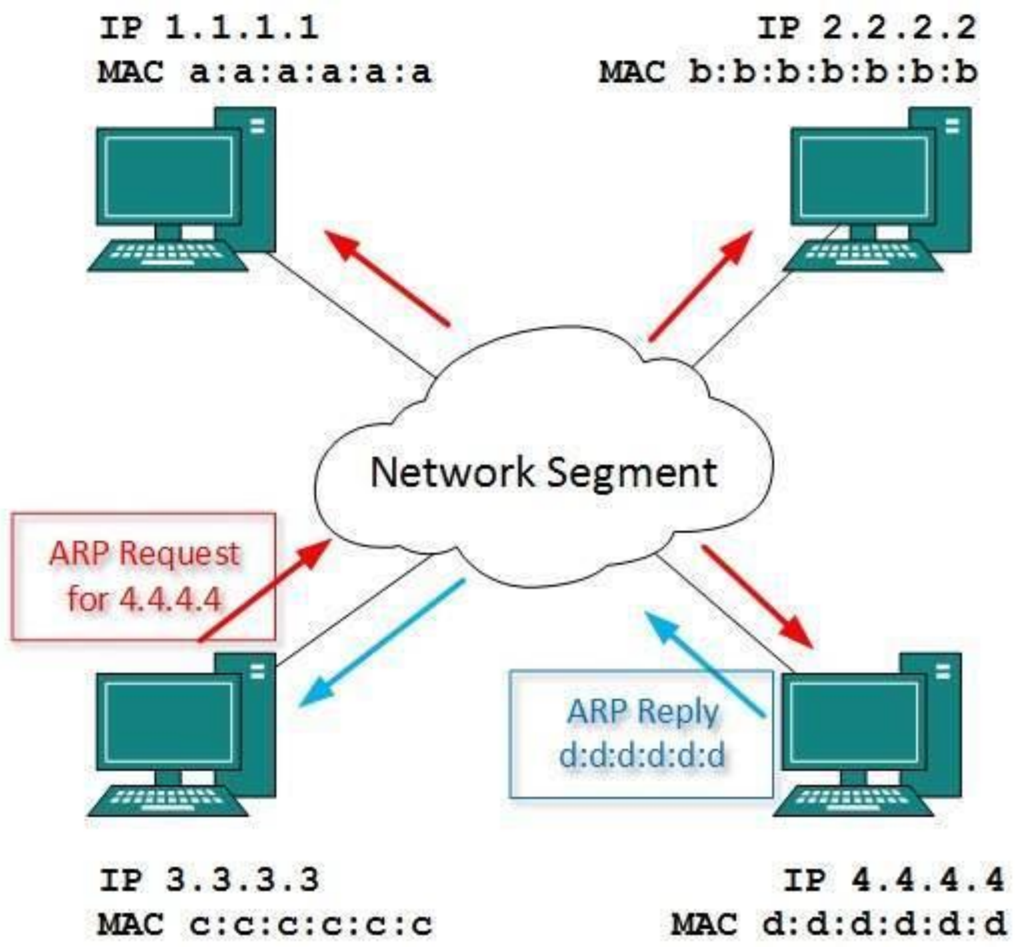
Every computer in a network has an IP address by which it can be uniquely identified and addressed. An IP address is Layer-3 (Network Layer) logical address. This address may change every time a computer restarts. A computer can have one IP at one instance of time and another IP at some different time.

Address Resolution Protocol(ARP)

- While communicating, a host needs Layer-2 (MAC) address of the destination machine which belongs to the same broadcast domain or network. A MAC address is physically burnt into the Network Interface Card (NIC) of a machine and it never changes.
- On the other hand, IP address on the public domain is rarely changed. If the NIC is changed in case of some fault, the MAC address also changes. This way, for Layer-2 communication to take place, a mapping between the two is required.
- To know the MAC address of remote host on a broadcast domain, a computer wishing to initiate communication sends out an ARP broadcast message asking, “Who has this IP address?” Because it is a broadcast, all hosts on the network segment (broadcast domain) receive this packet and process it. ARP packet contains the IP address of destination host, the sending host wishes to talk to. When a host receives an ARP packet destined to it, it replies back with its own MAC address.



Once the host gets destination MAC address, it can communicate with remote host using Layer-2 link protocol. This MAC to IP mapping is saved into ARP cache of both sending and receiving hosts. Next time, if they require to communicate, they can directly refer to their respective ARP cache.

Reverse ARP is a mechanism where host knows the MAC address of remote host but requires to know IP address to communicate.

Internet Control Message Protocol (ICMP)

- ICMP is network diagnostic and error reporting protocol. ICMP belongs to IP protocol suite and uses IP as carrier protocol. After constructing ICMP packet, it is encapsulated in IP packet. Because IP itself is a best-effort non-reliable protocol, so is ICMP.
- Any feedback about network is sent back to the originating host. If some error in the network occurs, it is reported by means of ICMP. ICMP contains dozens of diagnostic and error reporting messages.
- ICMP-echo and ICMP-echo-reply are the most commonly used ICMP messages to check the reachability of end-to-end hosts. When a host receives an ICMP-echo request, it is bound to send back an ICMP-echo-reply. If there is any problem in the transit network, the ICMP will report that problem.

Internet Protocol Version 4 (IPv4)

- IPv4 is 32-bit addressing scheme used as TCP/IP host addressing mechanism. IP addressing enables every host on the TCP/IP network to be uniquely identifiable.
- IPv4 provides hierarchical addressing scheme which enables it to divide the network into sub-networks, each with well-defined number of hosts. IP addresses are divided into many categories:
- **Class A** - it uses first octet for network addresses and last three octets for host addressing
- **Class B** - it uses first two octets for network addresses and last two for host addressing
- **Class C** - it uses first three octets for network addresses and last one for host addressing
- **Class D** - it provides flat IP addressing scheme in contrast to hierarchical structure for above three.
- **Class E** - It is used as experimental.
- IPv4 also has well-defined address spaces to be used as private addresses (not routable on internet), and public addresses (provided by ISPs and are routable on internet).
- Though IP is not reliable one; it provides 'Best-Effort-Delivery' mechanism.

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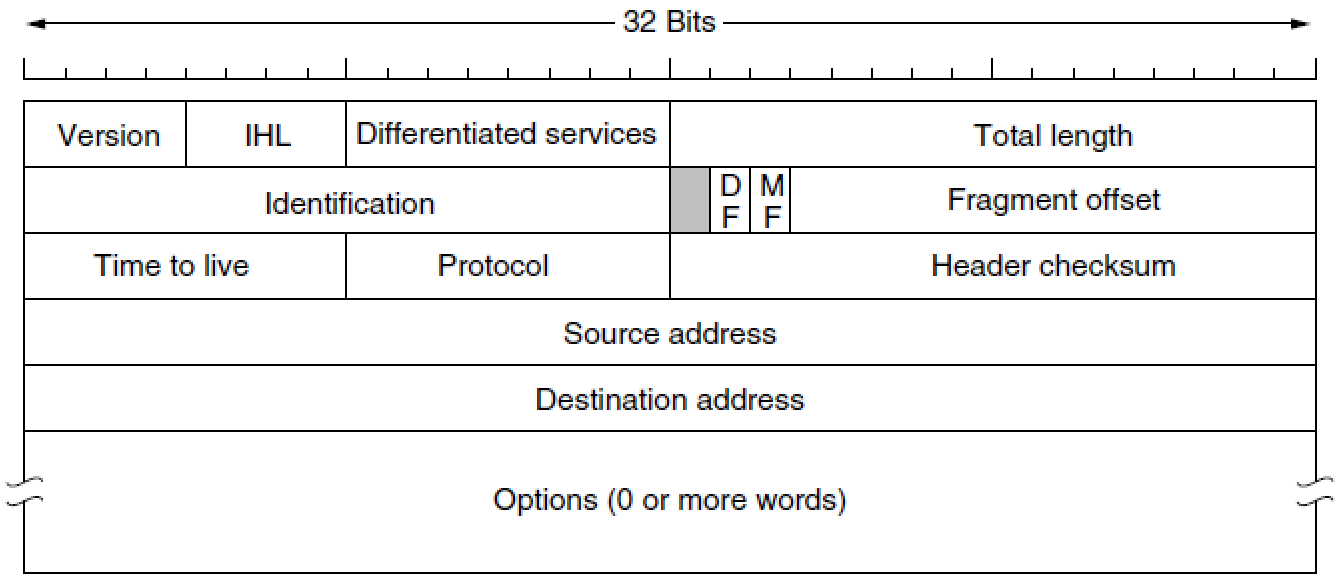
The IPv4 (Internet Protocol) header.



- The *Version* field keeps track of which version of the protocol the datagram belongs to.
- *IHL*, (Internet Header Length)is provided to tell how long the header is, in 32-bit words.
- The *Differentiated services* field specifies *Type of service*
- The *Total length* includes everything in the datagram—both header and data. The maximum length is 65,535 bytes.
- The *Identification* field is needed to allow the destination host to determine which packet a newly arrived fragment belongs to.
- Then come two 1-bit fields related to fragmentation. *DF* stands for Don’t Fragment. *MF* stands for More Fragments.

- The *Fragment offset* tells where in the current packet this fragment belongs.

- The *TtL (Time to live)* field is a counter used to limit packet lifetimes.
- The *Protocol* field tells it which transport process to give the packet to.
- *Header checksum:* Since the header carries vital information such as addresses, it rates its own checksum for protection.
- The *Source address* and *Destination address* indicate the IP address of the source and destination network interfaces.
- The *Options* field was designed to provide an escape to allow subsequent versions of the protocol to include information not present in the original design.

Internet Protocol Version 6 (IPv6)

- Exhaustion of IPv4 addresses gave birth to a next generation Internet Protocol version 6. IPv6 addresses its nodes with 128-bit wide address providing plenty of address space for future to be used on entire planet or beyond.
- IPv6 has introduced Anycast addressing but has removed the concept of broadcasting. IPv6 enables devices to self-acquire an IPv6 address and communicate within that subnet. This auto-configuration removes the dependability of Dynamic Host Configuration Protocol (DHCP) servers. This way, even if the DHCP server on that subnet is down, the hosts can communicate with each other.
  - Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to dynamically assign an IP address to any device, or node, on a network so they can communicate using IP (Internet Protocol). There is no need to manually assign IP addresses to new devices.
- IPv6 provides new feature of IPv6 mobility. Mobile IPv6 equipped machines can roam around without the need of changing their IP addresses.
- IPv6 is still in transition phase and is expected to replace IPv4 completely in coming years.

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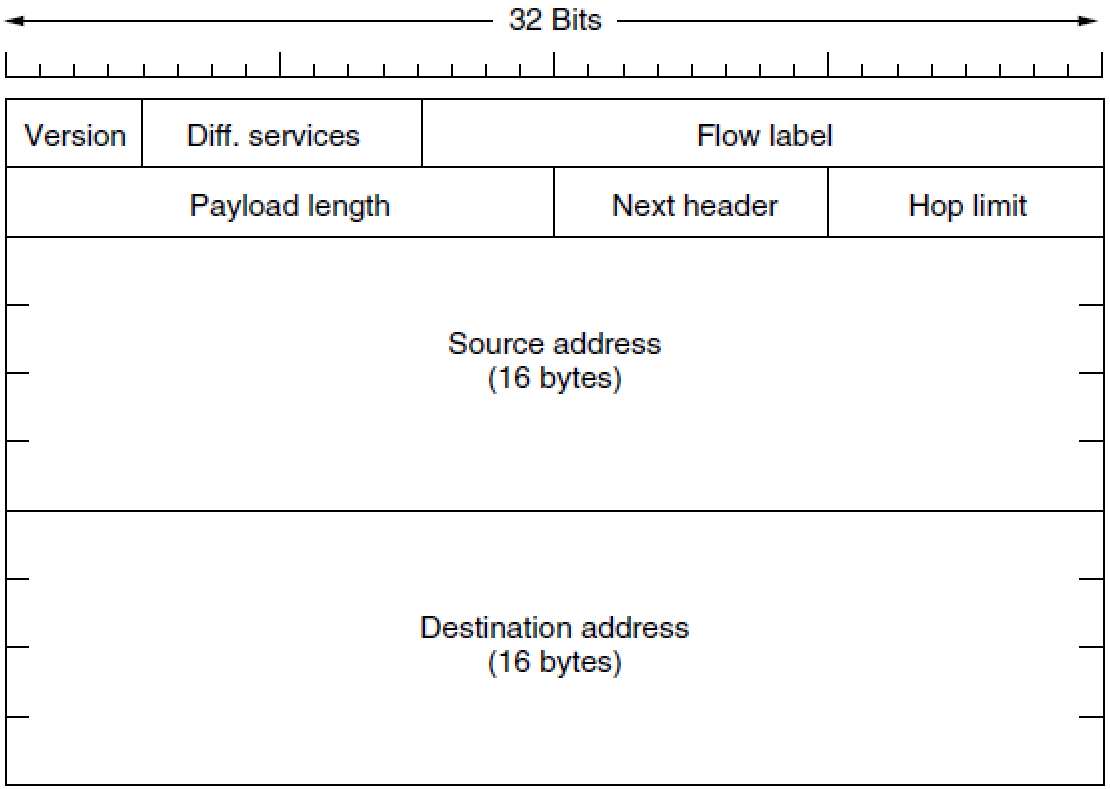
The IPv6 fixed header (required).



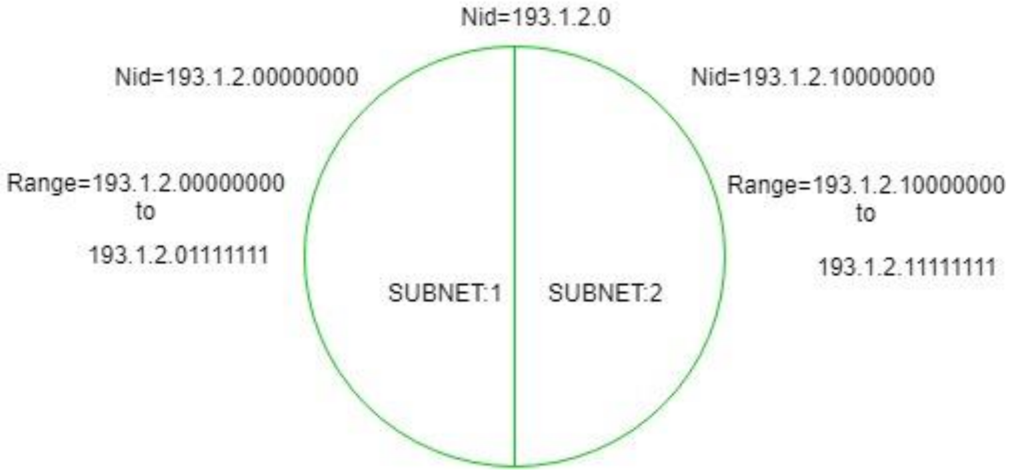
- The *Flow label* field provides a way for a source and destination to mark groups of packets that have the same requirements and should be treated in the same way by the network
- The *Payload length* field tells how many bytes follow the byte header
- The *Next header* field tells which transport protocol handler (e.g., TCP, UDP) to pass the packet to.
- The *Hop limit* field is same as the *Time to live* field in IPv4

Subnets

- When a bigger network is divided into smaller networks, to maintain security, then that is known as Subnetting. So, maintenance is easier for smaller networks. For example, if we consider a class A address, the possible number of hosts is 2^{24} for each network, it is obvious that it is difficult to maintain such a huge number of hosts, but it would be quite easier to maintain if we divide the network into small parts.

Now, let's divide a network into two parts:

To divide a network into two parts, you need to choose one bit for each Subnet from the host ID part.



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- In the figure, there are two Subnets.
  - **Note:** It is a class C IP so, there are 24 bits in the network id part and 8 bits in the host id part.
- *Subnetting for a network should be done in such a way that it does not affect the network bits.* In class C the first 3 octets are network bits so it remains as it is.

For Subnet-1: The first bit which is chosen from the host id part is zero and the range will be from (193.1.2.00000000 till you get all 1's in the host ID part i.e, 193.1.2.01111111) except for the first bit which is chosen zero for subnet id part.

Thus, the range of subnet-1: ***193.1.2.0 to 193.1.2.127***

- Subnet id of Subnet-1 is : 193.1.2.0
- Direct Broadcast id of Subnet-1 is : 193.1.2.127
- Total number of host possible is : 126 (Out of 128, 2 id's are used for Subnet id & Direct Broadcast id)
- Subnet mask of Subnet- 1 is : 255.255.255.128

For Subnet-2: The first bit chosen from the host id part is one and the range will be from (193.1.2.100000000 till you get all 1's in the host ID part i.e, 193.1.2.11111111).

Thus, the range of subnet-2: ***193.1.2.128 to 193.1.2.255***

- Subnet id of Subnet-2 is : 193.1.2.128
- Direct Broadcast id of Subnet-2 is : 193.1.2.255
- Total number of host possible is : 126 (Out of 128, 2 id's are used for Subnet id & Direct Broadcast id)
- Subnet mask of Subnet- 2 is : 255.255.255.192

Finally, after using the subnetting the total number of usable hosts are reduced from 254 to 252.

Note:

- To divide a network into four (2^2) parts you need to choose two bits from the host id part for each subnet i.e, (00, 01, 10, 11).
- To divide a network into eight (2^3) parts you need to choose three bits from the host id part for each subnet i.e, (000, 001, 010, 011, 100, 101, 110, 111) and so on.
- We can say that if the total number of subnets in a network increases the total number of usable hosts decreases.

Along with the advantage there is a small disadvantage for subnetting that is, before subnetting to find the IP address first network id is found then host id followed by process id, but after subnetting first network id is found then subnet id then host id and finally process id by this the computation increases.

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Subnet mask

- The second item, which is required for TCP/IP to work, is the subnet mask. The subnet mask is used by the TCP/IP protocol to determine whether a host is on the local subnet or on a remote network.
- In TCP/IP, the parts of the IP address that are used as the network and host addresses aren't fixed. Unless you have more information, the network and host addresses above can't be determined. This information is supplied in another 32-bit number called a subnet mask. The subnet mask is 255.255.255.0 in this example. It isn't obvious what this number means unless you know 255 in binary notation equals 11111111. So, the subnet mask is 11111111.11111111.11111111.00000000.
- Lining up the IP address and the subnet mask together, the network, and host portions of the address can be separated:
  - 11000000.10101000.01111011.10000100 - IP address (192.168.123.132)
  - 11111111.11111111.11111111.00000000 - Subnet mask (255.255.255.0)
  - The first 24 bits (the number of ones in the subnet mask) are identified as the network address. The last 8 bits (the number of remaining zeros in the subnet mask) are identified as the host address. It gives you the following addresses:
  - 11000000.10101000.01111011.00000000 - Network address (192.168.123.0)
  - 00000000.00000000.00000000.10000100 - Host address (000.000.000.132)
- So now you know, for this example using a 255.255.255.0 subnet mask, that the network ID is 192.168.123.0, and the host address is 0.0.0.132. When a packet arrives on the 192.168.123.0 subnet (from the local subnet or a remote network), and it has a destination address of 192.168.123.132, your computer will receive it from the network and process it.

- Almost all decimal subnet masks convert to binary numbers that are all ones on the left and all zeros on the right. Some other common subnet masks are:

Decimal	Binary
255.255.255.192	1111111.11111111.1111111.11000000
255.255.255.224	1111111.11111111.1111111.11100000

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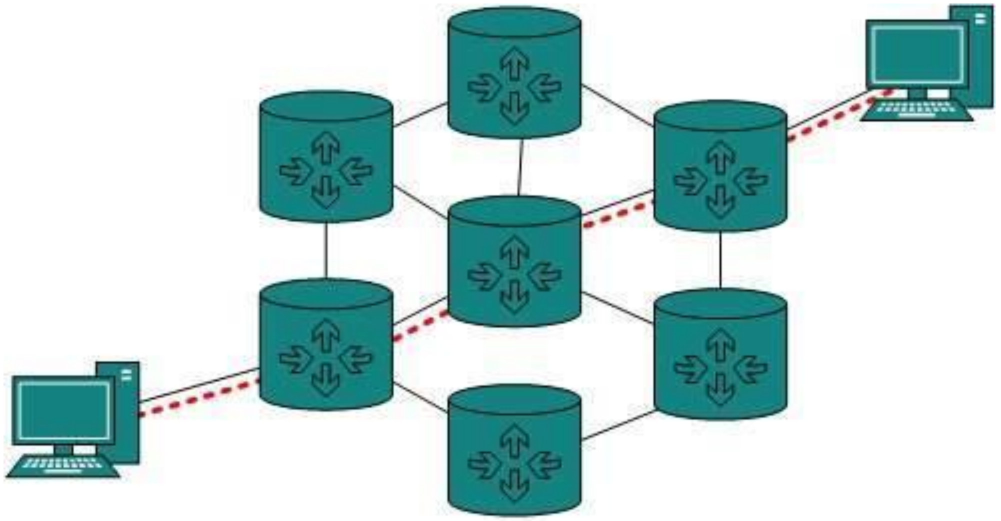
Internetworking

- Routing between two networks is called internetworking or more simply an **internet**.
- Networks can be considered different based on various parameters such as, Protocol, topology, Layer-2 network and addressing scheme.
- In internetworking, routers have knowledge of each other's address and addresses beyond them. They can be statically configured on different network or they can learn by using internetworking routing protocol.
- Routing protocols which are used within an organization or administration are called Interior Gateway Protocols or IGP.
- Routing between different organizations or administrations may have Exterior Gateway Protocol, and there is only one EGP i.e. Border Gateway Protocol.



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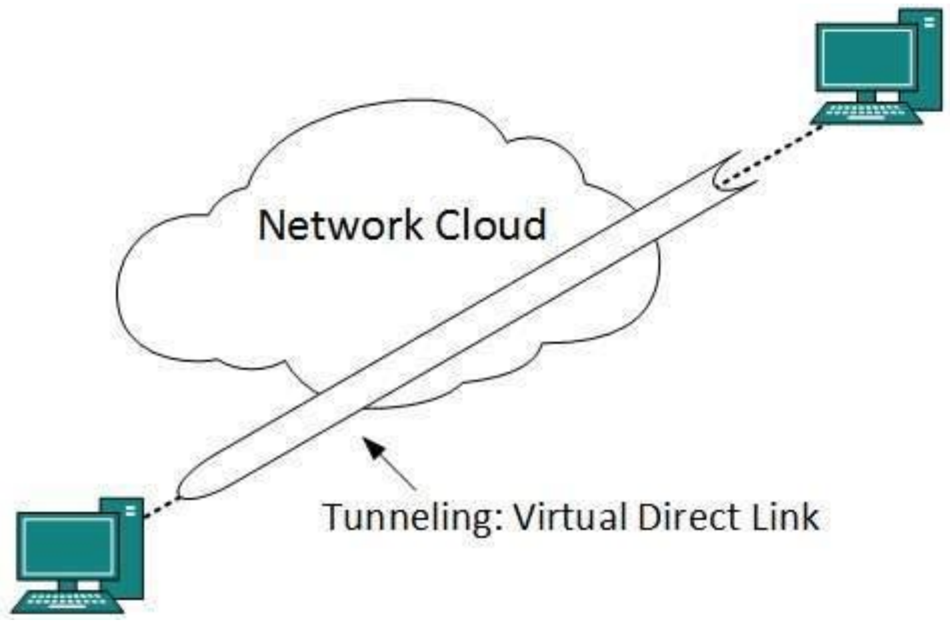
Tunneling

- If they are two geographically separate networks, which want to communicate with each other, they may deploy a dedicated line between or they have to pass their data through intermediate networks.
- Tunneling is a mechanism by which two or more same networks communicate with each other, by passing intermediate networking complexities. Tunneling is configured at both ends.



- When the data enters from one end of Tunnel, it is tagged. This tagged data is then routed inside the intermediate or transit network to reach the other end of Tunnel. When data exists the Tunnel its tag is removed and delivered to the other part of the network.
- Both ends seem as if they are directly connected and tagging makes data travel through transit network without any modifications.

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Packet Fragmentation

- Most Ethernet segments have their maximum transmission unit (MTU) fixed to 1500 bytes. A data packet can have more or less packet length depending upon the application. Devices in the transit path also have their hardware and software capabilities which tell what amount of data that device can handle and what size of packet it can process.
- If the data packet size is less than or equal to the size of packet the transit network can handle, it is processed neutrally. If the packet is larger, it is broken into smaller pieces and then forwarded. This is called packet fragmentation. Each fragment contains the same destination and source address and routed through transit path easily. At the receiving end it is assembled again.
- If a packet with DF (don't fragment) bit set to 1 comes to a router which can not handle the packet because of its length, the packet is dropped.
- When a packet is received by a router has its MF (more fragments) bit set to 1, the router then knows that it is a fragmented packet and parts of the original packet is on the way.
- If packet is fragmented too small, the overhead is increases. If the packet is fragmented too large, intermediate router may not be able to process it and it might get dropped.

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That's all about

UNIT 3

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Unit 4

Transport Layer

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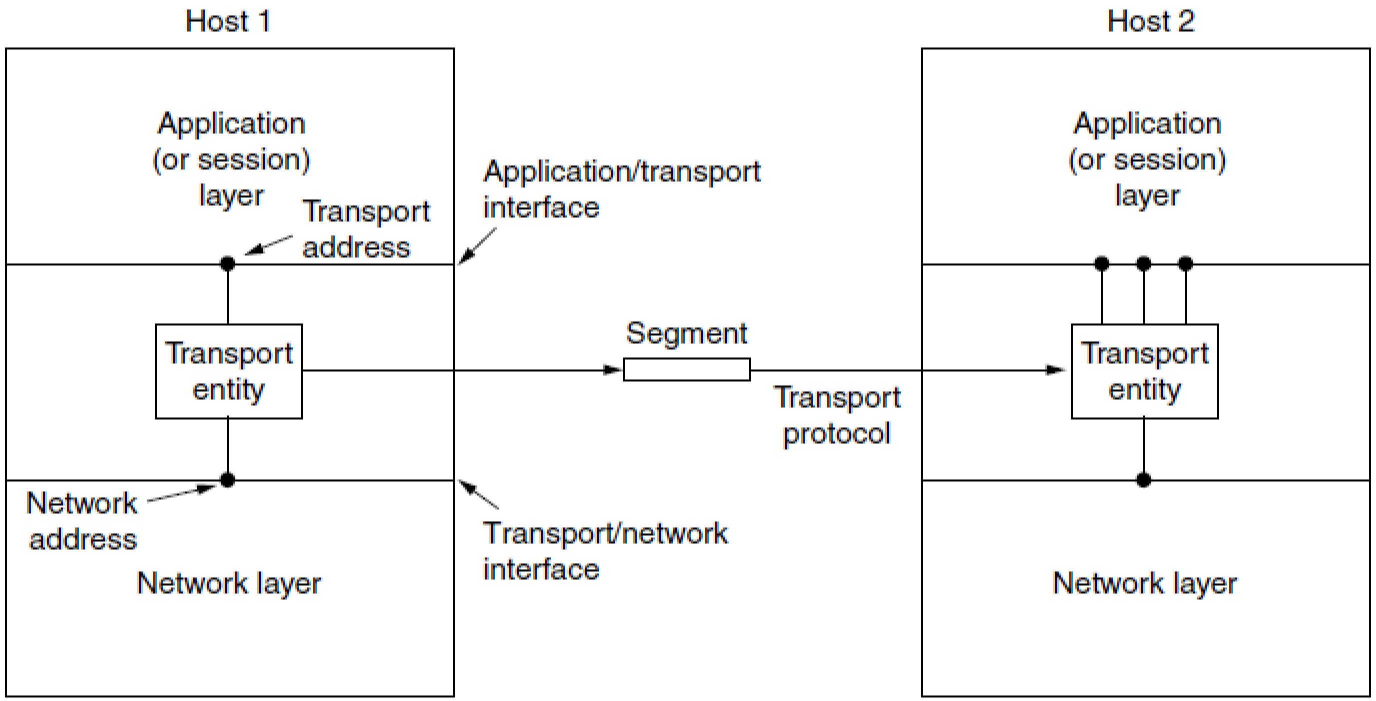
- The transport layer builds on the network layer to provide data transport from a process on a source machine to a process on a destination machine with a desired level of reliability that is independent of the physical networks currently in use where as The network layer provides end-to-end packet delivery using datagrams or virtual circuits.
- It provides the abstractions that applications need to use the network. Without the transport layer, the whole concept of layered protocols would make little sense.
- Transport layer implementations are contained in both the TCP/IP model (RFC 1122), which is the foundation of the Internet, and the Open Systems Interconnection (OSI) model of general networking
- In the Open Systems Interconnection model the transport layer is most often referred to as **Layer 4** or **L4**.
- The best-known transport protocol is the Transmission Control Protocol (TCP). It lent its name to the title of the entire Internet Protocol Suite, TCP/IP, used for connection- oriented transmissions, whereas the connectionless User Datagram Protocol (UDP) is used for simpler messaging transmissions.
- The software and/or hardware within the transport layer that does the work is called the **transport entity**. The transport entity can be located in the operating system kernel, in a library package bound into network applications, in a separate user process, or even on the network interface card.

- The (logical) relationship of the network, transport, and application layers is illustrated in Figure



- Just as there are two types of network service, connection-oriented and connectionless, there are also two types of transport service.
- In both cases, connections have three phases: establishment, data transfer, and release.

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Services

Transport layer services are conveyed to an application via a programming interface to the transport layer protocols. The services may include the following features:

- End-to-end delivery:
  - The transport layer transmits the entire message to the destination. Therefore, it ensures the end-to-end delivery of an entire message from a source to the destination.
- Reliable delivery:
  - The transport layer provides reliability services by retransmitting the lost and damaged packets.
  - **The reliable delivery has four aspects:**
    - ***Error control:*** The primary role of reliability is **Error Control**. Transport layer protocols are designed to provide error-free transmission.
    - ***Sequence control:*** On the sending end, the transport layer is responsible for ensuring that the packets received from the upper layers can be used by the lower layers. On the receiving end, it ensures that the various segments of a transmission can be correctly reassembled.
    - ***Loss control :*** The transport layer ensures that all the fragments of a transmission arrive at the destination, not some of them.
    - ***Duplication control :*** The transport layer guarantees that no duplicate data arrive at the destination. Sequencing allows the receiver to identify and discard duplicate segments.

- Flow Control:
  - Flow control is used to prevent the sender from overwhelming the receiver. If the receiver is overloaded with too much data, then the receiver discards the packets and asking for the retransmission of packets.
- Multiplexing:
  - The transport layer uses the multiplexing to improve transmission efficiency.
  - **Multiplexing can occur in two ways:**
    - ***Upward multiplexing:*** Upward multiplexing means multiple transport layer connections use the same network connection. To make more cost-effective, the transport layer sends several transmissions bound for the same destination along the same path; this is achieved through upward multiplexing.
    - ***Downward multiplexing:*** Downward multiplexing means one transport layer connection uses the multiple network connections. Downward multiplexing allows the transport layer to split a connection among several paths to improve the throughput. This type of multiplexing is used when networks have a low or slow capacity.

Elements of Transport Protocols

- Addressing
- Connection Establishment/Release
- Error Control
- Flow Control
- Multiplexing/De multiplexing
- Fragmentation and re-assembly
- Crash Recovery

Addressing

- Transport Layer deals with addressing or labelling a frame. It also differentiates between a connection and a transaction. Connection identifiers are ports or sockets that label each frame, so the receiving device knows which process it has been sent from. This helps in keeping track of multiple-message conversations. Ports acts as end points in the Internet.
- We will use the generic term **TSAP** (**Transport Service Access Point**) to mean a specific endpoint in the transport layer. The analogous endpoints in the network layer (i.e., network layer addresses) are not-surprisingly called **NSAPs** (**Network Service Access Points**). IP addresses are examples of NSAPs.
- The purpose of having TSAPs is that in some networks, each computer has a single NSAP, so some way is needed to distinguish multiple transport endpoints that share that NSAP.

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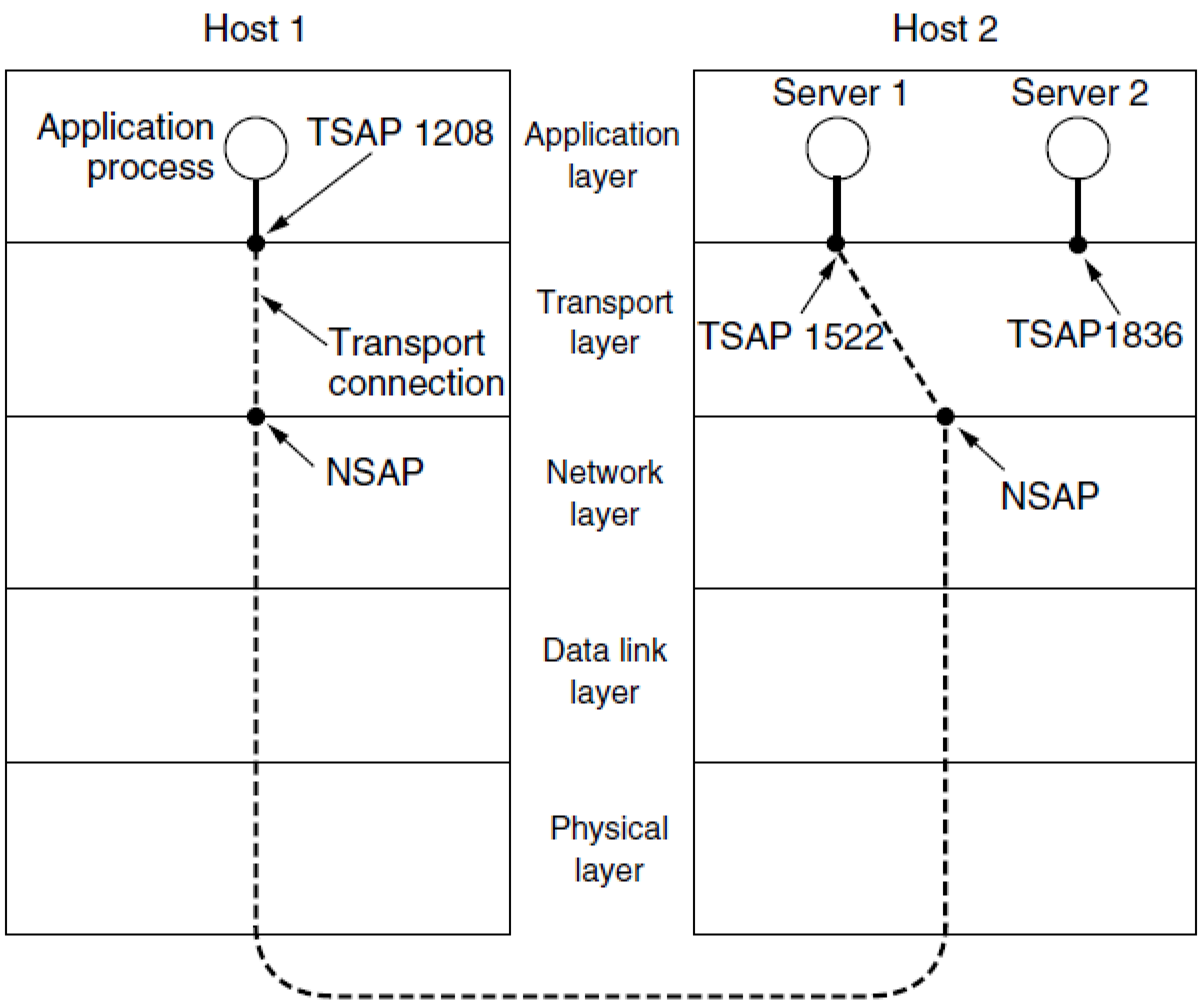
- The following Figure illustrates the relationship between the NSAPs, the TSAPs, and a transport connection.



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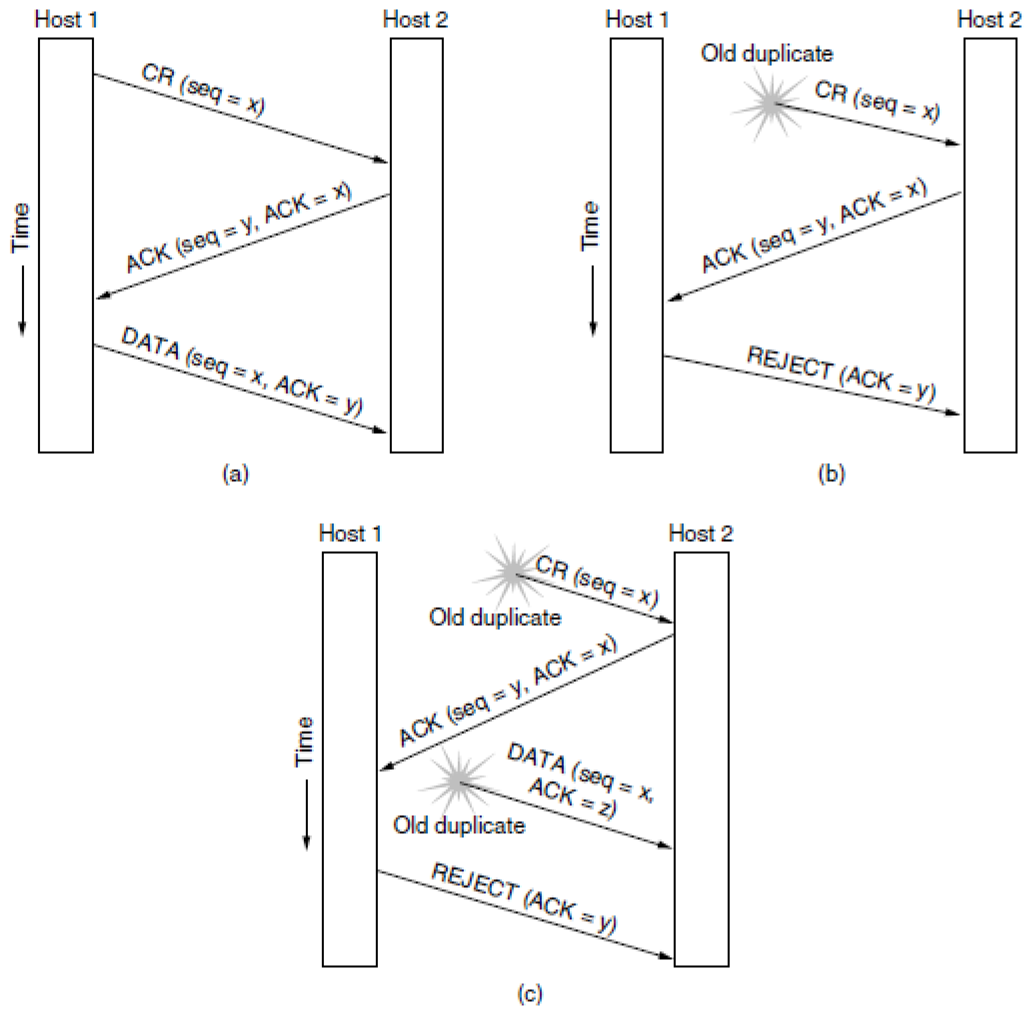
Connection Establishment/Release

- The transport layer creates and releases the connection across the network. This includes a naming mechanism so that a process on one machine can indicate with whom it wishes to communicate. The transport layer enables us to establish and delete connections across the network to multiplex several message streams onto one communication channel.
- At first glance, it would seem sufficient for one transport entity to just send a ***CONNECTION REQUEST*** segment to the destination and wait for a ***CONNECTIONACCEPTED*** reply. The problem occurs when the network can lose, delay, corrupt,and duplicate packets. This behavior causes serious complications.



Three protocol scenarios for establishing a connection using a
three-way handshake. CR denotes CONNECTION REQUEST.
(a) Normal operation.
(b) Old duplicate CONNECTION REQUEST appearing out of nowhere.
(c) Duplicate CONNECTION REQUEST and duplicate ACK.

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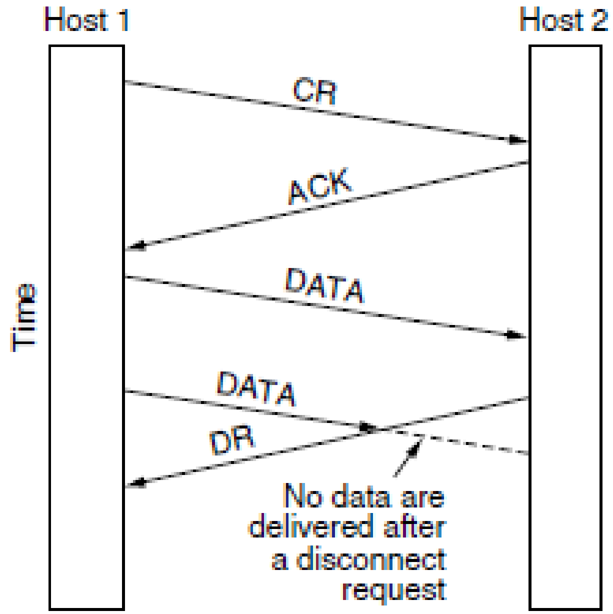
- Releasing a connection is easier than establishing one. Nevertheless, there are more pitfalls than one might expect here. As we mentioned earlier, there are two styles of terminating a connection: asymmetric release and symmetric release.
- Asymmetric release is the way the telephone system works: when one party hangs up, the connection is broken. Symmetric release treats the connection as two separate unidirectional connections and requires each one to be released separately.
  - Asymmetric release is abrupt and may result in data loss. Consider the scenario of Figure. After the connection is established, host 1 sends a segment that arrives properly at host 2. Then host 1 sends another segment. Unfortunately, host 2 issues a DISCONNECT before the second segment arrives. The result is that the connection is released and data are lost.
  - Symmetric release does the job when each process has a fixed amount of data to send and clearly knows when it has sent it.



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Error Control

- Error detection and error recovery are an integral part of reliable service, and therefore they are necessary to perform error control mechanisms on an end-to-end basis. To control errors from lost or duplicate segments, the transport layer enables unique segment sequence numbers to the different packets of the message, creating virtual circuits, allowing only one virtual circuit per session.

Flow Control

- The underlying rule of flow control is to maintain a synergy between a fast process and a slow process. The transport layer enables a fast process to keep pace with a slow one. Acknowledgements are sent back to manage end-to-end flow control. Go back N algorithms are used to request retransmission of packets starting with packet number N. Selective Repeat is used to request specific packets to be retransmitted.

Multiplexing/De multiplexing

- The transport layer establishes a separate network connection for each transport connection required by the session layer. To improve throughput, the transport layer establishes multiple network connections. When the issue of throughput is not important, it multiplexes several transport connections onto the same network connection, thus reducing the cost of establishing and maintaining the network connections.
- When several connections are multiplexed, they call for demultiplexing at the receiving end. In the case of the transport layer, the communication takes place only between two processes and not between two machines. Hence, communication at the transport layer is also known as peer-to-peer or process-to-process communication.

Fragmentation and re-assembly

- When the transport layer receives a large message from the session layer, it breaks the message into smaller units depending upon the requirement. This process is called fragmentation. Thereafter, it is passed to the network layer. Conversely, when the transport layer acts as the receiving process, it reorders the pieces of a message before reassembling them into a message.

Crash Recovery

- If hosts and routers are subject to crashes or connections are long-lived (e.g., large software or media downloads), recovery from these crashes becomes an issue. If the transport entity is entirely within the hosts, recovery from network and router crashes is straightforward. The transport entities expect lost segments all the time and know how to cope with them by using retransmissions.
- A more troublesome problem is how to recover from host crashes. In particular, it may be desirable for clients to be able to continue working when servers crash and quickly reboot.

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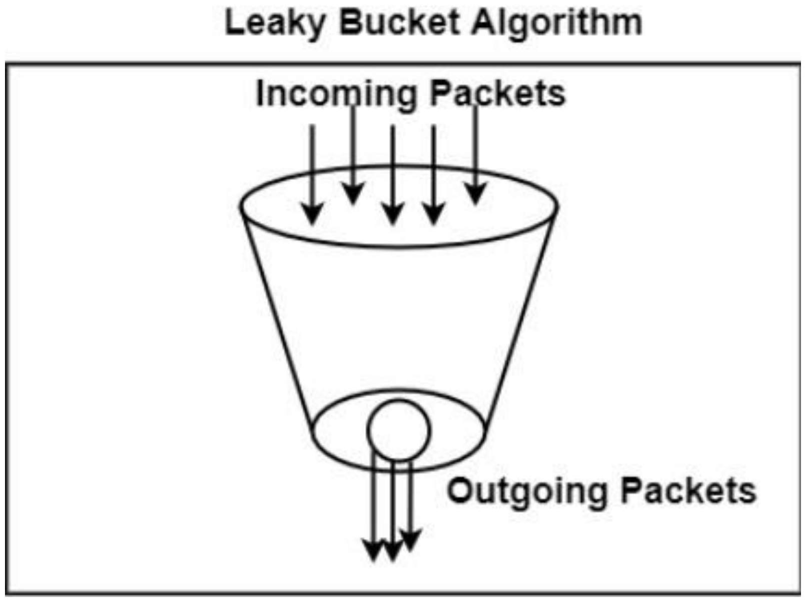
CONGESTION CONTROL

- If the transport entities on many machines send too many packets into the network too quickly, the network will become congested, with performance degraded as packets are delayed and lost. Controlling congestion to avoid this problem is the combined responsibility of the network and transport layers. Congestion occurs at routers, so it is detected at the network layer.
- However, congestion is ultimately caused by traffic sent into the network by the transport layer. The only effective way to control congestion is for the transport protocols to send packets into the network more slowly.
- Congestive-Avoidance Algorithms (CAA) are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network.
- There are two congestion control algorithm which are as follows:

Leaky Bucket

- The leaky bucket algorithm discovers its use in the context of network traffic shaping or rate-limiting. The algorithm allows controlling the rate at which a record is injected into a network and managing burstiness in the data rate.
- In this algorithm, a bucket with a volume of ‘b’ bytes and a hole in the bottom is considered. If the bucket is null, it means b bytes are available as storage. A packet with a size smaller than b bytes arrives at the bucket and will forward it. If the packet's size increases by more than b bytes, it will either be discarded or queued. It is also considered that the bucket leaks through the hole in its bottom at a constant rate of r bytes per second.



- The outflow is considered constant when there is any packet in the bucket and zero when it is empty. This defines that if data flows into the bucket faster than data flows out through the hole, the bucket overflows.
- The disadvantages compared with the leaky-bucket algorithm are the inefficient use of available network resources. The leak rate is a fixed parameter. In the case of the traffic, volume is deficient, the large area of network resources such as bandwidth is not being used effectively.

The leaky-bucket algorithm does not allow individual flows to burst up to port speed to effectively consume network resources when there would not be resource contention in the network.

Token Bucket Algorithm

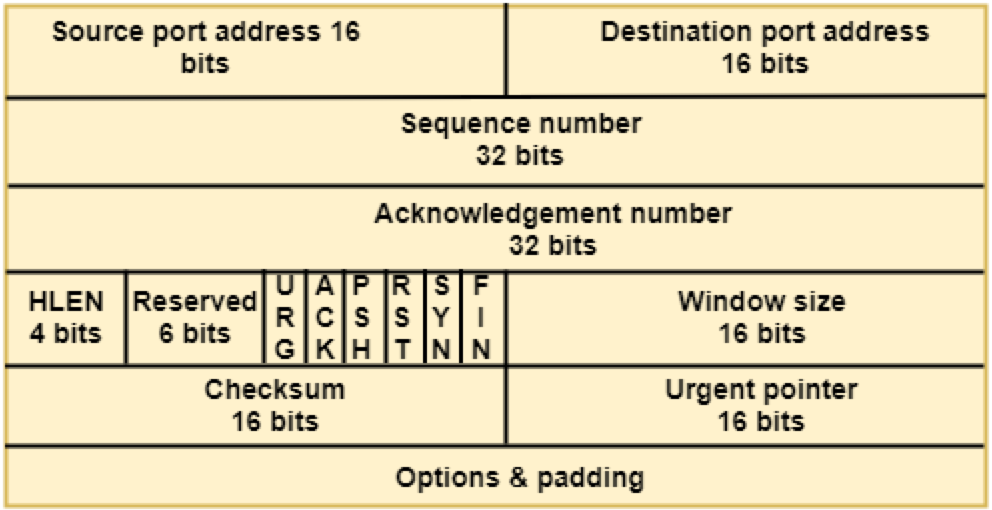
- The leaky bucket algorithm has a rigid output design at the average rate independent of the bursty traffic. In some applications, when large bursts arrive, the output is allowed to speed up. This calls for a more flexible algorithm, preferably one that never loses information. Therefore, a token bucket algorithm finds its uses in network traffic shaping or rate-limiting.
- It is a control algorithm that indicates when traffic should be sent. This order comes based on the display of tokens in the bucket. The bucket contains tokens. Each of the tokens defines a packet of predetermined size. Tokens in the bucket are deleted for the ability to share a packet.
- When tokens are shown, a flow to transmit traffic appears in the display of tokens. No token means no flow sends its packets. Hence, a flow transfers traffic up to its peak burst rate in good tokens in the bucket.
- Thus, the token bucket algorithm adds a token to the bucket each 1 / r seconds. The volume of the bucket is ‘b’ tokens. When a token appears, and the bucket is complete, the token is discarded. If a packet of n bytes appears and n tokens are deleted from the bucket, the packet is forwarded to the network.
- When a packet of n bytes appears but fewer than n tokens are available. No tokens are removed from the bucket in such a case, and the packet is considered non-conformant. The non-conformant packets can either be dropped or queued for subsequent transmission when sufficient tokens have accumulated in the bucket.
- They can also be transmitted but marked as being non-conformant. The possibility is that they may be dropped subsequently if the network is overloaded.

TCP and UDP Protocols

The transport layer is represented by two protocols: TCP and UDP.

TCP

- TCP stands for Transmission Control Protocol.
- It provides full transport layer services to applications.
- It is a connection-oriented protocol means the connection established between both the ends of the transmission. For creating the connection, TCP generates a virtual circuit between sender and receiver for the duration of a transmission.



TCP Segment Format

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- **Source port address:** It is used to define the address of the application program in a source computer. It is a 16-bit field.
- **Destination port address:** It is used to define the address of the application program in a destination computer. It is a 16-bit field.
- **Sequence number:** A stream of data is divided into two or more TCP segments. The 32-bit sequence number field represents the position of the data in an original data stream.
- **Acknowledgement number:** A 32-field acknowledgement number acknowledge the data from other communicating devices. If ACK field is set to 1, then it specifies the sequence number that the receiver is expecting to receive.
- **Header Length (HLEN):** The size of TCP header is 20-bytes (16-bits for source port, 16-bits for the destination port, 32-bits for seq number, 32-bits for ack number, 4-bits header length)
- **Reserved:** It is a six-bit field which is reserved for future use.
- **Control bits:** Each bit of a control field functions individually and independently. A control bit defines the use of a segment or serves as a validity check for other fields.
  - **URG:** The URG field indicates that the data in a segment is urgent.
  - **ACK:** When ACK field is set, then it validates the acknowledgement number.
  - **PSH:** The PSH field is used to inform the sender that higher throughput is needed so if possible, data must be pushed with higher throughput.
  - **RST:** The reset bit is used to reset the TCP connection when there is any confusion occurs in the sequence numbers.
  - **SYN:** The SYN field is used to synchronize the sequence numbers in three types of segments: connection request, connection confirmation (with the ACK bit set), and confirmation acknowledgement.
  - **FIN:** The FIN field is used to inform the receiving TCP module that the sender has finished sending data. It is used in connection termination in three types of segments: termination request, termination confirmation, and acknowledgement of termination confirmation.

- **Window Size:** The window is a 16-bit field that defines the size of the window.
- **Checksum:** The checksum is a 16-bit field used in error detection.
- **Urgent pointer:** If URG flag is set to 1, then this 16-bit field is an offset from the sequence number indicating that it is a last urgent data byte.
- **Options and padding:** It defines the optional fields that convey the additional information to the receiver.

UDP

- UDP stands for **User Datagram Protocol**.
- UDP is a simple protocol and it provides non sequenced transport functionality.
- UDP is a connectionless protocol.
- This type of protocol is used when reliability and security are less important than speed and size.
- UDP is an end-to-end transport level protocol that adds transport-level addresses, checksum error control, and length information to the data from the upper layer.
- The packet produced by the UDP protocol is known as a user datagram.

Source port address 16 bits	Destination port address 16 bits
Total Length 16 bits	Checksum 16 bits
Data	

User Datagram Format

- **Source port address:** It defines the address of the application process that has delivered a message. The source port address is of 16 bits address.
- **Destination port address:** It defines the address of the application process that will receive the message. The destination port address is of a 16-bit address.
- **Total length:** It defines the total length of the user datagram in bytes. It is a 16-bit field.
- **Checksum:** The checksum is a 16-bit field which is used in error detection.

The size of the UDP header is 8-bytes (16-bits for source port, 16-bits for destination port, 16-bits for length, 16-bits for checksum); it’s significantly smaller than the TCP header.

<u>Quality of Service Model</u>

- Quality-of-Service (QoS) refers to traffic control mechanisms that seek to either differentiate performance based on application or network-operator requirements or provide predictable or guaranteed performance to applications, sessions, or traffic aggregates. Basic phenomenon for QoS means in terms of packet delay and losses of various kinds.
- The QoS is primarily used to control resources like bandwidth, equipment, wide-area facilities etc. It can get more efficient use of network resources.

QoS requirements can be specified as below:

- ***Congestion Management***: QoS allows a router to put packets into different queues.
- ***Queue Management***: The queues in a buffer can fill and overflow. A packet would be dropped if a queue is complete, and the router cannot prevent it from being dropped
- ***Elimination of overhead bits***: It can also increase efficiency by removing too many overhead bits.
- ***Traffic shaping and policing***: Shaping can prevent the overflow problem in buffers by limiting the full bandwidth potential of the applications packets. Sometimes, many network topologies with a high bandwidth link connected with a low-bandwidth link in remote sites can overflow low bandwidth connections. Therefore, shaping is used to provide the traffic flow from the high bandwidth link closer to the low bandwidth link to avoid the low bandwidth link's overflow.

PERFORMANCE ISSUES

- Performance issues are very important in computer networks. When hundreds or thousands of computers are interconnected, complex interactions, with unforeseen consequences, are common.

we will look at six aspects of network performance:

1. Performance problems.
2. Measuring network performance.
3. Host design for fast networks.
4. Fast segment processing.
5. Header compression.
6. Protocols for ‘‘long fat’’ networks.

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1. Performance problems

- Some performance problems, such as congestion, are caused by temporary resource overloads. If more traffic suddenly arrives at a router than the router can handle, congestion will build up and performance will suffer.
- Performance also degrades when there is a structural resource imbalance. For example, if a gigabit communication line is attached to a low-end PC, the poor host will not be able to process the incoming packets fast enough and some will be lost. These packets will eventually be retransmitted, adding delay, wasting bandwidth, and generally reducing performance.

2. Measuring network performance

Measurements can be made in different ways and at many locations. Few of them are:

- **Make Sure That the Sample Size Is Large Enough**
- **Be Sure That Nothing Unexpected Is Going On during Your Tests**
- **Be Careful When Using a Coarse-Grained Clock**
- **Be Careful about Extrapolating(like prediction) the Results**

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3. Host design for fast networks

There are some rules of thumb for software implementation of network protocols on hosts.

- **Host Speed Is More Important Than Network Speed**
- **Reduce Packet Count to Reduce Overhead**
- **Minimize Data Touching**
- **Minimize Context Switches** (e.g., from kernel mode to user mode)

4. Fast segment processing

- The key to fast segment processing is to separate out the normal, successful case (one-way data transfer) and handle it specially.
- Many protocols tend to emphasize what to do when something goes wrong (e.g., a packet getting lost), but to make the protocols run fast, the designer should aim to minimize processing time when everything goes right. Minimizing processing time when an error occurs is secondary.

5. Header compression.

- When we consider performance on wireless and other networks in which bandwidth is limited. Reducing software overhead can help mobile computers run more efficiently, but it does nothing to improve performance when the network links are the bottleneck.
- **Header compression** is used to reduce the bandwidth taken over links by higher-layer protocol headers. Specially designed schemes are used instead of general purpose methods. This is because headers are short, so they do not compress well individually, and decompression requires all prior data to be received.

6. Protocols for ‘‘long fat’’ networks.

- Since the 1990s, there have been gigabit networks that transmit data over large distances. Because of the combination of a fast network, or ‘‘fat pipe,’’ and long delay, these networks are called **long fat networks**.
  - The first problem is that many protocols use 32-bit sequence numbers. When the Internet began, the lines between routers were mostly 56-kbps leased lines, so a host blasting away at full speed took over 1 week to cycle through the sequence numbers.
  - A second problem is that the size of the flow control window must be greatly increased.
  - The conclusion that can be drawn here is that for good performance, the receiver's window must be at least as large as the ***bandwidth-delay product***, and preferably somewhat larger since the receiver may not respond instantly.
    - Bandwidth delay product is **a measurement of how many bits can fill up a network link**.

That's all about

UNIT 4

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Unit 5

Application Layer

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- Application layer is the top most layer in OSI and TCP/IP layered model. This layer exists in both layered Models because of its significance, of interacting with user and user applications. This layer is for applications which are involved in communication system.
- The layers below the application layer are there to provide transport services, but they do not do real work for users.
- However, even in the application layer there is a need for support protocols, to allow the applications to function.

Domain Name System

- DNS is a hostname for IP address translation service. DNS is a distributed database implemented in a hierarchy of name servers. It is an application layer protocol for message exchange between clients and servers.

Requirement: Every host is identified by the IP address but remembering numbers is very difficult for the people also the IP addresses are not static therefore a mapping is required to change the domain name to the IP address. So DNS is used to convert the domain name of the websites to their numerical IP address.

There are various kinds of DOMAIN:

- ***Generic domain***: .com(commercial) .edu(educational) .mil(military) .org(non profit organization) .net(similar to commercial) all these are generic domain.
- ***Country domain*** .in (india) .us .uk
- ***Inverse domain*** if we want to know what is the domain name of the website. Ip to domain name mapping. So DNS can provide both the mapping for example to find the ip addresses of goolge then we have to type ping google.com in command prompt.

Select Command Prompt

```
operable program or batch file.

C:\Users\MRUH>ping google.com

Pinging google.com [142.250.196.14] with 32 bytes of data:
Reply from 142.250.196.14: bytes=32 time=25ms TTL=117
Reply from 142.250.196.14: bytes=32 time=64ms TTL=117
Reply from 142.250.196.14: bytes=32 time=29ms TTL=117
Reply from 142.250.196.14: bytes=32 time=28ms TTL=117

Ping statistics for 142.250.196.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 64ms, Average = 36ms

C:\Users\MRUH>ping mallareddyuniversity.ac.in

Pinging mallareddyuniversity.ac.in [185.230.63.107] with 32 bytes of data:
Reply from 185.230.63.107: bytes=32 time=229ms TTL=241
Reply from 185.230.63.107: bytes=32 time=255ms TTL=241
Reply from 185.230.63.107: bytes=32 time=229ms TTL=241
Reply from 185.230.63.107: bytes=32 time=238ms TTL=241

Ping statistics for 185.230.63.107:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 229ms, Maximum = 255ms, Average = 237ms

C:\Users\MRUH>
```

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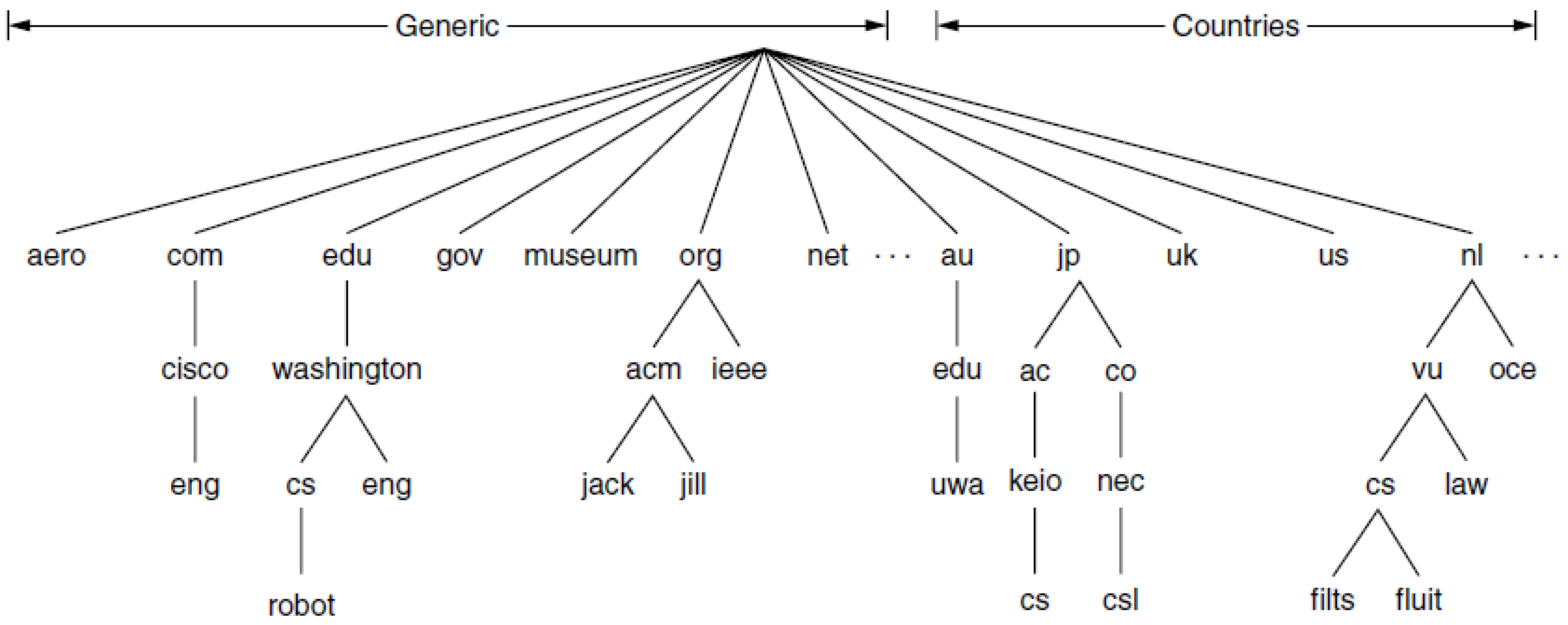
DNS Name Space

- Managing a large and constantly changing set of names is a nontrivial problem.
- In the postal system, name management is done by requiring letters to specify (implicitly or explicitly) the country, state or province, city, street address, and name of the addressee.
- Using this kind of hierarchical addressing ensures that there is no confusion between the ***Marvin Anderson on Main St. in White Plains*,*N.Y.*** and the ***Marvin Anderson on Main St. in Austin, Texas.*** DNS works the same way.
- For the Internet, the top of the naming hierarchy is managed by an organization called **ICANN** (**Internet Corporation for Assigned Names and Numbers**).
- ICANN was created for this purpose in 1998, as part of the maturing of the Internet to a worldwide, economic concern.
- Conceptually, the Internet is divided into over 250 **top-level domains**, where each domain covers many hosts. Each domain is partitioned into subdomains, and these are further partitioned, and so on.
- All these domains can be represented by a tree, as shown in following Figure. The leaves of the tree represent domains that have no subdomains (but do contain machines, of course). A leaf domain may contain a single host, or it may represent a company and contain thousands of hosts.



A portion of the Internet domain name space.

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Domain	Intended use	Start date	Restricted?
com	Commercial	1985	No
edu	Educational institutions	1985	Yes
gov	Government	1985	Yes
int	International organizations	1988	Yes
mil	Military	1985	Yes
net	Network providers	1985	No
org	Non-profit organizations	1985	No
aero	Air transport	2001	Yes
biz	Businesses	2001	No
coop	Cooperatives	2001	Yes
info	Informational	2002	No
museum	Museums	2002	Yes
name	People	2002	No
pro	Professionals	2002	Yes
cat	Catalan	2005	Yes
jobs	Employment	2005	Yes
mobi	Mobile devices	2005	Yes
tel	Contact details	2005	Yes
travel	Travel industry	2005	Yes

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Generic top-level domains

Working of DNS

- DNS is a client/server network communication protocol. DNS clients send requests to the. server while DNS servers send responses to the client.
- Client requests contain a name which is converted into an IP address known as a forward DNS lookups while requests containing an IP address which is converted into a name known as reverse DNS lookups.
- DNS implements a distributed database to store the name of all the hosts available on the internet.
- If a client like a web browser sends a request containing a hostname, then a piece of software such as **DNS resolver** sends a request to the DNS server to obtain the IP address of a hostname. If DNS server does not contain the IP address associated with a hostname, then it forwards the request to another DNS server. If IP address has arrived at the resolver, which in turn completes the request over the internet protocol.

SNMP

- SNMP stands for **Simple Network Management Protocol**. It is an Internet-standard protocol for handling devices on IP networks.
- If an organization has 1000 devices then to check all devices, one by one every day, are working properly or not is a hectic task. To ease these up, Simple Network Management Protocol (SNMP) is used. Devices that typically provide SNMP include routers, switches, servers, workstations, printers, modem racks, and more.

SNMP components –

There are 3 components of SNMP:

- **SNMP Manager –**
  It is a centralized system used to monitor network. It is also known as Network Management Station (NMS)
- **SNMP agent –**
  It is a software management software module installed on a managed device. Managed devices can be network devices like PC, routers, switches, servers, etc.
- **Management Information Base –**
  MIB consists of information on resources that are to be managed. This information is organized hierarchically. It consists of objects instances which are essentially variables.

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SNMP messages –
Different variables are:

- **GetRequest –**
  SNMP manager sends this message to request data from the SNMP agent. In response to this, the SNMP agent responds with the requested value through a response message.
- **GetNextRequest –**
  This message can be sent to discover what data is available on an SNMP agent. The SNMP manager can request data continuously until no more data is left. In this way, the SNMP manager can take knowledge of all the available data on SNMP agents.
- **GetBulkRequest –**
  This message is used to retrieve large data at once by the SNMP manager from the SNMP agent. It is introduced in SNMPv2c.
- **SetRequest –**
  It is used by the SNMP manager to set the value of an object instance on the SNMP agent.
- **Response –**
  It is a message sent from the agent upon a request from the manager. When sent in response to Get messages, it will contain the data requested. When sent in response to the Set message, it will contain the newly set value as confirmation that the value has been set.
- **Trap –**
  These are the message sent by the agent without being requested by the manager. It is sent when a fault has occurred.
- **InformRequest –**
  It was introduced in SNMPv2c, used to identify if the trap message has been received by the manager or not. The agents can be configured to send trap message continuously until it receives an Inform message. It is the same as a trap but adds an acknowledgement that the trap doesn't provide.

ELECTRONIC MAIL

- Electronic mail, or more commonly **email**, has been around for over three decades. Faster and cheaper than paper mail, email has been a popular application
- Since the early days of the Internet. Before 1990, it was mostly used in academia. During the 1990s, it became known to the public at large and grew exponentially

The basic components of an email system are :

1. User Agent (UA) : The UA is normally a program which is used to send and receive mail. Sometimes, it is called as mail reader. It accepts variety of commands for composing, receiving and replying to messages as well as for manipulation of the mailboxes.

2. Message Transfer Agent (MTA) : MTA is actually responsible for transfer of mail from one system to another. To send a mail, a system must have client MTA and system MTA. It transfer mail to mailboxes of recipients if they are connected in the same machine. It delivers mail to peer MTA if destination mailbox is in another machine. The delivery from one MTA to another MTA is done by Simple Mail Transfer Protocol.

3. Mailbox : It is a file on local hard drive to collect mails. Delivered mails are present in this file. The user can read it delete it according to his/her requirement.

4. Spool file : This file contains mails that are to be sent. User agent appends outgoing mails in this file using SMTP. MTA extracts pending mail from spool file for their delivery.

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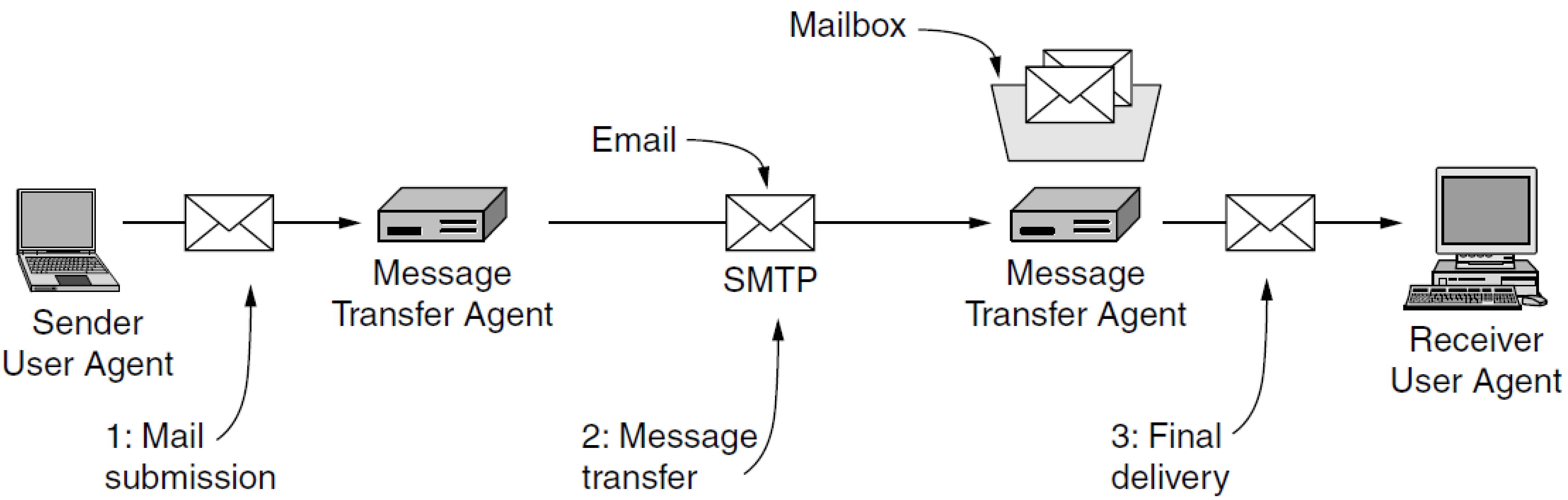

Architecture of the email system.

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Types of Email Protocols:

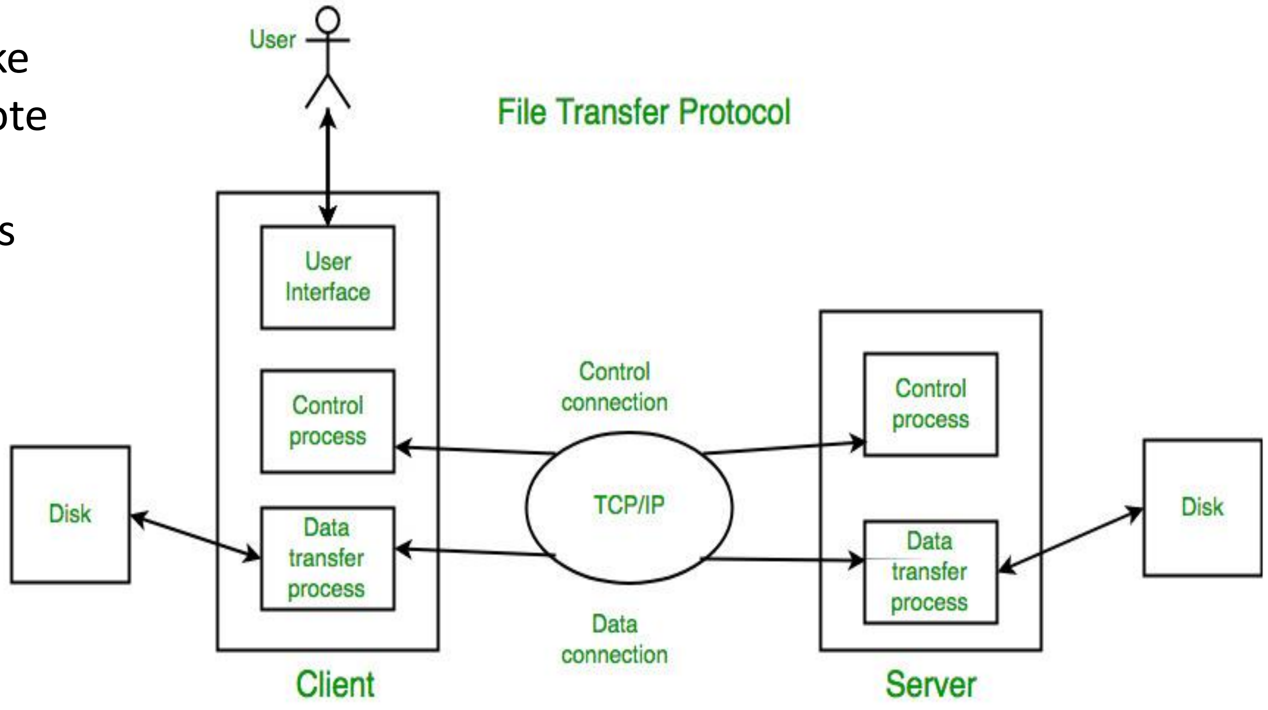
Three basic types of email protocols involved for sending and receiving mails are:

- ***SMTP:*** Simple Mail Transfer Protocol is used to send mails over the internet. SMTP is an application layer and connection-oriented protocol. SMTP is efficient and reliable for sending emails.
- ***POP3***: Post Office Protocol is used to retrieve email for a single client. POP3 version is the current version of POP used. It is an application layer protocol. It allows to access mail offline and thus, needs less internet time. To access the message it has to be downloaded.
- ***IMAP:*** Internet Message Access Protocol is used to retrieve mails for multiple clients. There are several IMAP versions: IMAP, IMAP2, IMAP3, IMAP4, etc. IMAP is an application layer protocol. IMAP allows to access email without downloading them and also supports email download. The emails are maintained by the remote server. It enables all email operations such as creating, manipulating, delete the email without reading it.

File Transfer Protocol

- File Transfer Protocol(FTP) is an application layer protocol that moves files between local and remote file systems. It runs on the top of TCP, like HTTP. To transfer a file, 2 TCP connections are used by FTP in parallel: control connection and data connection.

control connection is used for sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc.,
data connection is used for sending the actual file, FTP makes use of a data connection



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Some of the FTP commands are :

USER – This command sends the user identification to the server.
PASS – This command sends the user password to the server.
CWD – This command allows the user to work with a different directory or dataset for file storage or retrieval without altering his login or accounting information.
RMD – This command causes the directory specified in the path name to be removed as a directory.
MKD – This command causes the directory specified in the pathname to be created as a directory.
PWD – This command causes the name of the current working directory to be returned in the reply.
RETR – This command causes the remote host to initiate a data connection and to send the requested file over the data connection.
STOR – This command causes to store of a file into the current directory of the remote host.
LIST – Sends a request to display the list of all the files present in the directory.
ABOR – This command tells the server to abort the previous FTP service command and any associated transfer of data.
QUIT – This command terminates a USER and if file transfer is not in progress, the server closes the control connection.

Some of the FTP replies are :

200 Command okay.
530 Not logged in.
331 User name okay, need a password.
225 Data connection open; no transfer in progress.
221 Service closing control connection.
551 Requested action aborted: page type unknown.
502 Command not implemented.
503 Bad sequence of commands.
504 Command not implemented for that parameter.

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TFTP

- ***TFTP*** is an abbreviation for ***"Trivial File Transfer Protocol"***. It is a sample protocol that is commonly used for file transmission. TFTP employs the ***User Datagram Protocol (UDP)*** to transport data from one end to the other.
- In comparison to the FTP, it is very simple in design and has limited functionalities (FTP).
- TFTP doesn't provide authentication or security while transferring data. As a result, boot data or configuration files are typically shared between computer systems in a local setup.
- It aids in booting devices and systems that lack storage devices or hard disk drives because utilizing a small amount of memory may be easily installed. It is primarily used for booting systems that store configuration on a remote TFTP server.
- It operates on Port number 69, and its service is given by UDP.

Difference between FTP and TFTP

S.NO	FTP	TFTP
1.	FTP stands for File Transfer Protocol.	TFTP stands for Trivial File Transfer Protocol.
2.	The software of FTP is larger than TFTP.	Software of TFTP is smaller than FTP.
3.	FTP works on two ports: 20 and 21.	TFTP works on 69 Port number.
4.	FTP services are provided by TCP.	TFTP services are provided by UDP.
5.	The complexity of FTP is higher than TFTP.	The complexity of TFTP is less than FTP complexity.
6.	There are many commands or messages in FTP.	There are only 5 messages in TFTP.
7.	FTP need authentication for communication.	TFTP does not need authentication for communication.
8.	FTP is generally suited for uploading and downloading of files by remote users.	TFTP is mainly used for transmission of configurations to and from network devices.
9.	FTP is a reliable transfer protocol.	TFTP is an unreliable transfer protocol.
10.	FTP is based on TCP.	TFTP is based on UDP.
11.	FTP is slower.	TFTP is faster as compared to FTP.
12.	FTP is specified in RFC959 document.	TFTP is specified in RFC783 document.
13.	FTP requires more programming effort.	TFTP requires less programming effort.
14.	FTP requires more memory.	TFTP requires less memory.
15.	FTP uses the robust control commands.	TFTP uses the simple control commands.

BOOTP (Bootstrap Protocol)

- Bootstrap Protocol (BOOTP) is a basic protocol that automatically provides each participant in a network connection with a unique IP address for identification and authentication as soon as it connects to the network. This helps the server to speed up data transfers and connection requests.
- BOOTP uses a unique IP address algorithm to provide each system on the network with a completely different IP address in a fraction of a second.
- This shortens the connection time between the server and the client. It starts the process of downloading and updating the source code even with very little information.
- BOOTP uses a combination of **TFTP** (Trivial File Transfer Protocol) and **UDP** (User Datagram Protocol) to request and receive requests from various network-connected participants and to handle their responses.
- In a BOOTP connection, the server and client just need an IP address and a gateway address to establish a successful connection. Typically, in a BOOTP network, the server and client share the same LAN, and the routers used in the network must support BOOTP bridging.
- A great example of a network with a TCP / IP configuration is the Bootstrap Protocol network. Whenever a computer on the network asks for a specific request to the server, BOOTP uses its unique IP address to quickly resolve them.

Working of Bootstrap Protocol :

- At the very beginning, each network participant does not have an IP address. The network administrator then provides each host on the network with a unique IP address using the IPv4 protocol.
- The client installs the BOOTP network protocol using TCP / IP Intervention on its computer system to ensure compatibility with all network protocols when connected to this network.
- The BOOTP network administrator then sends a message that contains a valid unicast address. This unicast address is then forwarded to the BOOTP client by the master server.

HTTP Protocol

- HTTP is a simple request-response protocol that normally runs over TCP. It specifies what messages clients may send to servers and what responses they get back in return.
- The request and response headers are given in ASCII, just like in SMTP. The contents are given in a MIME(Multipurpose Internet Mail Extensions)-like format, also like in SMTP. This simple model was partly responsible for the early success of the Web because it made development and deployment straightforward.

Features

- **HTTP is connectionless:** The HTTP client, i.e., a browser initiates an HTTP request and after a request is made, the client waits for the response. The server processes the request and sends a response back after which client disconnect the connection. So client and server knows about each other during current request and response only. Further requests are made on new connection like client and server are new to each other.
- **HTTP is media independent:** It means, any type of data can be sent by HTTP as long as both the client and the server know how to handle the data content. It is required for the client as well as the server to specify the content type using appropriate MIME-type.
- **HTTP is stateless:** As mentioned above, HTTP is connectionless and it is a direct result of HTTP being a stateless protocol. The server and client are aware of each other only during a current request. Afterwards, both of them forget about each other.

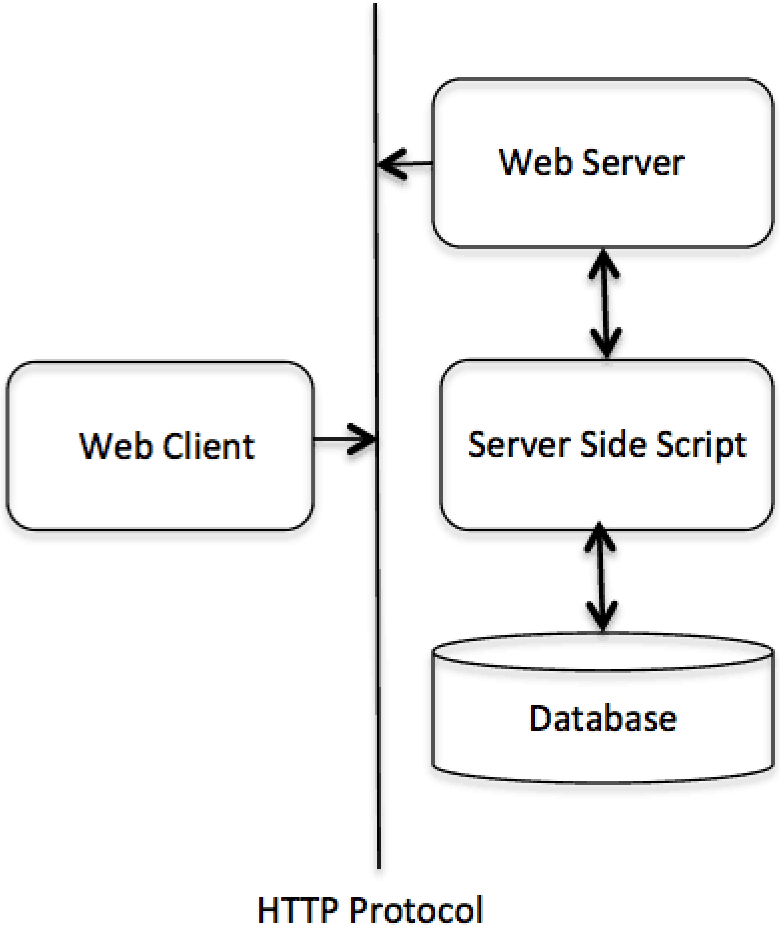
The HTTP protocol is a request/response protocol based on the client/server based architecture

Client

- The HTTP client sends a request to the server in the form of a request method, URI, and protocol version, followed by a MIME-like message containing request modifiers, client information, and possible body content over a TCP/IP connection.

Server

- The HTTP server responds with a status line, including the message's protocol version and a success or error code, followed by a MIME-like message containing server information, entity meta information, and possible entity-body content.



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HTTP Methods

Method	Description
GET	Read a Web page
HEAD	Read a Web page's header
POST	Append to a Web page
PUT	Store a Web page
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Connect through a proxy
OPTIONS	Query options for a page

Note: The names are case sensitive, so *GET* is allowed but not *get*.

HTTP status code response groups

Code	Meaning	Examples
1xx	Information	100 = server agrees to handle client's request
2xx	Success	200 = request succeeded; 204 = no content present
3xx	Redirection	301 = page moved; 304 = cached page still valid
4xx	Client error	403 = forbidden page; 404 = page not found
5xx	Server error	500 = internal server error; 503 = try again later

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World Wide Web

- World Wide Web, which is also known as a Web, is a collection of websites or web pages stored in web servers and connected to local computers through the internet. These websites contain text pages, digital images, audios, videos, etc. Users can access the content of these sites from any part of the world over the internet using their devices such as computers, laptops, cell phones, etc. The WWW, along with internet, enables the retrieval and display of text and media over a device.
- The Web began in 1989 at CERN, the European Center for Nuclear Research by Tim Berners-Lee.
- In 1994, CERN and M.I.T. signed an agreement setting up the **W3C** (**World Wide Web Consortium**), an organization devoted to further developing the Web, standardizing protocols, and encouraging interoperability between sites. Berners- Lee became the director. Since then, several hundred universities and companies have joined the consortium.
- The consortium's home page is at *www.w3.org*.

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Features of WWW:

- Hyper Text Information System
- Cross-Platform
- Distributed
- Open Standards and Open Source
- Uses Web Browsers to provide a single interface for many services
- Dynamic, Interactive and Evolving.

Components of the Web: There are 3 components of the web:

- *Uniform Resource Locator (URL)*
- *Hyper Text Transfer Protocol (HTTP)*
- *Hyper Text Markup Language (HTML)*

Uniform Resource Locater (URL)

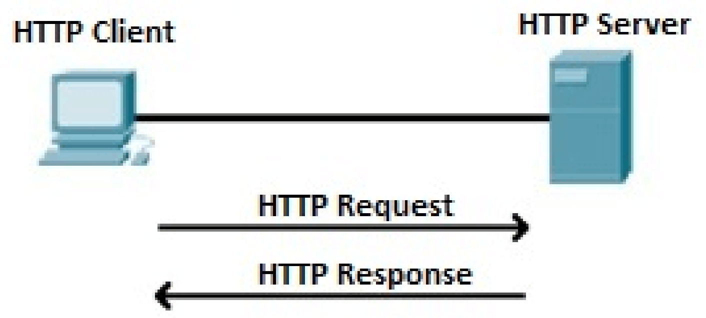
- The URL is a standard for specifying any kind of information on the Internet.
- The URL consists of four parts: protocol, host computer, port and path.
  - The protocol is the client or server program which is used to retrieve the document or file. The protocol can be ftp or http.
  - The host is the name of computer on which the information is located.
  - The URL can optionally contain the port number and it is separated from the host name by a colon.
  - Path is the pathname of the file where the file is stored.

Name	Used for	Example
http	Hypertext (HTML)	http://www.ee.uwa.edu/~rob/
https	Hypertext with security	https://www.bank.com/accounts/
ftp	FTP	ftp://ftp.cs.vu.nl/pub/minix/README
file	Local file	file:///usr/suzanne/prog.c
mailto	Sending email	mailto:JohnUser@acm.org
rtsp	Streaming media	rtsp://youtube.com/montypython.mpg
sip	Multimedia calls	sip:eve@adversary.com
about	Browser information	about:plugins

Some common URL schemes

Hypertext Transfer Protocol (HTTP)

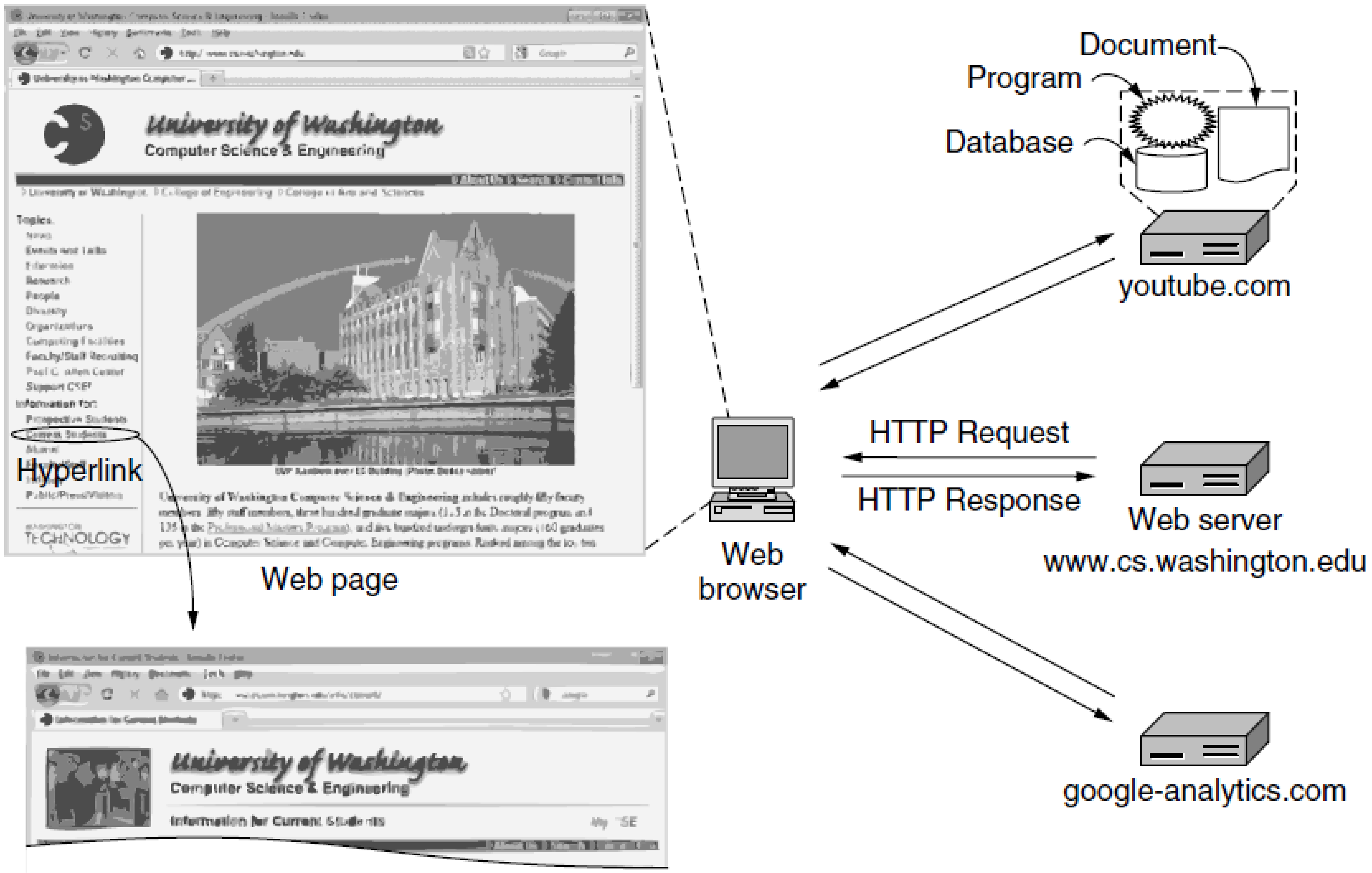
- Hyper Text Transfer Protocol (HTTP) is an application layer protocol which enables WWW to work smoothly and effectively.
- It is based on a client-server model. The client is a web browser which communicates with the web server which hosts the website.
- This protocol defines how messages are formatted and transmitted and what actions the Web Server and browser should take in response to different commands.



- When you enter a URL in the browser, an HTTP command is sent to the Web server, and it transmits the requested Web Page.

Hypertext Markup Language (HTML):

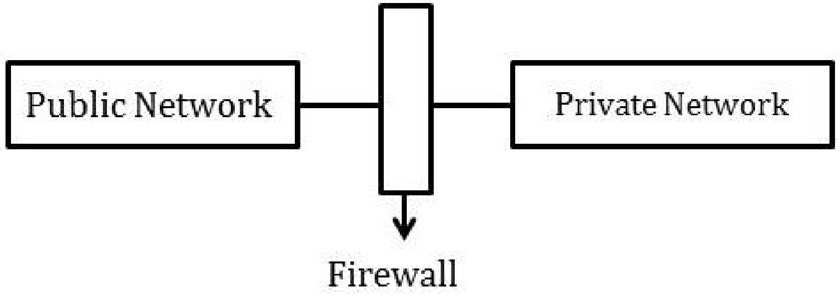
- HTML is a standard markup language which is used for creating web pages.
- It describes the structure of web pages through HTML elements or tags.
- These tags are used to organize the pieces of content such as 'heading,' 'paragraph,' 'table,' 'Image,' and more.
- We don't see HTML tags when we open a webpage as browsers don't display the tags and use them only to render the content of a web page.
- In simple words, HTML is used to display text, images, and other resources through a Web browser.



Architecture of the Web

Firewalls

- A firewall is a network security device; it is a protective layer for the server that monitors and filters all the incoming and outgoing network traffic. It uses a set of rules to determine whether to allow or block a specific network traffic. Firewalls can prevent unauthorized use before reaching the servers. Firewalls can be hardware or software-based.
- To protect private networks and individual machines, a firewall can be employed to filter incoming or outgoing traffic based on a predefined set of rules known as *firewall policies*.



Packet flowing through a firewall can have one of the following three outcomes −

- **Accepted** − Permitted through the firewall.
- **Dropped** − Not allowed through with no reply
- **Rejected** − Not allowed through accompanied by an attempt to inform the source that the packet was rejected.

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Types of Firewall

1. Packet Filters (Stateless Firewall)

- A packet filter firewall can forward or block packets based on the information in the network layer and transport layer headers source and destination IP addresses, source, and destination port address, and type of protocol (TCP and UDP).
- A packet filter firewall is a router that uses a filtering table to decide which packet must be discarded or not to forward. It filters at the network or transport layer.

	Source IP	Dest. IP	Source Port	Dest. Port	Action
1	192.168.21.0	--	--	--	deny
2	--	--	--	23	deny
3	--	192.168.21.3	--	--	deny
4	--	192.168.21.0	--	>1023	Allow

Sample Packet Filter Firewall Rule

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2. Stateful firewall filters − It is also known as a network firewall; this filter maintains a record of all the connections passing through. It can determine if a packet is either the start of a new connection or a part of an existing connection or is an invalid packet.

3. Application firewall − A web application firewall is used for HTTP applications. There are sets of rules that are applied to monitor or block data packets from HTTP network traffic. For example, these rules can help block cross-site scripting (XSS) and SQL injections.

Firewalls are either *Host-based* or *Network-based*

- Host-based firewall is installed on each network node which controls each incoming and outgoing packet. It is a software application or suite of applications, comes as a part of the operating system. Host firewall protects each host from attacks and unauthorized access.
- Network -based firewall function on network level. In other words, these firewalls filter all incoming and outgoing traffic across the network.

That's all about

UNIT 5

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